

Since y denotes $f(x)$, where f is a function, the Product Rule must be applied to $D_x(4xy^3)$ and $D_x(x^2y)$. Thus

$$\begin{aligned} D_x(4xy^3) &= 4xD_x(y^3) + y^3D_x(4x) \\ &= 4x(3y^2y') + y^3(4) \\ &= 12xy^2y' + 4y^3 \end{aligned}$$

and

$$\begin{aligned} D_x(x^2y) &= x^2D_xy + yD_x(x^2) \\ &= x^2y' + y(2x). \end{aligned}$$

Substituting these in the first equation of the solution and differentiating the other terms leads to

$$(12xy^2y' + 4y^3) - (x^2y' + 2xy) + 3x^2 - 5 = 0.$$

Collecting the terms containing y' and transposing the remaining terms to the right side of the equation gives us

$$(12xy^2 - x^2)y' = 5 - 3x^2 + 2xy - 4y^3.$$

Consequently,
$$y' = \frac{5 - 3x^2 + 2xy - 4y^3}{12xy^2 - x^2}.$$
 ■

Example 6 Find y' if $y = x^2 \sin y$.

Solution Differentiating both sides of the given equation with respect to x and using the Product Rule, we obtain

$$D_x y = x^2 D_x(\sin y) + (\sin y) D_x(x^2).$$

Since $y = f(x)$ for some (implicit) function f we have, by Theorem (3.33),

$$D_x \sin y = (\cos y) D_x y.$$

Using this, and the fact that $D_x(x^2) = 2x$, the first equation of our solution may be written

$$D_x y = (x^2 \cos y) D_x y + (\sin y)(2x)$$

or equivalently,

$$y' = (x^2 \cos y)y' + 2x \sin y.$$

Finally, we solve for y' as follows:

$$y' - (x^2 \cos y)y' = 2x \sin y$$

$$(1 - x^2 \cos y)y' = 2x \sin y$$

$$y' = \frac{2x \sin y}{1 - x^2 \cos y}. \quad \blacksquare$$

3.7 Exercises

Find at least one implicit function f determined by each of the equations in Exercises 1–8 and state the domain of f .

- 1 $3x - 2y + 4 = 2x^2 + 3y - 7x$
- 2 $x^3 - xy + 4y = 1$
- 3 $x^2 + y^2 - 16 = 0$
- 4 $3x^2 - 4y^2 = 12$
- 5 $x^2 - 2xy + y^2 = x$
- 6 $3x^2 - 4xy + y^2 = 0$
- 7 $\sqrt{x} + \sqrt{y} = 1$
- 8 $|x - y| = 2$

Assuming that each of the equations in Exercises 9–20 determines a function f such that $y = f(x)$, find y' .

- 9 $8x^2 + y^2 = 10$
- 10 $4x^3 - 2y^3 = x$
- 11 $2x^3 + x^2y + y^3 = 1$
- 12 $5x^2 + 2x^2y + y^2 = 8$
- 13 $5x^2 - xy - 4y^2 = 0$
- 14 $x^4 + 4x^2y^2 - 3xy^3 + 2x = 0$
- 15 $\sin^2 3y = x + y - 1$
- 16 $x = \sin xy$
- 17 $x^2y^3 + 4xy + x - 6y = 2$
- 18 $4 - 7xy = (y^2 + 4)^5$
- 19 $y^2 = x \cos y$

20 $\cos(x - y) = y \cos x$

In Exercises 21–24 find an equation of the tangent line to the graph of the given equation at the indicated point P .

- 21 $xy + 16 = 0$, $P(-2, 8)$
- 22 $y^2 - 4x^2 = 5$, $P(-1, 3)$
- 23 $2x^3 - x^2y + y^3 - 1 = 0$, $P(2, -3)$
- 24 $3y^4 + 4x - x^2 \sin y - 4 = 0$, $P(1, 0)$
- 25 Show that the equation $x^2 + y^2 + 1 = 0$ does not determine a function f such that $y = f(x)$.
- 26 Show that the equation $y^2 = x$ determines an infinite number of implicit functions. Sketch the graphs of four such functions.
- 27 How many implicit functions are determined by the following equations?
 - (a) $x^4 + y^4 - 1 = 0$
 - (b) $x^4 + y^4 = 0$
 - (c) $\cos x + \sin y = 3$
- 28 Use implicit differentiation to prove that if P is any point on the circle $x^2 + y^2 = a^2$, then the tangent line at P is perpendicular to OP .
- 29 Suppose that $3x^2 - x^2y^3 + 4y = 12$ determines a differentiable function f such that $y = f(x)$. If $f(2) = 0$, use differentials to approximate the change in $f(x)$ if x changes from 2 to 1.97.
- 30 Suppose that $x^3 + xy + y^4 = 19$ determines a differentiable function f such that $y = f(x)$. If $P(1, 2)$ is a point on the graph of f , use differentials to approximate the y -coordinate b of the point $Q(1.10, b)$ on the graph.

3.8 Derivatives Involving Powers of Functions

The Power Rule (3.13) can be extended to the case where the exponent is a rational number. Thus, if

$$y = x^{m/n}$$

where m and n are integers and $n \neq 0$, then

$$y^n = x^m.$$

Assuming that y' exists and no zero denominators occur we have, by implicit differentiation,

$$ny^{n-1} \frac{dy}{dx} = mx^{m-1},$$

or
$$\frac{dy}{dx} = \frac{m}{n} x^{m-1} y^{1-n} = \frac{m}{n} x^{m-1} (x^{m/n})^{1-n}.$$

Simplifying the right side by means of laws of exponents gives us the following **Power Rule for Rational Exponents**: If m and n are integers and $n \neq 0$, then

$$D_x(x^{m/n}) = \frac{m}{n} x^{(m/n)-1}.$$

Our proof of this formula is incomplete in the sense that we have assumed the existence of y' . To give a complete proof the definition of derivative (3.6) could be used.

Example 1 Find y' if $y = 6\sqrt[3]{x^4} + 4/\sqrt{x}$.

Solution Introducing rational exponents, $y = 6x^{4/3} + 4x^{-1/2}$. Using the Power Rule,

$$\begin{aligned} y' &= 6(4/3)x^{(4/3)-1} + 4(-1/2)x^{-(1/2)-1} \\ &= 8x^{1/3} - 2x^{-3/2} \\ &= 8x^{1/3} - 2/x^{3/2} \end{aligned}$$

or, in terms of radicals,

$$y' = 8\sqrt[3]{x} - 2/\sqrt{x^3}$$

for all $x > 0$. ■

It can also be shown that the Power Rule for Functions (3.31) is true for every *rational* exponent. The following examples illustrate uses of this rule.

Example 2 Find $f'(x)$ if $f(x) = \sqrt[3]{5x^2 - x + 4}$.

Solution Writing $f(x) = (5x^2 - x + 4)^{1/3}$ and using the Power Rule for Functions with $n = 1/3$,

$$\begin{aligned} f'(x) &= \frac{1}{3}(5x^2 - x + 4)^{-2/3} D_x(5x^2 - x + 4) \\ &= \left(\frac{1}{3}\right) \frac{1}{(5x^2 - x + 4)^{2/3}} (10x - 1) \\ &= \frac{10x - 1}{3\sqrt[3]{(5x^2 - x + 4)^2}}. \end{aligned} \quad \blacksquare$$

Example 3 Find dy/dx if $y = (3x + 1)^6\sqrt{2x - 5}$.

Solution Since $y = (3x + 1)^6(2x - 5)^{1/2}$, we have, by the Product and Power Rules,

$$\begin{aligned}\frac{dy}{dx} &= (3x + 1)^6 \frac{1}{2}(2x - 5)^{-1/2}(2) + (2x - 5)^{1/2} 6(3x + 1)^5(3) \\ &= \frac{(3x + 1)^6}{\sqrt{2x - 5}} + 18(3x + 1)^5\sqrt{2x - 5} \\ &= \frac{(3x + 1)^6 + 18(3x + 1)^5(2x - 5)}{\sqrt{2x - 5}} \\ &= \frac{(3x + 1)^5(39x - 89)}{\sqrt{2x - 5}}. \quad \blacksquare\end{aligned}$$

Example 4 Find y' if $y = \sqrt{\sin 6x}$.

Solution Writing $y = (\sin 6x)^{1/2}$ and using the Power Rule for Functions we obtain

$$y' = \frac{1}{2}(\sin 6x)^{-1/2} D_x \sin 6x.$$

Next, by Theorem (3.33),

$$D_x \sin 6x = (\cos 6x) D_x(6x) = (\cos 6x)(6) = 6 \cos 6x.$$

Consequently,

$$\begin{aligned}y' &= \frac{1}{2}(\sin 6x)^{-1/2}(6 \cos 6x) \\ &= \frac{3 \cos 6x}{(\sin 6x)^{1/2}} = \frac{3 \cos 6x}{\sqrt{\sin 6x}}. \quad \blacksquare\end{aligned}$$

The next example is of interest because it illustrates the fact that after the Power Rule is applied to $[g(x)]'$, it may be necessary to apply it again in order to find $g'(x)$.

Example 5 Find $f'(x)$ if $f(x) = (7x + \sqrt{x^2 + 6})^4$.

Solution Applying the Power Rule,

$$\begin{aligned}f'(x) &= 4(7x + \sqrt{x^2 + 6})^3 D_x(7x + \sqrt{x^2 + 6}) \\ &= 4(7x + \sqrt{x^2 + 6})^3 [D_x(7x) + D_x\sqrt{x^2 + 6}].\end{aligned}$$

Again applying the Power Rule,

$$\begin{aligned}D_x\sqrt{x^2 + 6} &= D_x(x^2 + 6)^{1/2} = \frac{1}{2}(x^2 + 6)^{-1/2} D_x(x^2 + 6) \\ &= \frac{1}{2\sqrt{x^2 + 6}}(2x) = \frac{x}{\sqrt{x^2 + 6}}.\end{aligned}$$

Therefore,

$$f'(x) = 4(7x + \sqrt{x^2 + 6})^3 \left[7 + \frac{x}{\sqrt{x^2 + 6}} \right]. \quad \blacksquare$$

3.8 Exercises

Differentiate the functions defined in Exercises 1–26.

- 1 $f(x) = \sqrt[3]{x^2} + 4\sqrt{x^3}$
- 2 $f(x) = 10\sqrt[5]{x^3} + \sqrt[3]{x^5}$
- 3 $k(r) = \sqrt[3]{8r^3 + 27}$
- 4 $h(z) = (2z^2 - 9z + 8)^{-2/3}$
- 5 $F(v) = 5\sqrt[5]{v^5 - 32}$
- 6 $k(s) = 1/\sqrt{3s - 4}$
- 7 $f(x) = \sqrt{2x}$
- 8 $g(x) = \sqrt[5]{1/x}$
- 9 $f(z) = 10\sqrt{z^3} + 3/\sqrt[3]{z}$
- 10 $f(t) = \sqrt[3]{t^2} - (1/\sqrt{t^3})$
- 11 $g(w) = (w^2 - 4w + 3)/w^{3/2}$
- 12 $K(x) = 8x^2\sqrt{x} + 3x\sqrt[3]{x}$
- 13 $f(x) = \sin \sqrt{x}$
- 14 $f(x) = \cos(6x - 5)^{3/2}$
- 15 $k(\theta) = \cos^2 \sqrt{3 - 8\theta}$
- 16 $r(t) = \sqrt{\sin 2t - \cos 2t}$
- 17 $g(x) = \sqrt{x^2 + 1} \sin \sqrt{x^2 + 1}$
- 18 $h(\phi) = \frac{\sin 4\phi}{\sqrt{\phi^2 + 4}}$
- 19 $M(x) = \sqrt{4x^2 - 7x + 4}$
- 20 $F(s) = \sqrt[3]{5s - 8}$
- 21 $f(t) = 4/(9t^2 + 16)^{2/3}$
- 22 $k(v) = 1/\sqrt{(v^4 + 7v^2)^3}$
- 23 $H(u) = \sqrt{\frac{3u + 8}{2u + 5}}$
- 24 $G(x) = \left(\frac{x}{x^2 + 1} \right)^{5/2}$
- 25 $k(s) = \sqrt[4]{s^2 + 9(4s + 5)^4}$

$$26 \quad g(y) = (15y + 2)(y^2 - 2)^{3/4}$$

$$27 \quad h(x) = \sqrt{4 + \cos^2 3x}$$

$$28 \quad f(t) = \sin^2 2t \sqrt{\cos 2t}$$

$$29 \quad g(\theta) = \sqrt{\sin 2\theta}/\theta$$

$$30 \quad H(x) = (2x + 3)/\sqrt{4x^2 + 9}$$

$$31 \quad f(x) = (7x + \sqrt{x^2 + 3})^6$$

$$32 \quad p(z) = \sqrt{1 + \sqrt{1 + 2z}}$$

In Exercises 33–36 find an equation of the tangent line to the graph of the equation at the point P .

$$33 \quad y = \sqrt{2x^2 + 1}, P(-1, \sqrt{3})$$

$$34 \quad y = (5x - 8)^{1/3}, P(7, 3)$$

$$35 \quad y = 4x^{2/3} + 2x^{-1/3} - 10, P(-8, 5)$$

$$36 \quad y = (4x + 1)/\sqrt{\cos x}, P(0, 1)$$

37 Find the point P on the graph of $y = \sqrt{2x - 4}$ such that the tangent line at P passes through the origin.

38 Find the points on the graph of $y = x^{5/3} + x^{1/3}$ at which the tangent line is perpendicular to the line $2y + x = 7$.

Assuming that each of the equations in Exercises 39–44 determines a function f such that $y = f(x)$, find y' .

$$39 \quad \sqrt{x} + \sqrt{y} = 100$$

$$40 \quad x^{2/3} + y^{2/3} = 4$$

$$41 \quad 6x + \sqrt{xy} - 3y = 4$$

$$42 \quad xy^2 + \sqrt[3]{xy} + x^4 = 7$$

$$43 \quad x^2 + \sqrt{\sin y} - y^2 = 1$$

$$44 \quad 2x - \sqrt{xy} + y^3 = 16$$

In Exercises 45 and 46 find dy .

$$45 \quad y = \sqrt[3]{6x + 11}$$

$$46 \quad y = 1/\sqrt{x^2 + 1}$$

47 Use differentials to approximate $\sqrt[3]{65}$.

- 48 Use differentials to approximate $\sqrt{99}$.
- 49 Use differentials to approximate the change in $f(x) = (4x^2 + 9)^{3/2}$ if x changes from 2 to 1.998.
- 50 Use differentials to approximate
- $$N = 5(1.01)^{3/5} - 3(1.01)^{1/5} + 7.$$
- 51 The curved surface area S of a right circular cone having altitude h and base radius r is given by $S = \pi r \sqrt{r^2 + h^2}$. For a certain cone, $r = 6$ cm and the altitude is measured as 8 cm, with a maximum error in measurement of 0.1 cm. Calculate S from these measurements and use differentials

to estimate the maximum error in the calculation. What is the approximate percentage error?

- 52 The period T of a simple pendulum of length l may be calculated by means of the formula $T = 2\pi\sqrt{l/g}$, where g is a constant. Use differentials to approximate the change in l that will increase T by 1%.

In Exercises 53–56, use the fact that $|a| = \sqrt{a^2}$ to find $f'(x)$. In addition, find the domain of f' and sketch the graph of f .

- 53 $f(x) = |1 - x|$ 54 $f(x) = \sin |(\pi/2) - x|$
- 55 $f(x) = |1 - x^2|$ 56 $f(x) = x/|x|$

3.9 Higher Order Derivatives

The derivative of a function f leads to another function f' . If f' has a derivative, it is denoted by f'' and is called the **second derivative** of f . Thus

$$f''(x) = D_x(f'(x)) = D_x(D_x(f(x)))$$

for every x such that the indicated derivatives exist. The expression on the right in this equation is abbreviated $D_x^2 f(x)$. In like manner the **third derivative** f''' of f is the derivative of the second derivative. Specifically,

$$f'''(x) = D_x(f''(x)) = D_x(D_x^2 f(x)) = D_x^3 f(x).$$

In general, if n is a positive integer, then $f^{(n)}$ denotes the n th derivative of f and is found by starting with f and differentiating, successively, n times. Using operator notation, $f^{(n)}(x) = D_x^n f(x)$. The integer n is called the **order** of the derivative $f^{(n)}(x)$.

If $y = f(x)$, then the first n derivatives are denoted by

$$D_x y, \quad D_x^2 y, \quad D_x^3 y, \quad \dots, \quad D_x^n y$$

or

$$y', \quad y'', \quad y''', \quad \dots, \quad y^{(n)}.$$

If the differential notation is used we write

$$\frac{dy}{dx}, \quad \frac{d^2 y}{dx^2}, \quad \frac{d^3 y}{dx^3}, \quad \dots, \quad \frac{d^n y}{dx^n}.$$

In this case, the symbols used for higher derivatives should *not* be interpreted as quotients.

Example 1 If $f(x) = 4x^2 - 5x + 8 - (3/x)$, find the first four derivatives of $f(x)$.

Solution Since $f(x) = 4x^2 - 5x + 8 - 3x^{-1}$,

$$f'(x) = 8x - 5 + 3x^{-2} = 8x - 5 + \frac{3}{x^2}$$

$$f''(x) = 8 - 6x^{-3} = 8 - \frac{6}{x^3}$$

$$f'''(x) = 18x^{-4} = \frac{18}{x^4}$$

$$f^{(4)}(x) = -72x^{-5} = -\frac{72}{x^5}. \quad \blacksquare$$

Example 2 Find the first eight derivatives of $f(x) = \sin x$.

Solution Applying Theorem (3.20),

$$f'(x) = D_x \sin x = \cos x$$

$$f''(x) = D_x \cos x = -\sin x$$

$$f'''(x) = D_x(-\sin x) = -D_x(\sin x) = -\cos x$$

$$f^{(4)}(x) = D_x(-\cos x) = -D_x(\cos x) = -(-\sin x) = \sin x$$

Since $f^{(4)}(x) = \sin x$, it follows that if we continue differentiating, the same pattern repeats, that is,

$$f^{(5)}(x) = \cos x, \quad f^{(6)}(x) = -\sin x,$$

$$f^{(7)}(x) = -\cos x \quad f^{(8)}(x) = \sin x. \quad \blacksquare$$

The next example illustrates a technique for finding higher derivatives of implicit functions.

Example 3 Find y'' if $y^4 + 3y - 4x^3 = 5x + 1$.

Solution The given equation was investigated in Example 3 of Section 3.7, where we found that

$$y' = (12x^2 + 5)/(4y^3 + 3).$$

Hence,
$$y'' = D_x(y') = D_x\left(\frac{12x^2 + 5}{4y^3 + 3}\right).$$

We now use the quotient rule, differentiating implicitly as follows:

$$\begin{aligned} y'' &= \frac{(4y^3 + 3)D_x(12x^2 + 5) - (12x^2 + 5)D_x(4y^3 + 3)}{(4y^3 + 3)^2} \\ &= \frac{(4y^3 + 3)(24x) - (12x^2 + 5)(12y^2y')}{(4y^3 + 3)^2} \end{aligned}$$

Substituting for y' yields

$$\begin{aligned} y'' &= \frac{(4y^3 + 3)(24x) - (12x^2 + 5) \cdot 12y^2 \left(\frac{12x^2 + 5}{4y^3 + 3} \right)}{(4y^3 + 3)^2} \\ &= \frac{(4y^3 + 3)^2(24x) - 12y^2(12x^2 + 5)^2}{(4y^3 + 3)^3}. \end{aligned}$$

■

There are numerous mathematical and physical applications of higher derivatives. Some of them will be discussed in the next chapter.

3.9 Exercises

Find the first and second derivatives of the functions defined in Exercises 1–14.

1 $f(x) = 3x^4 - 4x^2 + x - 2$

2 $g(x) = 3x^8 - 2x^5$

3 $H(s) = \sqrt[3]{s} + (2/s^2)$

4 $F(t) = t^{3/2} - 2t^{1/2} + 4t^{-1/2}$

5 $g(z) = \sqrt{3z + 1}$

6 $k(s) = (s^2 + 4)^{2/3}$

7 $k(r) = (4r + 7)^5$

8 $f(x) = \sqrt[5]{10x + 7}$

9 $f(x) = \sqrt{x^2 + 4}$

10 $h(x) = 1$

11 $f(x) = \sin^3 x$

12 $g(x) = x^2 \cos 2x$

13 $f(x) = 1/(1 - \cos 3x)$

14 $H(\theta) = \sqrt{1 + \sin \theta}$

In Exercises 15–22 find $D_x^3 y$.

15 $y = 2x^5 + 3x^3 - 4x + 1$

16 $y = \sqrt{2 - 5x}$

17 $y = \cos 3x$

18 $y = \sin(x^2)$

19 $y = (2x - 3)/(3x + 1)$

20 $y = 1/(x^2 + 4)$

21 $y = \sqrt[3]{2 - 9x}$

22 $y = (3x + 1)^4$

Assuming that each of the equations in Exercises 23–26 determines a function f , such that $y = f(x)$, find y'' .

23 $x^3 - y^3 = 1$

24 $x^2 y^3 = 1$

25 $x^2 - 3xy + y^2 = 4$

26 $\sqrt{xy} - y + x = 0$

27 Find all the nonzero derivatives of

$$f(x) = x^6 - 2x^4 + 3x^3 - x + 2.$$

28 Find all the nonzero derivatives of $f(x) = (x^2 - 1)^3$.

29 If $f(x) = 1/x$, find a formula for $f^{(n)}(x)$ where n is any positive integer. What is $f^{(n)}(1)$?

30 If $f(x) = \sqrt{x}$, find a formula for $f^{(n)}(x)$ where n is any positive integer.

31 If $f(x)$ is a polynomial of degree n , show that $f^{(k)}(x) = 0$ if $k > n$.

32 Find a polynomial $f(x)$ of degree 2 such that $f(1) = 5$, $f'(1) = 3$, and $f''(1) = -4$.

33 If $f(x) = 2x^3 + 3x^2 - 12x + 7$, find the value of f'' at each zero of f' .

34 If $f(x) = x^4 - x^3 - 6x^2 + 7x$, find an equation of the tangent line to the graph of f' at the point $P(2, 3)$.

35 If $f(x) = x^4 - 10x^2 + x + 2$, use differentials to approximate the change in $f'(x)$ if x changes from 2 to 2.005.

- 36 Suppose $u = f(x)$ and $v = g(x)$ where f and g have derivatives of all orders. If $y = uv$, prove that

$$y' = u'v + 2u'v' + uv''$$

$$y''' = u'''v + 3u''v' + 3u'v'' + uv'''$$

$$y^{(4)} = u^{(4)}v + 4u'''v' + 6u''v'' + 4u'v''' + uv^{(4)}.$$

Formulate a conjecture for $y^{(n)}$.

- 37 If $y = f(g(x))$ and f'' and g'' exist, use the Chain Rule to express $D_x^2 y$ in terms of the first and second derivatives of f and g .

- 38 Suppose f has a second derivative. Prove that (a) if f is an even function, then f'' is even; and (b) if f is an odd function, then f'' is odd. Illustrate these facts by using polynomial functions and the sine and cosine functions. (Hint: See Exercise 60 of Section 3.6.)

In Exercises 39 and 40, use mathematical induction to prove that the given statement is true for every positive integer n .

39 If $f(x) = \sin x$, then $f^{(n)}(x) = \sin [x + (n\pi)/2]$.

40 If $f(x) = \cos x$, then $f^{(n)}(x) = \cos [x + (n\pi)/2]$.

3.10 Newton's Method

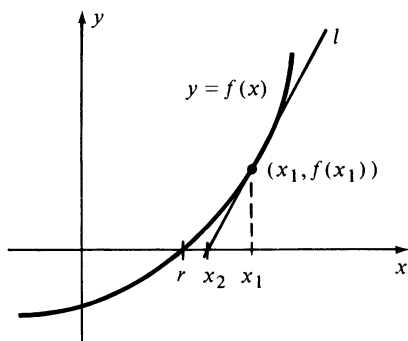


FIGURE 3.15

In this section we shall describe a method for approximating a real zero of a differentiable function f , that is, a real number r such that $f(r) = 0$. To use the method we begin by making a first approximation x_1 to the zero r . Since r is an x -intercept of the graph of f , a suitable number x_1 can usually be found by referring to a rough sketch of the graph. If we consider the tangent line l to the graph of f at the point $(x_1, f(x_1))$, and if x_1 is sufficiently close to r , then as illustrated in Figure 3.15, the x -intercept x_2 of l should be a better approximation to r .

Since the slope of l is $f'(x_1)$, an equation of the tangent line is

$$y - f(x_1) = f'(x_1)(x - x_1).$$

The x -intercept x_2 is given by

$$0 - f(x_1) = f'(x_1)(x_2 - x_1).$$

If $f'(x_1) \neq 0$, the preceding equation is equivalent to

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}.$$

If we take x_2 as a second approximation to r , then the process may be repeated by using the tangent line at $(x_2, f(x_2))$. If $f'(x_2) \neq 0$, this leads to a third approximation x_3 , given by

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}.$$

The process is continued until the desired degree of accuracy is obtained. This technique of successive approximations of real zeros is referred to as **Newton's Method**, which we state as follows.

Newton's Method (3.35)

Let f be a differentiable function and suppose r is a real zero of f . If x_n is an approximation to r , then the next approximation is given by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

provided $f'(x_n) \neq 0$.

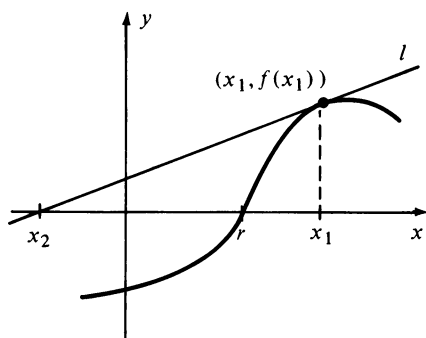


FIGURE 3.16

Newton's Method does not guarantee that x_{n+1} is a better approximation to r than x_n for every n . In particular, some care must be exercised in choosing the first approximation x_1 . Indeed, if x_1 is not sufficiently close to r , it is possible for the second approximation x_2 to be worse than x_1 , as illustrated in Figure 3.16. It is evident that we should not choose a number x_n such that $f'(x_n)$ is close to 0, for then the tangent line l is almost horizontal.

We shall use the following rule when applying Newton's Method. If an approximation to k decimal places is required, we shall approximate each of the numbers $x_2, x_3, \dots$ to k decimal places, continuing the process until two consecutive approximations are the same. This is illustrated in the following examples.

Example 1 Use Newton's Method to approximate $\sqrt{7}$ to five decimal places.

Solution The stated problem is equivalent to that of approximating the positive real zero r of $f(x) = x^2 - 7$. Since $f(2) = -3$ and $f(3) = 2$, it follows from the continuity of f that $2 < r < 3$. Moreover, since f is increasing there can be only one zero in the interval $(2, 3)$.

If x_n is any approximation to r , then by (3.35) the next approximation x_{n+1} is given by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 7}{2x_n}.$$

Let us choose $x_1 = 2.5$ as a first approximation. Using the formula for x_{n+1} with $n = 1$ gives us

$$x_2 = 2.5 - \frac{(2.5)^2 - 7}{2(2.5)} = 2.65000.$$

Again using the formula (with $n = 2$) we obtain the next approximation

$$x_3 = 2.65000 - \frac{(2.65000)^2 - 7}{2(2.65000)} \approx 2.64575.$$

Repeating the procedure (with $n = 3$),

$$x_4 = 2.64575 - \frac{(2.64575)^2 - 7}{2(2.64575)} \approx 2.64575.$$

Since two consecutive values of x_n are the same (to the desired degree of accuracy) we have $\sqrt{7} \approx 2.64575$. It can be shown that to nine decimal places, $\sqrt{7} \approx 2.645751311$. ■

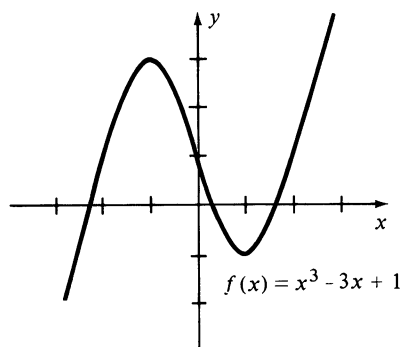


FIGURE 3.17

Example 2 Find, to four decimal places, the largest positive real root of the equation $x^3 - 3x + 1 = 0$.

Solution If we let $f(x) = x^3 - 3x + 1$, then the problem is equivalent to finding the largest positive real zero of f . The graph of f is sketched in Figure 3.17. Note that f has three real zeros. We wish to find the zero that lies between 1 and 2. Since $f'(x) = 3x^2 - 3$, the formula for x_{n+1} in Newton's Method is

$$x_{n+1} = x_n - \frac{x_n^3 - 3x_n + 1}{3x_n^2 - 3}.$$

Referring to the graph we take, as a first approximation, $x_1 = 1.5$ and proceed as follows:

$$x_2 = 1.5 - \frac{(1.5)^3 - 3(1.5) + 1}{3(1.5)^2 - 3} \approx 1.5333$$

$$x_3 = 1.5333 - \frac{(1.5333)^3 - 3(1.5333) + 1}{3(1.5333)^2 - 3} \approx 1.5321$$

$$x_4 = 1.5321 - \frac{(1.5321)^3 - 3(1.5321) + 1}{3(1.5321)^2 - 3} \approx 1.5321.$$

Thus the desired approximation is 1.5321. The remaining two real roots can be approximated in similar fashion (see Exercise 15). ■

Example 3 Approximate, to three decimal places, the real root of $x - \cos x = 0$.

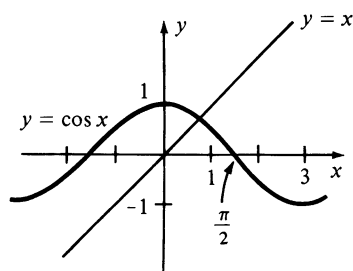


FIGURE 3.18

Solution We wish to find a value of x such that $\cos x = x$. This coincides with the x -coordinate of the point of intersection of the graphs of $y = \cos x$ and $y = x$. It appears, from Figure 3.18, that $x_1 = 0.8$ is a reasonable first approximation. (Note that the figure also indicates that there is only one real root of the given equation.)

If we let $f(x) = x - \cos x$, then $f'(x) = 1 + \sin x$ and the formula in Newton's Method is

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n}.$$

Following the usual procedure gives us the following, where a calculator should be used to evaluate the sine and cosine functions.

$$x_2 = 0.8 - \frac{0.8 - \cos 0.8}{1 + \sin 0.8} \approx 0.740$$

$$x_3 = 0.740 - \frac{0.740 - \cos 0.740}{1 + \sin 0.740} \approx 0.739$$

$$x_4 = 0.739 - \frac{0.739 - \cos 0.739}{1 + \sin 0.739} \approx 0.739$$

Hence 0.739 is the desired approximation. ■

3.10 Exercises

In Exercises 1 and 2 use Newton's Method to approximate the given number to four decimal places.

1 $\sqrt[3]{2}$

2 $\sqrt[5]{3}$

In Exercises 3–10 use Newton's Method to approximate the indicated real root to four decimal places.

3 The positive root of $x^3 + 5x - 3 = 0$

4 The largest root of $2x^3 - 4x^2 - 3x + 1 = 0$

5 The root of $x^4 + 2x^3 - 5x^2 + 1 = 0$ which is between 1 and 2

6 The root of $x^4 - 5x^2 + 2x - 5 = 0$ which is between 2 and 3

7 The root of $x^5 + x^2 - 9x - 3 = 0$ between -2 and -1 .

8 The root of $\cos x + x = 2$

9 The positive root of $2x - 3 \sin x = 0$

10 The root of $\sin x + x \cos x = \cos x$ that is between 0 and 1.

In Exercises 11–18 use Newton's Method to find all real roots of the given equation to two decimal places.

11 $x^4 = 125$

12 $10x^2 - 1 = 0$

13 $x^4 - x - 2 = 0$

14 $x^5 - 2x^2 + 4 = 0$

15 $x^3 - 3x + 1 = 0$

16 $x^3 + 2x^2 - 8x - 3 = 0$

17 $2x - 5 - \sin x = 0$

18 $x^2 - \cos 2x = 0$

3.11 Review: Concepts

Define or explain each of the following.

1 The slope of the tangent line to the graph of a function

2 The velocity of a point moving rectilinearly

3 The derivative of a function

4 Differentiable function

5 Increment notation

6 Differentials

7 Right-hand derivative of a function

8 Left-hand derivative of a function

9 Differentiability on an interval

10 The relationship between continuity and differentiability of a function

11 The Product Rule

12 The Quotient Rule

13 Derivatives of the sine and cosine functions

14 The Chain Rule

15 The Power Rule for Functions

16 Implicit differentiation

17 Higher order derivatives of functions

18 Newton's Method

Review: Exercises

In Exercises 1 and 2 find $f'(x)$ directly from the definition of derivative.

1 $f(x) = 4/(3x^2 + 2)$

2 $f(x) = \sqrt{5 - 7x}$

Find the first derivatives in Exercises 3–36.

3 $f(x) = 2x^3 - 7x + 2$

4 $k(x) = 1/(x^4 - x^2 + 1)$

5 $g(t) = \sqrt{6t + 5}$

6 $h(t) = 1/\sqrt{6t + 5}$

7 $F(z) = \sqrt[3]{7z^2 - 4z + 3}$

8 $f(w) = \sqrt[5]{3w^2}$

9 $G(x) = 6/(3x^2 - 1)^4$

10 $H(x) = (3x^2 - 1)^4/6$

11 $F(y) = (y^2 - y^{-2})^{-2}$

12 $h(z) = [(z^2 - 1)^5 - 1]^5$

13 $g(x) = \sqrt[5]{(3x + 2)^4}$

14 $P(x) = (x + x^{-1})^2$

15 $r(s) = \left(\frac{8s^2 - 4}{1 - 9s^3} \right)^4$

16 $g(w) = \frac{(w - 1)(w - 3)}{(w + 1)(w + 3)}$

17 $F(x) = (x^6 + 1)^5(3x + 2)^3$

18 $k(z) = (z^2 + (z^2 + 9)^{1/2})^{1/2}$

19 $g(y) = \sqrt{1 + \cos 2y}$

20 $p(x) = (2x^4 + 3x^2 - 1)/x^2$

21 $f(x) = \sin^2(4x^3)$

22 $H(t) = (1 + \sin 3t)^3$

23 $h(x) = \sqrt{x + \sqrt{x + \sqrt{x}}}$

24 $K(r) = \sqrt{r}\sqrt{r+1}\sqrt{r+2}$

25 $f(x) = \sqrt[3]{2x + 3}/\sqrt{3x + 2}$

26 $f(x) = 6x^2 - (5/x) + (2/\sqrt[3]{x^2})$

27 $g(z) = (9z^{5/3} - 5z^{3/5})^3$

28 $F(t) = (5t^2 - 7)/(t^2 + 2)$

29 $k(s) = (2s^2 - 3s + 1)(9s - 1)^4$

30 $H(x) = |\cos x|$

31 $f(w) = \sqrt{(2w + 5)/(7w - 9)}$

32 $S(t) = \sqrt{t^2 + t + 1}\sqrt[3]{4t - 9}$

33 $P(\theta) = \theta^2 \cos^2 \theta^2$

34 $g(v) = 1/(1 + \cos^2 2v)$

35 $g(x) = (\cos \sqrt[3]{x} - \sin \sqrt[3]{x})^3$

36 $f(x) = \sin(\cos 5x)$

Assuming that each of the equations in Exercises 37–42 determines a function f such that $y = f(x)$, find y' .

37 $5x^3 - 2x^2y^2 + 4y^3 - 7 = 0$

38 $3x^2 - xy^2 + y^{-1} = 1$

39 $(\sqrt{x} + 1)/(\sqrt{y} + 1) = y$

40 $y^2 - \sqrt{xy} + 3x = 2$

41 $xy^2 = \sin(x + 2y)$

42 $y = \cos(xy)$

In Exercises 43 and 44, find an equation of the tangent line to the graph of the given equation at the indicated point P .

43 $y = 2x - (4/\sqrt{x})$, $P(4, 6)$

44 $x^2y - y^3 = 8$, $P(-3, 1)$

45 Find the x -coordinates of all points on the graph of $y = 3x - \cos 2x$ at which the tangent line is perpendicular to the line $2x + 4y = 5$.

46 Given $y = \sin x - \cos x$, (a) find all points on the graph at which the tangent line is horizontal; (b) find the slope of the tangent line at the points where the graph crosses the x -axis.

In Exercises 47 and 48, find y' , y'' , and y''' .

47 $y = 5x^3 + 4\sqrt{x}$

48 $y = 2x^2 - 3x - \cos 5x$

49 If $x^2 + 4xy - y^2 = 8$, find y'' by implicit differentiation.

50 Given $f(x) = x^3 - x^2 - 5x + 2$,

(a) Find the x -coordinates of all points on the graph of f at which the tangent line is parallel to the line through $A(-3, 2)$ and $B(1, 14)$.

(b) Find the value of f'' at each zero of f' .

51 If $f(x) = 1/(1 - x)$, find a formula for $f^{(n)}(x)$, where n is any positive integer.

- 52 If $y = 5x/(x^2 + 1)$, find dy and use it to approximate the change in y if x changes from 2 to 1.98. What is the exact change in y ?
- 53 The side of an equilateral triangle is estimated to be 4 in. with a maximum error of 0.03 in. Use differentials to estimate the maximum error in the calculated area of the triangle. What is the approximate percentage error?
- 54 If $s = 3r^2 - 2\sqrt{r + 1}$ and $r = t^3 + t^2 + 1$, use the Chain Rule to find the value of ds/dt at $t = 1$.
- 55 If $f(x) = 2x^3 + x^2 - x + 1$ and $g(x) = x^5 + 4x^3 + 2x$, use differentials to approximate the change in $g(f(x))$ if x changes from -1 to -1.01 .
- 56 Use differentials to approximate $\sqrt[3]{64.2}$.
- 57 Suppose f and g are functions such that $f(2) = -1$, $f'(2) = 4$, $f''(2) = -2$, $g(2) = -3$, $g'(2) = 2$ and $g''(2) = 1$. Find the values of each of the following at $x = 2$.
- (a) $(2f - 3g)'$ (b) $(2f - 3g)''$ (c) $(fg)'$
- (d) $(fg)''$ (e) $\left(\frac{f}{g}\right)'$ (f) $\left(\frac{f}{g}\right)''$
- 58 Let V and S denote the volume and surface area, respectively, of a spherical balloon. If the diameter is 8 cm and the volume increases by 12 cm^3 , use differentials to approximate the change in S .
- 59 According to *Stefan's Law*, the radiant energy emitted from the surface of a body is given by $R = kT^4$, where R denotes the rate of emission per unit area, T is the temperature (in degrees Kelvin) and k is a constant. If there is a 0.5% error in the measurement of T , find the resulting percentage error in the calculated value of R .
- 60 The intensity of illumination from a source of light is inversely proportional to the square of the distance from the source. If a student works at a desk that is a certain distance from a lamp, use differentials to find the percentage change in distance that will increase the intensity by 10%.
- 61 Use Newton's Method to approximate the root of $\sin x - x \sin x = 0$ that is between π and $3\pi/2$ to three decimal places.
- 62 Use Newton's Method to approximate $\sqrt[4]{5}$ to three decimal places.

4 Applications of the Derivative

In this chapter the derivative is used as a tool to investigate some mathematical and physical problems. Included are methods for determining maximum and minimum values of functions, applications of extrema to applied situations, and the graphical concepts of *concavity* and *points of inflection*. We shall also examine how the derivative may be used to find rates at which

quantities change, not only with respect to time, but also with respect to other variables. The concept of *antiderivative* is introduced near the end of the chapter and applied to problems on rectilinear motion. The final section contains a discussion of how derivatives may be used in economics.

4.1 Local Extrema of Functions

Suppose that the graph illustrated in Figure 4.1 was made by a recording instrument used to measure the variation of a physical quantity with respect to time. In this case the x -axis represents time and the y -coordinates of points on the graph denote magnitudes of the quantity. For example, y -values might represent measurements such as temperature, resistance in an

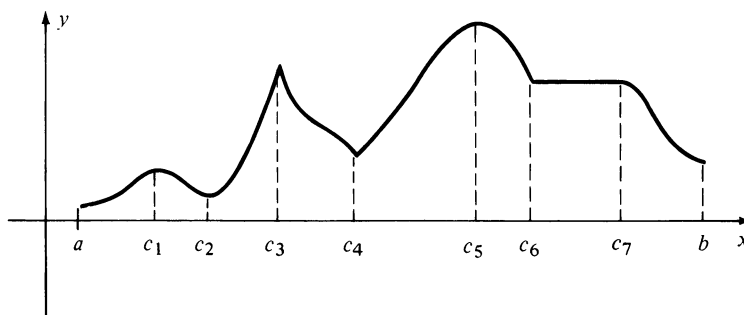


FIGURE 4.1

electrical circuit, blood pressure of an individual, the amount of chemical in a solution, or the bacteria count in a culture.

Referring to the graph we see that the quantity increased in the time interval $[a, c_1]$, decreased in $[c_1, c_2]$, increased in $[c_2, c_3]$, etc. If we restrict our attention to the interval $[c_1, c_4]$, the quantity had its largest (or maximum) value at c_3 and its smallest (or minimum) value at c_2 . In other intervals there were different largest or smallest values. For example, over the entire interval $[a, b]$, the maximum value occurred at c_5 and the minimum value at a .

If Figure 4.1 is a sketch of the graph of a function f , it is convenient to use similar terminology to describe the behavior of $f(x)$ as x varies. The mathematical terms that are employed are included in the next two definitions.

Definition (4.1)

Let a function f be defined on an interval I and let x_1, x_2 denote numbers in I .

- (i) f is **increasing** on I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (ii) f is **decreasing** on I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.
- (iii) f is **constant** on I if $f(x_1) = f(x_2)$ for every x_1 and x_2 .

Geometric interpretations of the preceding definition are given in Figure 4.2, where the interval I is not indicated. Of course, if f is constant on I , then the graph is part of a horizontal line.

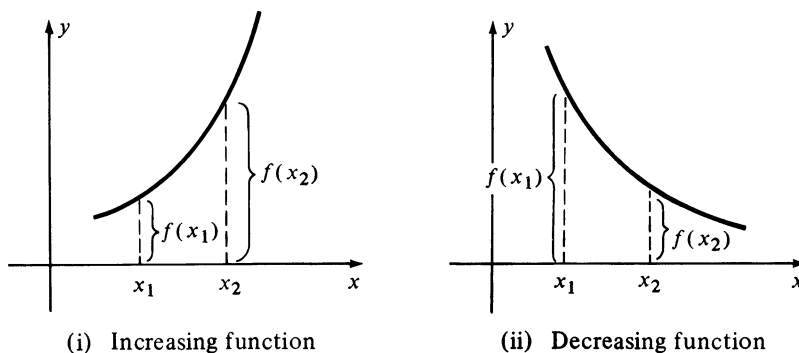


FIGURE 4.2

We shall use the phrases “ f is increasing” and “ $f(x)$ is increasing” interchangeably. This will also be done for the term “decreasing.” If a function is increasing, then the graph rises as x increases, as illustrated in (i) of Figure 4.2. If a function is decreasing, then the graph falls as x increases, as in (ii) of the figure. If the sketch in Figure 4.1 represents the graph of a function f , then f is increasing on the intervals $[a, c_1]$, $[c_2, c_3]$, and $[c_4, c_5]$. It is decreasing on $[c_1, c_2]$, $[c_3, c_4]$, $[c_5, c_6]$, and $[c_7, b]$. The function is constant on the interval $[c_6, c_7]$.

The next definition introduces terminology we shall use for largest and smallest values of functions on an interval.

Definition (4.2)

Let a function f be defined on an interval I and let c be a number in I .

- (i) $f(c)$ is the **maximum value** of f on I if $f(x) \leq f(c)$ for every x in I .
- (ii) $f(c)$ is the **minimum value** of f on I if $f(x) \geq f(c)$ for every x in I .

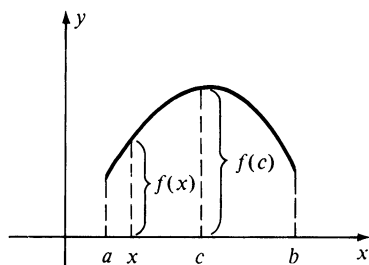
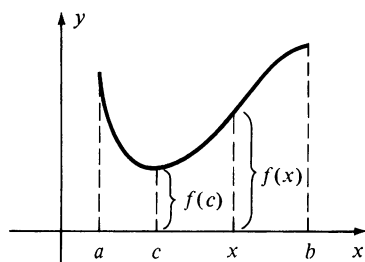
(i) Maximum value $f(c)$ (ii) Minimum value $f(c)$

FIGURE 4.3

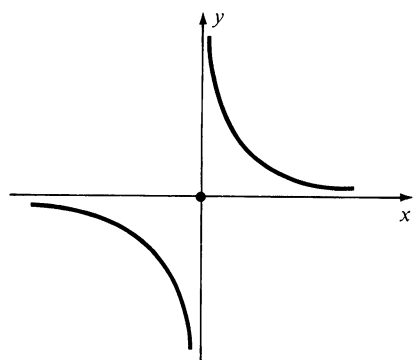


FIGURE 4.4

Maximum and minimum values are illustrated graphically in Figure 4.3. We have pictured I as a closed interval $[a, b]$; however, Definition (4.2) may be applied to any interval. Although the graphs in the figure have horizontal tangent lines at the point $(c, f(c))$, maximum and minimum values can also occur at points where graphs have corners or at endpoints of domains.

If $f(c)$ is the maximum value of f on I , we say that f takes on its maximum value at c . If we restrict our attention to points with x -coordinates in I , then the point $(c, f(c))$ is a highest point on the graph as illustrated in (i) of Figure 4.3. Similarly, if $f(c)$ is the minimum value of f on I we say that f takes on its minimum value at c , and, as in (ii) of the figure, $(c, f(c))$ is a lowest point on the graph. Maximum and minimum values are sometimes called **extreme values** or **extrema** of f . A function can take on a maximum or minimum value more than once. Indeed, if f is a constant function, then $f(c)$ is both a maximum and a minimum value of f for every real number c .

Certain functions may have a maximum value but no minimum value on an interval, or vice versa. Other functions may have neither a maximum nor minimum value. To illustrate, suppose $f(x) = 1/x$ if $x \neq 0$ and $f(0) = 0$. The graph of f is sketched in Figure 4.4.

Note that f is not continuous at 0. On the closed interval $[-1, 1]$, the function has neither a maximum value nor a minimum value. (Why?) The same is true on the open interval $(0, 1)$, for if $0 < a < 1$, then there always exist numbers x_1 and x_2 in $(0, 1)$ such that $f(x_1) > f(a)$ and $f(x_2) < f(a)$. On the half-open interval $(0, 1]$ there is a minimum value $f(1)$, but no maximum value. On $[-1, 0)$ there is a maximum value $f(-1)$, but no minimum value. On the closed interval $[1, 2]$ f has a maximum value $f(1)$ and a minimum value $f(2)$.

From the preceding illustration it appears that the existence of maximum or minimum values may depend on both the continuity of the function and whether the interval under consideration is open, closed, or half-open. The next theorem provides conditions under which a function takes on a maximum or minimum value on an interval. The interested reader may consult more advanced texts on calculus for the proof.

Theorem (4.3)

If a function f is continuous on a closed interval $[a, b]$, then f takes on a minimum value and a maximum value at least once in $[a, b]$.

The extrema are also called the **absolute minimum** and **absolute maximum values** for f on an interval. We shall also be interested in **local extrema** of f , defined as follows.

Definition (4.4)

Let c be a number in the domain of a function f .

- (i) $f(c)$ is a **local maximum** of f if there exists an open interval (a, b) containing c such that $f(x) \leq f(c)$ for all x in (a, b) .
- (ii) $f(c)$ is a **local minimum** of f if there exists an open interval (a, b) containing c such that $f(x) \geq f(c)$ for all x in (a, b) .

The term “local” is used because we *localize* our attention to a sufficiently small open interval containing c such that f takes on a largest (or smallest) value at c . Outside of that interval f may take on larger (or smaller) values. Sometimes the word “relative” is used in place of “local.” Each local maximum or minimum will be called a **local extremum** of f and the totality of such numbers are the **local extrema** of f . Several examples of local extrema are illustrated in Figure 4.5. As indicated in the figure, it is possible for a local minimum to be *larger* than a local maximum.

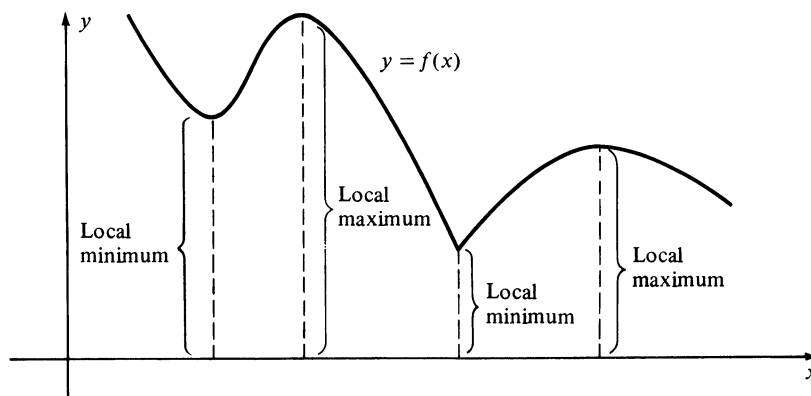


FIGURE 4.5

For the function whose graph is sketched in Figure 4.1, local maxima occur at c_1, c_3 , and c_5 whereas local minima occur at c_2 and c_4 . The functional values that correspond to numbers in the open interval (c_6, c_7) are *both* local maxima and local minima. (Why?) The local extrema may not include the absolute minimum or maximum values of f . For example, with reference to Figure 4.1, $f(a)$ is the minimum value of f on $[a, b]$, although it is not a local minimum since there is no *open* interval I contained in $[a, b]$ such that $f(a)$ is the least value of f on I . The number $f(c_5)$ is both a local maximum and the absolute maximum for f on $[a, b]$.

At a point corresponding to a local extremum of the function graphed in Figure 4.5, the tangent line is either horizontal or there is a corner. The x -coordinates of these points are numbers at which the derivative is zero or does not exist. The next theorem brings out the fact that this is generally true.

Theorem (4.5)

If a function f has a local extremum at a number c in an open interval, then either $f'(c) = 0$ or $f'(c)$ does not exist.

Proof Suppose f has a local extremum at c . If $f'(c)$ does not exist there is nothing more to prove. If $f'(c)$ exists, then precisely one of the following occurs: (i) $f'(c) > 0$, (ii) $f'(c) < 0$, or (iii) $f'(c) = 0$. We shall arrive at (iii) by proving that neither (i) nor (ii) can occur. Thus, suppose $f'(c) > 0$. Employing Theorem (3.7) for the derivative,

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} > 0$$

and hence by Theorem (2.7), there exists an open interval (a, b) containing c such that

$$\frac{f(x) - f(c)}{x - c} > 0$$

for all x in (a, b) different from c . The last inequality implies that if $a < x < b$ and $x \neq c$, then $f(x) - f(c)$ and $x - c$ are either both positive or both negative; that is,

$$\begin{cases} f(x) - f(c) < 0 & \text{whenever } x - c < 0, \text{ and} \\ f(x) - f(c) > 0 & \text{whenever } x - c > 0. \end{cases}$$

Another way of stating these facts is that if x is in (a, b) and $x \neq c$, then

$$\begin{cases} f(x) < f(c) & \text{whenever } x < c, \text{ and} \\ f(x) > f(c) & \text{whenever } x > c. \end{cases}$$

It follows that $f(c)$ is neither a local maximum nor a local minimum for f , contrary to hypothesis. Consequently (i) cannot occur. In like manner, the assumption that $f'(c) < 0$ leads to a contradiction. Hence (iii) must hold and the theorem is proved. $\square$

Corollary (4.6)

If $f'(c)$ exists and $f'(c) \neq 0$, then $f(c)$ is not a local extremum of the function f .

A result similar to Theorem (4.5) is true for the *absolute* maximum and minimum values of a function which is continuous on a closed interval $[a, b]$, provided the extrema occur on the *open* interval (a, b) . The theorem may be stated as follows.

Theorem (4.7)

If a function f is continuous on a closed interval $[a, b]$ and has its maximum or minimum value at a number c in the open interval (a, b) , then either $f'(c) = 0$ or $f'(c)$ does not exist.

The proof is exactly the same as that of Theorem (4.5) with the word “local” deleted.

It follows from Theorems (4.5) and (4.7) that the numbers at which the derivative either is zero or does not exist play a crucial role in the search for extrema of a function. Because of this, we give these numbers a special name in the next definition.

Definition (4.8)

A number c in the domain of a function f is a **critical number** of f if either $f'(c) = 0$ or $f'(c)$ does not exist.

Referring to Theorem (4.7) we see that if f is continuous on a closed interval $[a, b]$, then the absolute maximum and minimum values occur either at a critical number of f , or at the endpoints a or b of the interval. If either $f(a)$ or $f(b)$ is an absolute extremum of f on $[a, b]$ it is called an **end-point extremum**. The sketches in Figure 4.6 illustrate this concept.

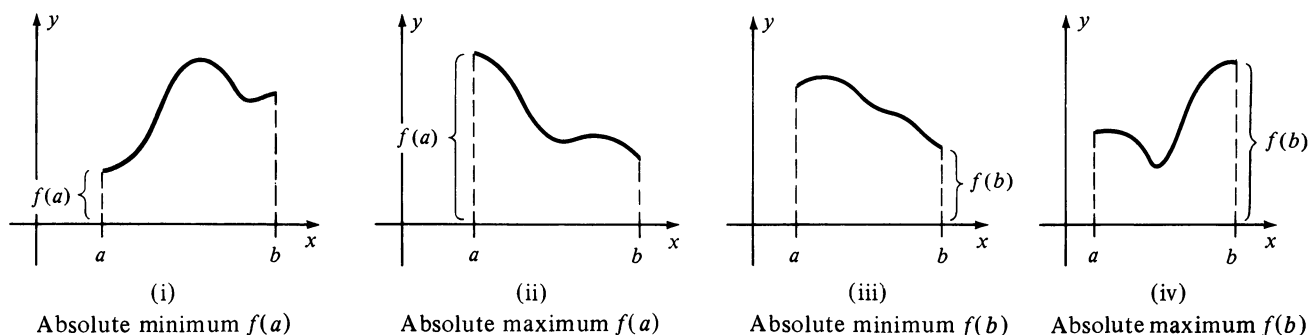


FIGURE 4.6 Endpoint extrema of f on $[a, b]$

The preceding discussion enables us to state the following.

(4.9) Guidelines for Finding the Absolute Extrema of a Continuous Function f on a Closed Interval $[a, b]$

- 1 Find all the critical numbers of f .
- 2 Calculate $f(c)$ for each critical number c .
- 3 Calculate $f(a)$ and $f(b)$.
- 4 The absolute maximum and minimum of f on $[a, b]$ are the largest and smallest of the functional values calculated in 2 and 3.

Example 1 If $f(x) = x^3 - 12x$, find the absolute maximum and minimum values of f on the closed interval $[-3, 5]$. Sketch the graph of f .

Solution Let us follow the Guidelines (4.9). Thus we begin by finding the critical numbers of f . Differentiating,

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x + 2)(x - 2).$$

Since the derivative exists everywhere, the only critical numbers are those for which the derivative is zero, that is, -2 and 2 . Since f is continuous on $[-3, 5]$, it follows from our discussion that the absolute maximum and minimum are among the numbers $f(-2)$, $f(2)$, $f(-3)$, and $f(5)$. Calculating these values (see Guidelines 2 and 3), we obtain

$$f(-2) = (-2)^3 - 12(-2) = -8 + 24 = 16$$

$$f(2) = 2^3 - 12(2) = 8 - 24 = -16$$

$$f(-3) = (-3)^3 - 12(-3) = -27 + 36 = 9$$

$$f(5) = 5^3 - 12(5) = 125 - 60 = 65.$$

Thus, by Guideline 4, the minimum value of f on $[-3, 5]$ is $f(2) = -16$ and the maximum value is the endpoint extremum $f(5) = 65$.

Using the functional values we have calculated and plotting several more points leads to the sketch in Figure 4.7, where for clarity, different scales have been used on the x - and y -axes.

The tangent line is horizontal at the points on the graph corresponding to the critical numbers -2 and 2 . It will follow from our work in Section 4.4 that $f(-2) = 16$ is a *local* maximum for f , as indicated by the graph. ■

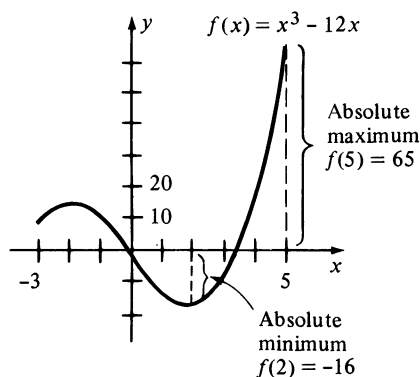


FIGURE 4.7

We see from Theorem (4.5) that if a function has a *local* extremum, then it *must* occur at a critical number; however, not every critical number leads to a local extremum, as illustrated by the following example.

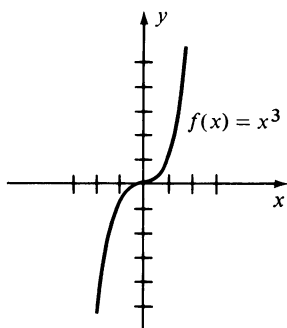


FIGURE 4.8

Example 2 If $f(x) = x^3$, prove that f has no local extremum.

Solution The graph of f is sketched in Figure 4.8.

The derivative of f is $f'(x) = 3x^2$, which exists for all x and is zero only if $x = 0$. Consequently 0 is the only critical number. However, if $x < 0$, then $f(x)$ is negative whereas if $x > 0$, then $f(x)$ is positive. Hence $f(0)$ is neither a local maximum nor a local minimum. Since a local extremum *must* occur at a critical number (see Theorem (4.5)) it follows that f has no local extrema. Note that the tangent line is horizontal at the point $P(0, 0)$, which has the critical number 0 as x -coordinate. ■

In later sections methods are developed for finding local extrema of functions. At that time it may be necessary to obtain critical numbers of functions that are defined by fairly complicated expressions, as illustrated in the next example.

Example 3 Find the critical numbers of f if $f(x) = (x + 5)^2 \sqrt[3]{x - 4}$.

Solution Differentiating $f(x) = (x + 5)^2(x - 4)^{1/3}$, we obtain

$$f'(x) = (x + 5)^2 \frac{1}{3}(x - 4)^{-2/3} + 2(x + 5)(x - 4)^{1/3}.$$

As an aid to finding the critical numbers we simplify $f'(x)$ as follows:

$$\begin{aligned} f'(x) &= \frac{(x+5)^2}{3(x-4)^{2/3}} + 2(x+5)(x-4)^{1/3} \\ &= \frac{(x+5)^2 + 6(x+5)(x-4)}{3(x-4)^{2/3}} \\ &= \frac{(x+5)[(x+5) + 6(x-4)]}{3(x-4)^{2/3}} \\ &= \frac{(x+5)(7x-19)}{3(x-4)^{2/3}}. \end{aligned}$$

Consequently, $f'(x) = 0$ if $x = -5$ or $x = \frac{19}{7}$. The derivative $f'(x)$ does not exist at $x = 4$. Thus f has three critical numbers, namely -5 , $\frac{19}{7}$, and 4 . ■

Example 4 If $f(x) = 2 \sin x + \cos 2x$, find the critical numbers of f that are in the interval $[0, 2\pi]$.

Solution Differentiating, we obtain

$$f'(x) = 2 \cos x - 2 \sin 2x.$$

Since $\sin 2x = 2 \sin x \cos x$, this may be rewritten

$$\begin{aligned} f'(x) &= 2 \cos x - 4 \sin x \cos x \\ &= 2 \cos x(1 - 2 \sin x). \end{aligned}$$

The derivative exists for all x , and $f'(x) = 0$ if either $\sin x = \frac{1}{2}$ or $\cos x = 0$. Hence the critical numbers of f in the interval $[0, 2\pi]$ are $\pi/6$, $5\pi/6$, $\pi/2$, and $3\pi/2$. We shall return to this function in Example 5 of Section 4.4 and sketch the graph of f . ■

4.1 Exercises

In Exercises 1–4, find the absolute maximum and minimum of f on the indicated closed interval.

1 $f(x) = 5 - 6x^2 - 2x^3$; $[-3, 1]$

2 $f(x) = 3x^2 - 10x + 7$; $[-1, 3]$

3 $f(x) = 1 - x^{2/3}$; $[-1, 8]$

4 $f(x) = x^4 - 5x^2 + 4$; $[0, 2]$

5 (a) If $f(x) = x^{1/3}$, prove that 0 is the only critical number of f , and that $f(0)$ is not a local extremum.

(b) If $f(x) = x^{2/3}$, prove that 0 is the only critical number of f , and that $f(0)$ is a local minimum of f .

6 If $f(x) = |x|$, prove that 0 is the only critical number of f ; that $f(0)$ is a local minimum of f ; and that the graph of f has no tangent line at the point $(0, 0)$.

In Exercises 7 and 8, prove that f has no local extrema. Sketch the graph of f . Prove that f is continuous on the interval $(0, 1)$, but f has neither a maximum nor minimum value on $(0, 1)$. Why doesn't this contradict Theorem (4.3)?

7 $f(x) = x^3 + 1$

8 $f(x) = 1/x^2$

Find the critical numbers of the functions defined in Exercises 9–26.

9 $f(x) = 4x^2 - 3x + 2$

10 $g(x) = 2x + 5$

11 $s(t) = 2t^3 + t^2 - 20t + 4$

12 $K(z) = 4z^3 + 5z^2 - 42z + 7$

13 $F(w) = w^4 - 32w$

14 $k(r) = r^5 - 2r^3 + r - 12$

15 $f(z) = \sqrt{z^2 - 16}$

16 $M(x) = \sqrt[3]{x^2 - x - 2}$

17 $g(t) = t^2 \sqrt[3]{2t - 5}$

18 $T(v) = (4v + 1)\sqrt{v^2 - 16}$

19 $G(x) = (2x - 3)/(x^2 - 9)$

20 $f(s) = s^2/(5s + 4)$

21 $f(t) = \sin^2 t - \cos t$

22 $g(t) = 4 \sin^3 t + 3\sqrt{2} \cos^2 t$

23 $K(\theta) = \sin 2\theta + 2 \cos \theta$

24 $f(x) = 8 \cos^3 x - 3 \sin 2x - 6x$

25 $f(x) = (1 + \sin x)/(1 - \sin x)$

26 $g(\theta) = 2\sqrt{3}\theta + \sin 4\theta$

27 Prove that a polynomial function of degree 1 has no local or absolute extrema on the interval $(-\infty, \infty)$. What is true on a closed interval $[a, b]$?

28 If f is a constant function and (a, b) is any open interval, prove that $f(c)$ is both a local and an absolute extremum of f for every number c in (a, b) .

29 If f is the greatest integer function, prove that every number is a critical number of f .

30 Let f be defined by the following conditions: $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Prove that every number is a critical number of f .

31 Prove that a quadratic function has exactly one critical number on $(-\infty, \infty)$.

32 Prove that a polynomial function of degree 3 has either two, one, or no critical numbers on $(-\infty, \infty)$. Sketch graphs that illustrate how each of these possibilities can occur.

33 If $f(x) = x^n$, where n is a positive integer, prove that f has either one or no local extrema on $(-\infty, \infty)$ according as n is even or odd, respectively. Sketch typical graphs illustrating each case.

34 Prove that a polynomial function of degree n can have at most $n - 1$ local extrema on $(-\infty, \infty)$.

4.2 Rolle's Theorem and the Mean Value Theorem

Sometimes it is extremely difficult to find the critical numbers of a function. As a matter of fact, there is no guarantee that critical numbers even exist. The next theorem, credited to the French mathematician Michel Rolle (1652–1719), provides sufficient conditions for the existence of a critical number. The theorem is stated for a function f that is continuous on an interval $[a, b]$, differentiable on (a, b) , and such that $f(a) = f(b)$. Some typical graphs of functions of this type are sketched in Figure 4.9.

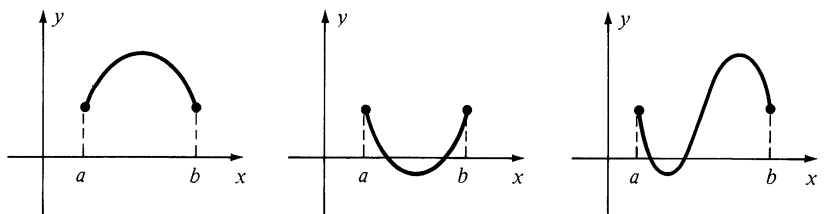


FIGURE 4.9

Referring to the sketches in Figure 4.9, it seems reasonable to expect that there is at least one number c between a and b such that the tangent line at the point $(c, f(c))$ is horizontal, or equivalently, such that $f'(c) = 0$. This is precisely the conclusion of the following theorem.

Rolle's Theorem (4.10)

If a function f is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , and if $f(a) = f(b)$, then $f'(c) = 0$ for at least one number c in (a, b) .

Proof The function f must fall into at least one of the following three categories.

- (i) $f(x) = f(a)$ for all x in (a, b) . In this case f is a constant function and hence $f'(x) = 0$ for all x . Consequently every number c in (a, b) is a critical number.
- (ii) $f(x) > f(a)$ for some x in (a, b) . In this case the maximum value of f in $[a, b]$ is greater than $f(a)$ or $f(b)$ and, therefore, must occur at some number c in the open interval (a, b) . Since the derivative exists throughout (a, b) , we conclude from Theorem (4.5) that $f'(c) = 0$.
- (iii) $f(x) < f(a)$ for some x in (a, b) . In this case the minimum value of f in $[a, b]$ is less than $f(a)$ or $f(b)$ and must occur at some number c in (a, b) . As in (ii), $f'(c) = 0$. □

Corollary (4.11)

If a function f is continuous on a closed interval $[a, b]$ and if $f(a) = f(b)$, then f has at least one critical number in the open interval (a, b) .

Proof On the one hand, if f' does not exist at some number c in (a, b) , then by Definition (4.8), c is a critical number. On the other hand, if f' exists throughout (a, b) , then a critical number exists by Rolle's Theorem. □

Before discussing the next theorem, which may be considered a generalization of Rolle's Theorem to the case in which $f(a) \neq f(b)$, let us consider the points $P(a, f(a))$ and $R(b, f(b))$ on the graph of f as illustrated by any of the sketches in Figure 4.10. If f' exists throughout the open interval (a, b) ,

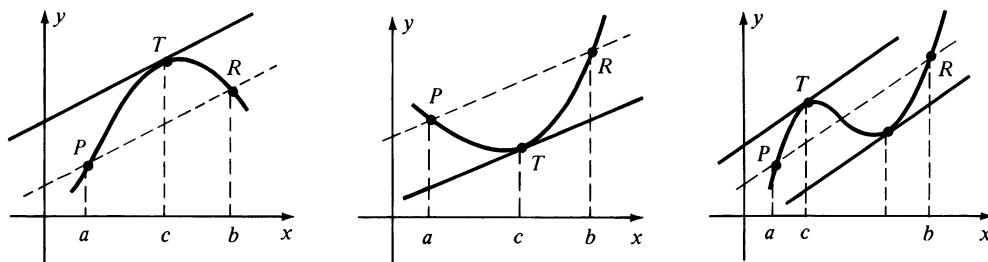


FIGURE 4.10

then it appears geometrically evident that there is at least one point $T(c, f(c))$ on the graph at which the tangent line is parallel to the secant line through P and R . This fact may be expressed in terms of slopes as follows:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

where the quotient on the right is obtained by using the formula for the slope of the line through P and R . If we multiply both sides of this equation by $b - a$, we obtain the formula stated in the next theorem.

The Mean Value Theorem (4.12)

If a function f is continuous on a closed interval $[a, b]$ and is differentiable on the open interval (a, b) , then there exists a number c in (a, b) such that

$$f(b) - f(a) = f'(c)(b - a).$$

Proof Let us define a function g as follows:

$$g(x) = f(x) - f(a) - \left[\frac{f(b) - f(a)}{b - a} \right](x - a)$$

for all x in $[a, b]$. Although we have seemingly “pulled g out of the air,” there is an interesting geometric interpretation for $g(x)$ (see Exercise 22). Since f is continuous on $[a, b]$ and differentiable on (a, b) , the same is true for g . Differentiating, we obtain

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Moreover, by direct substitution we see that $g(a) = g(b) = 0$, and hence the function g satisfies the hypotheses of Rolle’s Theorem. Consequently, there exists a number c in (a, b) such that $g'(c) = 0$, or equivalently,

$$f'(c) - \frac{f(b) - f(a)}{b - a} = 0.$$

The last equation may be written in the form stated in the conclusion of the theorem. □

The Mean Value Theorem is also called **The Theorem of the Mean**. It will be employed later to help establish several very important results. The following example provides a numerical illustration of the Mean Value Theorem.

Example Prove that the function f defined by $f(x) = x^3 - 8x - 5$ satisfies the hypotheses of the Mean Value Theorem on the interval $[1, 4]$,

and find a number c in the interval $(1, 4)$ that satisfies the conclusion of the theorem.

Solution Since f is a polynomial function it is continuous and differentiable for all real numbers. In particular, it is continuous on $[1, 4]$ and differentiable on the open interval $(1, 4)$. According to the Mean Value Theorem, there exists a number c in $(1, 4)$ such that

$$f(4) - f(1) = f'(c)(4 - 1).$$

Since $f'(x) = 3x^2 - 8$, this is equivalent to

$$27 - (-12) = (3c^2 - 8)(3).$$

We leave it to the reader to show that the last equation implies that $c = \pm\sqrt{7}$. Hence the desired number in the interval $(1, 4)$ is $c = \sqrt{7}$. ■

4.2 Exercises

- 1 If $f(x) = |x|$, show that $f(-1) = f(1)$ but $f'(c) \neq 0$ for every number c in the open interval $(-1, 1)$. Why doesn't this contradict Rolle's Theorem?
- 2 If $f(x) = 5 + 3(x - 1)^{2/3}$, show that $f(0) = f(2)$, but $f'(c) \neq 0$ for every number c in the open interval $(0, 2)$. Why doesn't this contradict Rolle's Theorem?
- 3 If $f(x) = 4/x$, prove that there is no number c such that $f(4) - f(-1) = f'(c)[4 - (-1)]$. Why doesn't this contradict the Mean Value Theorem applied to the interval $[-1, 4]$?
- 4 If f is the greatest integer function and if a and b are real numbers such that $b - a \geq 1$, prove that there is no number c such that $f(b) - f(a) = f'(c)(b - a)$. Why doesn't this contradict the Mean Value Theorem?
- 5 $f(x) = 3x^2 - 12x + 11$, $[0, 4]$
- 6 $f(x) = 5 - 12x - 2x^2$, $[-7, 1]$
- 7 $f(x) = x^4 + 4x^2 + 1$, $[-3, 3]$
- 8 $f(x) = x^3 - x$, $[-1, 1]$
- 9 $f(x) = x^3 + 1$, $[-2, 4]$
- 10 $f(x) = 5x^2 - 3x + 1$, $[1, 3]$
- 11 $f(x) = x + (4/x)$, $[1, 4]$
- 12 $f(x) = 3x^5 + 5x^3 + 15x$, $[-1, 1]$
- 13 $f(x) = x + \sin 2x$, $[0, 2\pi]$
- 14 $f(x) = 1/(x - 1)^2$, $[0, 2]$
- 15 $f(x) = 4 + \sqrt{x - 1}$, $[1, 5]$
- 16 $f(x) = 1 - 3x^{1/3}$, $[-8, -1]$
- 17 $f(x) = x^3 - 2x^2 + x + 3$, $[-1, 1]$
- 18 $f(x) = |x - 3|$, $[-1, 4]$
- 19 Prove that if f is a linear function, then f satisfies the hypotheses of the Mean Value Theorem on every closed interval $[a, b]$, and that every number c satisfies the conclusion of the theorem.
- 20 If f is a quadratic function and $[a, b]$ is any closed interval, prove that there is precisely one number c in the interval (a, b) that satisfies the conclusion of the Mean Value Theorem.
- 21 If f is a polynomial function of degree 3 and $[a, b]$ is any closed interval, prove that there are at most two numbers in (a, b) that satisfy the conclusion of the Mean Value Theorem. Sketch graphs that illustrate the various possibilities. What can be said of a polynomial function of degree 4? Illustrate with sketches. Generalize to polynomial functions of degree n , where n is any positive integer.
- 22 Prove that if $g(x)$ is the function defined in the proof of the Mean Value Theorem (4.12), then $|g(x)|$ is the distance (measured along a vertical line with x -intercept x) between the graph of f and the line through $P(a, f(a))$ and $R(b, f(b))$.

In Exercises 5–8 show that f satisfies the hypotheses of Rolle's Theorem on the indicated interval $[a, b]$ and find all numbers c in (a, b) such that $f'(c) = 0$.

In Exercises 9–18 determine whether the function f satisfies the hypotheses of the Mean Value Theorem on the indicated interval $[a, b]$ and if so, find all numbers c in (a, b) such that $f(b) - f(a) = f'(c)(b - a)$.

- 23 If f is continuous on $[a, b]$ and if $f'(x) = c$ for every x in (a, b) , use the Mean Value Theorem to prove that $f(x) = cx + d$ for some real number d .
- 24 If $f(x)$ is a polynomial of degree 3, use Rolle's Theorem to prove that f has at most three real zeros. Extend this result to polynomials of degree n .
- 25 A straight highway 50 miles long connects two cities A and B . Use the Mean Value Theorem to prove that it is impossible to travel from A to B by automobile in one hour without the speedometer reading 50 mi/hr at least once.
- 26 Use the Mean Value Theorem to prove that if u and v are any real numbers, then $|\sin u - \sin v| \leq |u - v|$.

4.3 The First Derivative Test

The following theorem indicates how the derivative may be used to determine intervals on which a function is increasing or decreasing.

Theorem (4.13)

Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) .

- (i) If $f'(x) > 0$ for all x in (a, b) , then f is increasing on $[a, b]$.
- (ii) If $f'(x) < 0$ for all x in (a, b) , then f is decreasing on $[a, b]$.

Proof (i) Suppose $f'(x) > 0$ for all x in (a, b) and consider any numbers x_1, x_2 in $[a, b]$ such that $x_1 < x_2$. We wish to show that $f(x_1) < f(x_2)$. Applying the Mean Value Theorem (4.12) to the interval $[x_1, x_2]$,

$$f(x_2) - f(x_1) = f'(w)(x_2 - x_1)$$

where w is in the open interval (x_1, x_2) . Since $x_2 - x_1 > 0$ and since, by hypothesis, $f'(w) > 0$, the right-hand side of the previous equation is positive; that is, $f(x_2) - f(x_1) > 0$. Hence $f(x_2) > f(x_1)$, which is what we wished to show. The proof of (ii) is similar and is left as an exercise. $\square$

Figures 4.11 and 4.12 give geometric illustrations of the preceding theorem. A typical tangent line l to the graph is shown at a point whose x -coordinate x is in the interval (a, b) . As illustrated in Figure 4.11, if $f'(x) > 0$

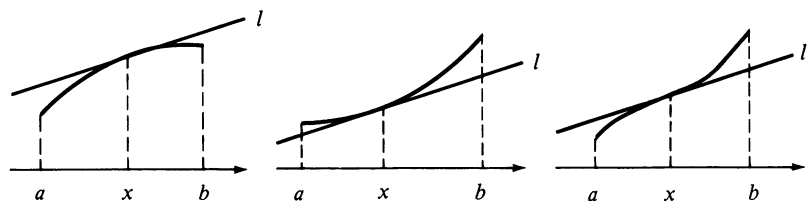


FIGURE 4.11 $f'(x) > 0$; f increasing on $[a, b]$

the tangent line rises and, as we have proved, so does the graph of f . If $f'(x) < 0$ both the tangent line and the graph of f fall, as illustrated in Figure 4.12.

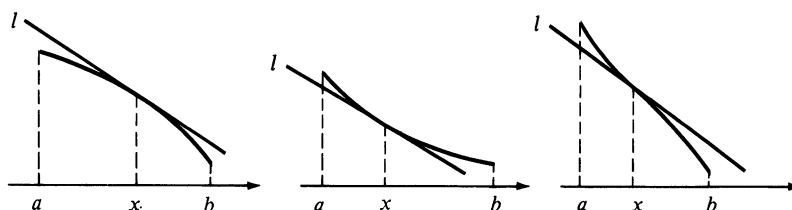


FIGURE 4.12 $f'(x) < 0$; f decreasing on $[a, b]$

The proof given for Theorem (4.13) may also be used to show that if $f'(x) > 0$ throughout an infinite interval of the form $(-\infty, a)$ or (b, ∞) , then f is increasing on $(-\infty, a]$ or $[b, \infty)$, respectively, provided f is continuous on those intervals. An analogous result holds for decreasing functions if $f'(x) < 0$.

To apply Theorem (4.13) it is necessary to determine intervals in which the derivative $f'(x)$ is either always positive, or always negative. Theorem (2.30) is useful in this respect. Specifically, if the derivative f' is continuous and has no zeros on an interval, then either $f'(x) > 0$ or $f'(x) < 0$ for all x in the interval. Thus, if we choose *any* number k in the interval, and if $f'(k) > 0$, then $f'(x) > 0$ for *all* x in the interval. Similarly, if $f'(k) < 0$, then $f'(x) < 0$ throughout the interval. We shall call $f'(k)$ a **test value** of $f'(x)$ for the interval. The use of test values is demonstrated in the following examples.

Example 1 If $f(x) = x^3 + x^2 - 5x - 5$, find the intervals on which f is increasing and the intervals on which f is decreasing. Sketch the graph of f .

Solution Differentiating, we obtain

$$f'(x) = 3x^2 + 2x - 5 = (3x + 5)(x - 1).$$

By Theorem (4.13) it is sufficient to find the intervals in which $f'(x) > 0$ and those in which $f'(x) < 0$. The factored form of $f'(x)$ and the critical numbers $-\frac{5}{3}$ and 1 suggest that we consider the open intervals $(-\infty, -\frac{5}{3})$, $(-\frac{5}{3}, 1)$, and $(1, \infty)$.

Note that on any one of these intervals, f' is continuous and has no zeros and, therefore, $f'(x)$ has the same sign throughout the interval. This sign can be determined by choosing a suitable test value for the interval.

If we choose -2 in the interval $(-\infty, -\frac{5}{3})$, we obtain the test value

$$f'(-2) = 3(-2)^2 + 2(-2) - 5 = 12 - 4 - 5 = 3.$$

Since 3 is positive, $f'(x)$ is positive throughout $(-\infty, -\frac{5}{3})$.

If we choose 0 in $(-\frac{5}{3}, 1)$, the test value is

$$f'(0) = 3(0)^2 + 2(0) - 5 = -5.$$

Since -5 is negative, $f'(x) < 0$ throughout $(-\frac{5}{3}, 1)$.

Finally, choosing 2 in $(1, \infty)$ we obtain

$$f'(2) = 3(2)^2 + 2(2) - 5 = 12 + 4 - 5 = 11.$$

Since $11 > 0$, it follows that $f'(x) > 0$ throughout $(1, \infty)$.

In future examples we will display our work in tabular form as follows, where the last column is a consequence of Theorem (4.13).

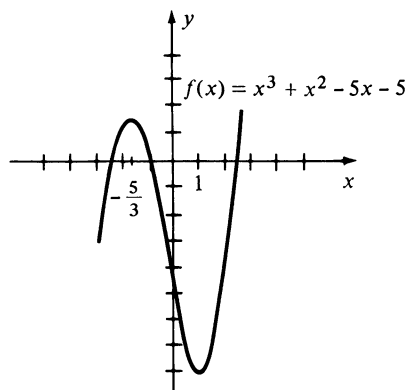


FIGURE 4.13

Interval	k	Test Value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, -\frac{5}{3})$	-2	3	+	increasing on $(-\infty, -\frac{5}{3}]$
$(-\frac{5}{3}, 1)$	0	-5	-	decreasing on $[-\frac{5}{3}, 1]$
$(1, \infty)$	2	11	+	increasing on $[1, \infty)$

Note that $f(x)$ may be written as $f(x) = x^2(x + 1) - 5(x + 1) = (x^2 - 5)(x + 1)$ and hence the x -intercepts of the graph are $\sqrt{5}$, $-\sqrt{5}$, and -1 . The y -intercept is $f(0) = -5$. The points corresponding to the critical numbers are $(-\frac{5}{3}, \frac{40}{27})$ and $(1, -8)$. Plotting these six points and using the information on where f is increasing or decreasing gives us the sketch shown in Figure 4.13. ■

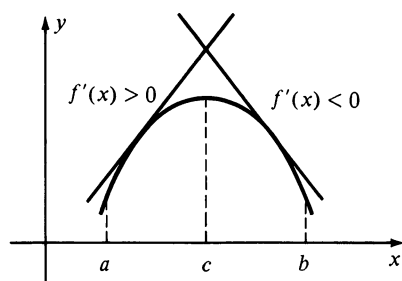
As we saw in Section 4.1, if a function has a local extremum, then it must occur at a critical number; however, not every critical number leads to a local extremum (see Example 2 of Section 4.1). To find the local extrema, we begin by locating all the critical numbers of the function. Next, each critical number is tested to determine whether or not a local extremum occurs. There are several methods for conducting this test. The following theorem is based on the sign of the first derivative of f . Roughly speaking, it states that if $f'(x)$ changes sign as x increases through a critical number c , then f has a local maximum or minimum at c . If $f'(x)$ does not change sign, then there is no extremum at c .

The First Derivative Test (4.14)

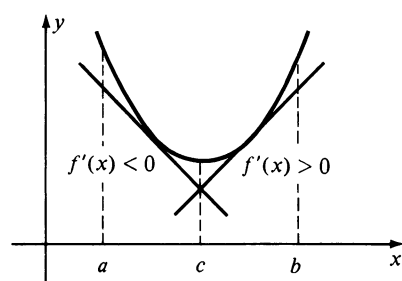
Suppose c is a critical number of a function f and (a, b) is an open interval containing c . Suppose further that f is continuous on $[a, b]$ and differentiable on (a, b) , except possibly at c .

- (i) If $f'(x) > 0$ for $a < x < c$ and $f'(x) < 0$ for $c < x < b$, then $f(c)$ is a local maximum of f .
- (ii) If $f'(x) < 0$ for $a < x < c$ and $f'(x) > 0$ for $c < x < b$, then $f(c)$ is a local minimum of f .
- (iii) If $f'(x) > 0$ or if $f'(x) < 0$ for all x in (a, b) except $x = c$, then $f(c)$ is not a local extremum of f .

Proof If the sign of $f'(x)$ varies as indicated in (i), then by Theorem (4.13) f is increasing on $[a, c]$ and decreasing on $[c, b]$, and thus $f(x) < f(c)$ for all x in (a, b) different from c . Hence, by Definition (4.4), $f(c)$ is a local maximum for f . Parts (ii) and (iii) may be proved in like manner. □



(i) Local maximum



(ii) Local minimum

FIGURE 4.14

A device that may be used to remember the first derivative test is to think of graphs of the type sketched in Figure 4.14. In the case of a local maximum, as illustrated in (i) of Figure 4.14, the slope of the tangent line at $P(x, f(x))$ is positive if $x < c$ and negative if $x > c$. In the case of a local minimum, as illustrated in (ii) of Figure 4.14, the opposite situation occurs. Similar drawings can be sketched if the graph has a corner at the point $(c, f(c))$.

Example 2 Find the local extrema of f if $f(x) = x^3 + x^2 - 5x - 5$.

Solution This function is the same as that considered in Example 1. The critical numbers of f are $-\frac{5}{3}$ and 1. We see from the table in Example 1 that the sign of $f'(x)$ changes from positive to negative as x increases through $-\frac{5}{3}$. Hence, by the First Derivative Test, f has a local maximum at $-\frac{5}{3}$. This maximum value is $f(-\frac{5}{3}) = \frac{40}{27}$. Similarly, a local minimum occurs at 1 since the sign of $f'(x)$ changes from negative to positive as x increases through 1. This minimum value is $f(1) = -8$. The reader should refer to Figure 4.13 for the geometric significance of these local extrema. ■

Example 3 Find the local maxima and minima of f if $f(x) = x^{1/3}(8 - x)$. Sketch the graph of f .

Solution By the Product Rule,

$$\begin{aligned} f'(x) &= x^{1/3}(-1) + (8 - x)\frac{1}{3}x^{-2/3} \\ &= \frac{-3x + (8 - x)}{3x^{2/3}} = \frac{4(2 - x)}{3x^{2/3}} \end{aligned}$$

and hence the critical numbers of f are 0 and 2. As in Example 1, these suggest that we consider the signs of $f'(x)$ corresponding to the intervals $(-\infty, 0)$, $(0, 2)$ and $(2, \infty)$. Since f' is continuous and has no zeros on each interval, we may determine the sign of $f'(x)$ by using a suitable test value $f'(k)$. The following table summarizes this work.

Interval	k	Test Value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, 0)$	-1	4	+	increasing on $(-\infty, 0]$
$(0, 2)$	1	$\frac{4}{3}$	+	increasing on $[0, 2]$
$(2, \infty)$	8	-2	-	decreasing on $[2, \infty)$

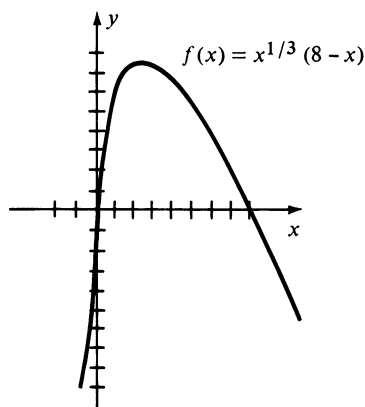


FIGURE 4.15

The number 8 was chosen in $(2, \infty)$ since $8^{2/3}$ is an integer. One may, of course, choose *any* number k in $(2, \infty)$.

By the First Derivative Test, f has a local maximum at 2 since the sign of $f'(x)$ changes from $+$ to $-$ as x increases through 2. This local maximum is $f(2) = 2^{1/3}(8 - 2) = 6\sqrt[3]{2} \approx 7.6$. The function does not have an extremum at 0 since the sign of $f'(x)$ does not change as x increases through 0. To sketch the graph we first plot points corresponding to the critical numbers. From the formula for $f(x)$ it is evident that the x -intercepts of the graph are 0 and 8. The graph is sketched in Figure 4.15. ■

Example 4 Find the local extrema of f if $f(x) = x^{2/3}(x^2 - 8)$. Sketch the graph of f .

Solution By the Product Rule,

$$\begin{aligned} f'(x) &= x^{2/3}(2x) + (x^2 - 8)\left(\frac{2}{3}x^{-1/3}\right) \\ &= \frac{6x^2 + 2(x^2 - 8)}{3x^{1/3}} = \frac{8(x^2 - 2)}{3x^{1/3}}. \end{aligned}$$

The critical numbers are $-\sqrt{2}$, 0 , and $\sqrt{2}$. These suggest an examination of the sign of $f'(x)$ in the intervals $(-\infty, -\sqrt{2})$, $(-\sqrt{2}, 0)$, $(0, \sqrt{2})$, and $(\sqrt{2}, \infty)$. Arranging our work in tabular form as in previous examples we obtain the following.

Interval	k	Test Value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, -\sqrt{2})$	-8	$-\frac{248}{3}$	$-$	decreasing on $(-\infty, -\sqrt{2})$
$(-\sqrt{2}, 0)$	-1	$\frac{8}{3}$	$+$	increasing on $[-\sqrt{2}, 0]$
$(0, \sqrt{2})$	1	$-\frac{8}{3}$	$-$	decreasing on $[0, \sqrt{2}]$
$(\sqrt{2}, \infty)$	8	$\frac{248}{3}$	$+$	increasing on $[\sqrt{2}, \infty)$

By the First Derivative Test, f has local minima at $-\sqrt{2}$ and $\sqrt{2}$ and a local maximum at 0 . The corresponding functional values are $f(0) = 0$ and

$$f(\sqrt{2}) = -6\sqrt[3]{2} = f(-\sqrt{2}).$$

Note that the derivative does not exist at 0 . The graph is sketched in Figure 4.16. ■

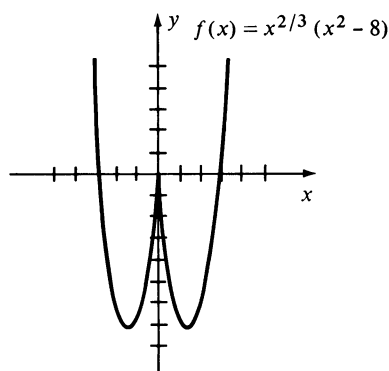


FIGURE 4.16

It was pointed out in Section 4.1 that an absolute maximum or minimum of a function may not be included among the local extrema. Recall from (4.9) that if a function f is continuous on a closed interval $[a, b]$ and if it is required to find the absolute maximum and minimum values, all the local extrema should be found and *in addition* the values $f(a)$ and $f(b)$ of f at the endpoints a and b of the interval $[a, b]$ should be calculated. The largest number among the local extrema and the values $f(a)$ and $f(b)$ is the absolute maximum of f on $[a, b]$, whereas the smallest of these numbers is the absolute minimum of f on $[a, b]$. To illustrate these remarks, let us consider the function discussed in the previous example, but restrict our attention to certain intervals.

Example 5 If $f(x) = x^{2/3}(x^2 - 8)$, find the absolute maximum and minimum values of f in each of the following intervals: (a) $[-1, \frac{1}{2}]$, (b) $[-1, 3]$, (c) $[-3, -2]$.

Solution The graph in Figure 4.16 indicates the local extrema and the intervals in which f is increasing or decreasing. Consequently, all we have to do is find the y -coordinates of the high and low points corresponding to each of the given intervals. We shall arrange our work in tabular form and leave it to the reader to check that the entries are correct.

Interval	Minimum	Maximum
$[-1, \frac{1}{2}]$	$f(-1) = -7$	$f(0) = 0$
$[-1, 3]$	$f(\sqrt{2}) = -6\sqrt[3]{2}$	$f(3) = \sqrt[3]{9}$
$[-3, -2]$	$f(-2) = -4\sqrt[3]{4}$	$f(-3) = \sqrt[3]{9}$

Note that on some intervals the maximum or minimum value of f is also a local extremum, whereas on other intervals this is not the case. ■

4.3 Exercises

In Exercises 1–16, find the local extrema of f . Describe the intervals in which f is increasing or decreasing, and sketch the graph of f .

- 1 $f(x) = 5 - 7x - 4x^2$
- 2 $f(x) = 6x^2 - 9x + 5$
- 3 $f(x) = 2x^3 + x^2 - 20x + 1$
- 4 $f(x) = x^3 - x^2 - 40x + 8$
- 5 $f(x) = x^4 - 8x^2 + 1$
- 6 $f(x) = x^3 - 3x^2 + 3x + 7$
- 7 $f(x) = x^{4/3} + 4x^{1/3}$
- 8 $f(x) = x^{2/3}(8 - x)$
- 9 $f(x) = x^2\sqrt[3]{x^2 - 4}$
- 10 $f(x) = x\sqrt{4 - x^2}$
- 11 $f(x) = x^{2/3}(x - 7)^2 + 2$
- 12 $f(x) = 4x^3 - 3x^4$
- 13 $f(x) = x^3 + (3/x)$
- 14 $f(x) = 8 - \sqrt[3]{x^2 - 2x + 1}$
- 15 $f(x) = 10x^3(x - 1)^2$
- 16 $f(x) = (x^2 - 10x)^4$

In Exercises 17–22, find the local extrema of f on the interval $[0, 2\pi]$, and determine where f is increasing or decreasing on $[0, 2\pi]$. Sketch the graph of f corresponding to $0 \leq x \leq 2\pi$.

- 17 $f(x) = \cos x + \sin x$
- 18 $f(x) = \cos x - \sin x$
- 19 $f(x) = \frac{1}{2}x - \sin x$
- 20 $f(x) = x + 2 \cos x$
- 21 $f(x) = 2 \cos x + \sin 2x$

22 $f(x) = 2 \cos x + \cos 2x$

In Exercises 23–28, find the local extrema of f .

- 23 $f(x) = \sqrt[3]{x^3 - 9x}$
- 24 $f(x) = x^2/\sqrt{x + 7}$
- 25 $f(x) = (x - 2)^3(x + 1)^4$
- 26 $f(x) = x^2(x - 5)^4$
- 27 $f(x) = (2x - 5)/(x + 3)$
- 28 $f(x) = (x^2 + 3)/(x - 1)$

29–32 For the functions defined in Exercises 1–4, find the absolute maximum and minimum values on each of the following intervals:

- (a) $[-1, 1]$ (b) $[-4, 2]$ (c) $[0, 5]$

In Exercises 33 and 34 sketch the graph of a differentiable function f that satisfies the given conditions.

- 33 $f'(-5) = 0$; $f'(0) = 0$; $f'(5) = 0$; $f'(x) > 0$ if $|x| > 5$; $f'(x) < 0$ if $0 < |x| < 5$.
- 34 $f'(a) = 0$ for $a = 1, 2, 3, 4, 5$, and $f'(x) > 0$ for all other values of x .

In Exercises 35 and 36 find the local extrema of f' . Describe the intervals in which f' is increasing or decreasing. Sketch the graph of f and study the variation of the slope of the tangent line as x increases through the domain of f .

- 35 $f(x) = x^4 - 6x^2$
- 36 $f(x) = 4x^3 - 3x^4$
- 37 If $f(x) = ax^3 + bx^2 + cx + d$, determine values for a, b, c , and d such that f has a local maximum 2 at $x = -1$ and a local minimum -1 at $x = 1$.
- 38 If $f(x) = ax^4 + bx^3 + cx^2 + dx + e$, determine values of a, b, c, d , and e such that f has a local maximum 2 at $x = 0$, and a local minimum -14 at $x = -2$ and $x = 2$, respectively.

4.4 Concavity and the Second Derivative Test

The concept of *concavity* is useful for describing the graph of a differentiable function f . If $f'(c)$ exists, then f has a tangent line with slope $f'(c)$ at the point $P(c, f(c))$. Figure 4.17 illustrates three possible situations that may occur if $f'(c) > 0$. Similar situations occur if $f'(c) < 0$ or $f'(c) = 0$. Note that in (i) of Figure 4.17 there is an open interval (a, b) containing c such that for all x in (a, b) , except $x = c$, the point $Q(x, f(x))$ on the graph of f lies above the point on the tangent line having x -coordinate x . In this case we say that on the interval (a, b) the graph of f is *above* the tangent line through P . In (ii) of Figure 4.17 we say that the graph of f is *below* the tangent line through P . In (iii), for every open interval (a, b) containing c , the tangent line is neither above nor below the graph but instead *crosses* the graph at that point.

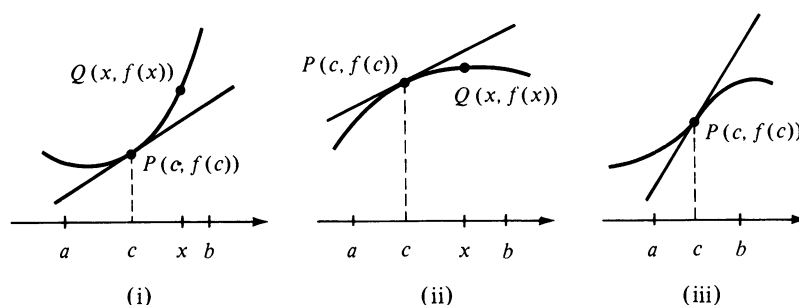


FIGURE 4.17

The terminology introduced in the next definition is used to describe the graphs illustrated in (i) and (ii) of Figure 4.17.

Definition (4.15)

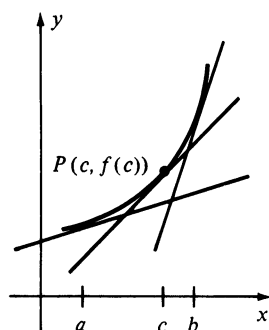
Let f be a function that is differentiable at a number c .

- (i) The graph of f is **concave upward** at the point $P(c, f(c))$ if there exists an open interval (a, b) containing c such that on (a, b) the graph of f is above the tangent line through P .
- (ii) The graph of f is **concave downward** at $P(c, f(c))$ if there exists an open interval (a, b) containing c such that on (a, b) the graph of f is below the tangent line through P .

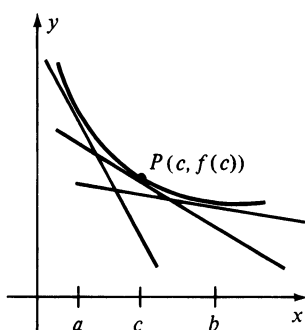
Consider a function f whose graph is concave upward at the point $P(c, f(c))$. Let (a, b) be an open interval containing c such that on (a, b) the graph of f is above the tangent line through P . If f is differentiable on (a, b) , then the graph of f has a tangent line at every point $Q(x, f(x))$ with x in (a, b) . If we consider a number of these tangent lines, as indicated in Figure 4.18,

it appears that as the x -coordinate x of the point of tangency increases, the slope $f'(x)$ of the tangent line also increases. Thus in (i) of Figure 4.18, a larger positive slope is obtained as P moves to the right. In (ii) of the figure, the slope of the tangent line also increases, becoming less negative as P moves to the right.

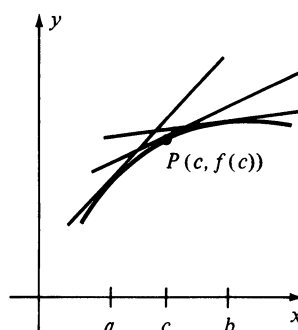
In like manner, if the graph of f is concave downward at $P(c, f(c))$, then the situations illustrated in Figure 4.19 may occur. Here the slope $f'(x)$ of the tangent line *decreases* as P moves to the right.



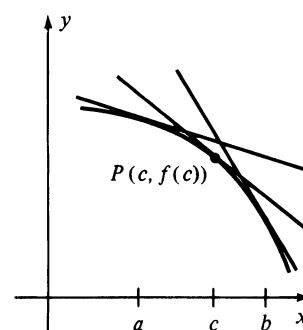
(i)



(ii)



(i)



(ii)

FIGURE 4.18 Upward concavity

FIGURE 4.19 Downward concavity

Conversely, we might expect that if the derivative $f'(x)$ increases as x increases through c , then the graph is concave upward at $P(c, f(c))$, whereas if $f'(x)$ decreases, then the graph is concave downward. According to Theorem (4.13), if the derivative of a function is positive on an interval, then the function is increasing. Consequently, if the values of the *second* derivative f'' are positive on an interval, then the *first* derivative f' is increasing. Similarly, if f'' is negative, then f' is decreasing. This suggests that the sign of f'' can be used to test concavity as stated in the next theorem.

Test for Concavity (4.16)

Suppose a function f is differentiable on an open interval containing c , and $f''(c)$ exists.

- (i) If $f''(c) > 0$, the graph is concave upward at $P(c, f(c))$.
- (ii) If $f''(c) < 0$, the graph is concave downward at $P(c, f(c))$.

Proof

Applying the formula in Theorem (3.7) to the function f' ,

$$f''(c) = \lim_{x \rightarrow c} \frac{f'(x) - f'(c)}{x - c}.$$

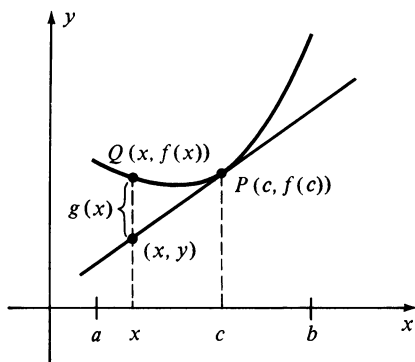


FIGURE 4.20

If $f''(c) > 0$, then by Theorem (2.7) there is an open interval (a, b) containing c such that

$$\frac{f'(x) - f'(c)}{x - c} > 0$$

for all x in (a, b) different from c . Hence $f'(x) - f'(c)$ and $x - c$ both have the same sign for all x in (a, b) different from c . We next show that this implies upward concavity.

For any x in the interval (a, b) , consider the point (x, y) on the tangent line through $P(c, f(c))$ and the point $Q(x, f(x))$ on the graph of f (see Figure 4.20). If we let $g(x) = f(x) - y$, then we can establish upward concavity by showing that $g(x)$ is positive for all x such that $x \neq c$.

Since an equation of the tangent line at P is $y - f(c) = f'(c)(x - c)$, the y -coordinate y of the point (x, y) on the tangent line is given by $y = f(c) + f'(c)(x - c)$. Consequently,

$$g(x) = f(x) - y = f(x) - f(c) - f'(c)(x - c).$$

Applying the Mean Value Theorem (4.12) to f and the interval $[x, c]$, we see that there exists a number w in the open interval (x, c) such that

$$f(x) - f(c) = f'(w)(x - c).$$

Substituting for $f(x) - f(c)$ in the preceding formula for $g(x)$ leads to

$$g(x) = f'(w)(x - c) - f'(c)(x - c)$$

which can be factored as follows:

$$g(x) = [f'(w) - f'(c)](x - c).$$

Since w is in the interval (a, b) we know, from the first paragraph of the proof, that $f'(w) - f'(c)$ and $w - c$ have the same sign. Moreover, since w is between x and c , $w - c$ and $x - c$ have the same sign. Thus $f'(w) - f'(c)$ and $x - c$ have the same sign for all x in (a, b) such that $x \neq c$. It follows from the factored form of $g(x)$ that $g(x)$ is positive if $x \neq c$, which is what we wished to prove. Part (ii) may be established in similar fashion. $\square$

If the second derivative $f''(x)$ changes sign as x increases through a number k , then by (4.16) the concavity changes from upward to downward, or from downward to upward. The point $(k, f(k))$ is called a *point of inflection* according to the next definition.

Definition (4.17)

A point $P(k, f(k))$ on the graph of a function f is a **point of inflection** if there exists an open interval (a, b) containing k such that one of the following statements holds.

- (i) $f''(x) > 0$ if $a < x < k$ and $f''(x) < 0$ if $k < x < b$; or
- (ii) $f''(x) < 0$ if $a < x < k$ and $f''(x) > 0$ if $k < x < b$.

The sketch in Figure 4.21 displays typical points of inflection on a graph. A graph is said to be **concave upward** (or **concave downward**) on an interval if it is concave upward (or downward) at every number in the interval. Intervals on which the graph in Figure 4.21 is concave upward or concave downward are abbreviated CU or CD, respectively. Observe that a corner may, or may not, be a point of inflection.

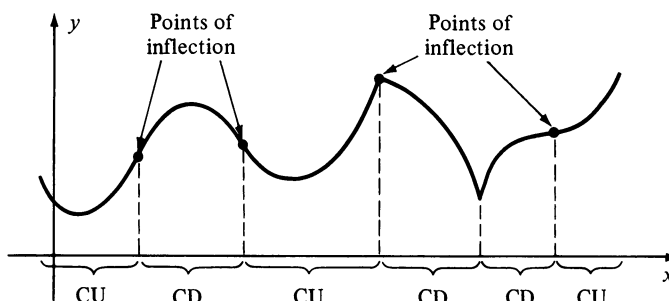


FIGURE 4.21

Example 1 If $f(x) = x^3 + x^2 - 5x - 5$, determine intervals on which the graph of f is concave upward and intervals on which the graph is concave downward.

Solution The function f was considered in Examples 1 and 2 of the preceding section. Since $f'(x) = 3x^2 + 2x - 5$, we have

$$f''(x) = 6x + 2 = 2(3x + 1).$$

Hence $f''(x) < 0$ if $3x + 1 < 0$, that is, if $x < -\frac{1}{3}$. It follows from the Test for Concavity (4.16) that the graph is concave downward on the infinite interval $(-\infty, -\frac{1}{3})$. Similarly $f''(x) > 0$ if $x > -\frac{1}{3}$ and, therefore, the graph is concave upward on $(-\frac{1}{3}, \infty)$. By Definition (4.17) the point $P(-\frac{1}{3}, -\frac{88}{27})$ at which $f''(x)$ changes sign (and the concavity changes from downward to upward) is a point of inflection. The graph of f obtained previously (see Figure 4.13) is sketched again in Figure 4.22 to show the point of inflection.

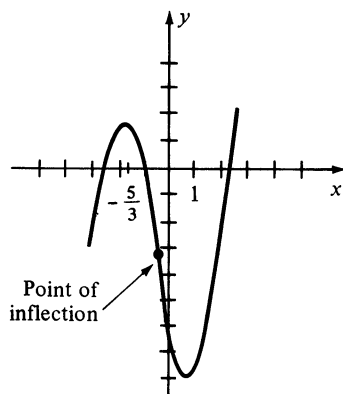


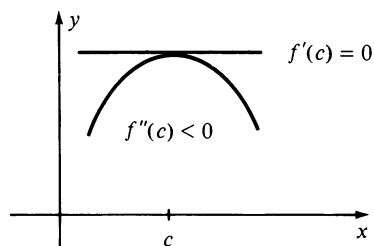
FIGURE 4.22

If $P(c, f(c))$ is a point of inflection on the graph of f and if f'' is continuous on an open interval containing c , then necessarily $f''(c) = 0$. This is a consequence of the Intermediate Value Theorem (2.29) applied to the function f'' . Thus, to locate points of inflection for a function whose second derivative is continuous, we begin by finding all numbers x such that $f''(x) = 0$. Each of these numbers is then tested to determine whether it is an x -coordinate of a point of inflection. Before giving an example of this technique, we state the following useful test for local maxima and minima.

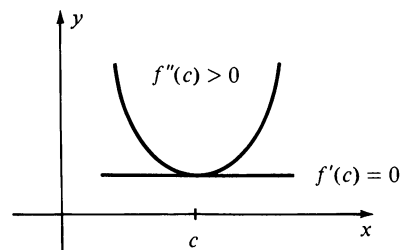
The Second Derivative Test (4.18)

Suppose a function f is differentiable on an open interval containing c and $f'(c) = 0$.

- (i) If $f''(c) < 0$, then f has a local maximum at c .
- (ii) If $f''(c) > 0$, then f has a local minimum at c .



(i) Local maximum



(ii) Local minimum

FIGURE 4.23

Proof If $f'(c) = 0$, then the tangent line to the graph at $P(c, f(c))$ is horizontal. If, in addition, $f''(c) < 0$, then the graph is concave downward at c and hence there is an interval (a, b) containing c such that the graph lies below the tangent line at P . It follows that $f(c)$ is a local maximum for f as illustrated in (i) of Figure 4.23. A similar proof may be given for part (ii), which is illustrated graphically in (ii) of Figure 4.23. $\square$

If $f''(c) = 0$, the Second Derivative Test is not applicable. In such cases the First Derivative Test should be employed (see Example 3).

Example 2 If $f(x) = 12 + 2x^2 - x^4$, use the Second Derivative Test to find the local maxima and minima of f . Discuss concavity, find the points of inflection, and sketch the graph of f .

Solution We begin by finding the first and second derivatives and factoring them as follows:

$$f'(x) = 4x - 4x^3 = 4x(1 - x^2)$$

$$f''(x) = 4 - 12x^2 = 4(1 - 3x^2).$$

The expression for $f'(x)$ is used to find the critical numbers 0, 1, and -1 . The values of f'' at these numbers are

$$f''(0) = 4 > 0, \quad f''(1) = -8 < 0 \quad \text{and} \quad f''(-1) = -8 < 0.$$

Hence, by the Second Derivative Test there is a local minimum at 0 and local maxima at 1 and -1 . The corresponding functional values are $f(0) = 12$ and $f(1) = 13 = f(-1)$. The following table summarizes the preceding discussion.

Critical number c	$f''(c)$	Sign of $f''(c)$	Conclusion
-1	-8	-	Local max: $f(-1) = 13$
0	4	+	Local min: $f(0) = 12$
1	-8	-	Local max: $f(1) = 13$

To locate the possible points of inflection we solve the equation $f''(x) = 0$, that is, $4(1 - 3x^2) = 0$. Evidently, the solutions are $-\sqrt{3}/3$ and $\sqrt{3}/3$. This suggests an examination of the sign of $f''(x)$ in each of the intervals $(-\infty, -\sqrt{3}/3)$, $(-\sqrt{3}/3, \sqrt{3}/3)$, and $(\sqrt{3}/3, \infty)$. Since f'' is continuous and has no zeros on each interval, we may use test values to determine the sign of $f''(x)$. Let us arrange our work in tabular form as follows, where the last column is a consequence of (4.18).

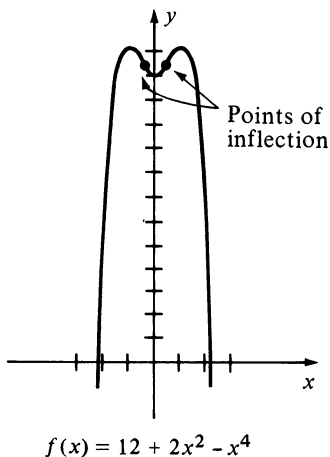


FIGURE 4.24

Interval	k	Test Value $f''(k)$	Sign of $f''(x)$	Concavity
$(-\infty, -\sqrt{3}/3)$	-1	-8	-	downward
$(-\sqrt{3}/3, \sqrt{3}/3)$	0	4	+	upward
$(\sqrt{3}/3, \infty)$	1	-8	-	downward

Since the sign of $f''(x)$ changes as x increases through $-\sqrt{3}/3$ and $\sqrt{3}/3$, the corresponding points $(\pm\sqrt{3}/3, 113/9)$ on the graph are points of inflection. These are the points at which the sense of concavity changes. Indeed, as shown in the table, the graph is concave upward on the open interval $(-\sqrt{3}/3, \sqrt{3}/3)$ and concave downward outside of $[-\sqrt{3}/3, \sqrt{3}/3]$. The graph is sketched in Figure 4.24. ■

Example 3 If $f(x) = x^5 - 5x^3$, use the Second Derivative Test to find the local extrema of f . Discuss concavity, find the points of inflection, and sketch the graph of f .

Solution Differentiating, we obtain

$$\begin{aligned} f'(x) &= 5x^4 - 15x^2 = 5x^2(x^2 - 3) \\ f''(x) &= 20x^3 - 30x = 10x(2x^2 - 3). \end{aligned}$$

Solving the equation $f'(x) = 0$ gives us the critical numbers 0, $-\sqrt{3}$, and $\sqrt{3}$. As in Example 2 we construct the following table (check the entries!).

Critical number c	$f''(c)$	Sign of $f''(c)$	Conclusion
$-\sqrt{3}$	$-30\sqrt{3}$	-	Local max: $f(-\sqrt{3}) = 6\sqrt{3}$
0	0	none	No conclusion
$\sqrt{3}$	$30\sqrt{3}$	+	Local min: $f(\sqrt{3}) = -6\sqrt{3}$

Since $f''(0) = 0$, the Second Derivative Test is not applicable at 0 and hence we turn to the First Derivative Test. The reader should check, using test values, that if $-\sqrt{3} < x < 0$, then $f'(x) < 0$, and if $0 < x < \sqrt{3}$, then $f'(x) < 0$. Since $f'(x)$ does not change sign, there is neither a maximum nor a minimum at $x = 0$.

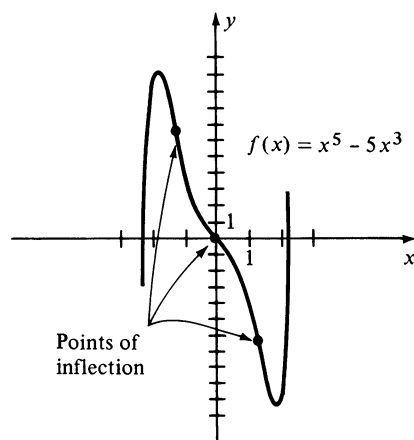


FIGURE 4.25

To find possible points of inflection we consider the equation $f''(x) = 0$, that is, $10x(2x^2 - 3) = 0$. The solutions of this equation, in order of magnitude, are $-\sqrt{6}/2$, 0 , and $\sqrt{6}/2$. As in Example 2 we construct the following table.

Interval	k	Test Value $f''(k)$	$f''(x)$	Concavity
$(-\infty, -\sqrt{6}/2)$	-2	-100	-	downward
$(-\sqrt{6}/2, 0)$	-1	10	+	upward
$(0, \sqrt{6}/2)$	1	-10	-	downward
$(\sqrt{6}/2, \infty)$	2	100	+	upward

Since the sign of $f''(x)$ changes as x increases through $-\sqrt{6}/2$, 0 , and $\sqrt{6}/2$, the points $(0, 0)$, $(-\sqrt{6}/2, 21\sqrt{6}/8)$, and $(\sqrt{6}/2, -21\sqrt{6}/8)$ are points of inflection. The graph is sketched in Figure 4.25 where, for clarity, we have used different scales on the x - and y -axes. ■

In all of the preceding examples f'' was continuous. It is also possible for $(c, f(c))$ to be a point of inflection if either $f'(c)$ or $f''(c)$ does not exist, as illustrated in the next example.

Example 4 If $f(x) = 1 - x^{1/3}$, find the local extrema, discuss concavity, find the points of inflection, and sketch the graph of f .

Solution Differentiating, we obtain

$$f'(x) = -\frac{1}{3}x^{-2/3} = -\frac{1}{3x^{2/3}}$$

$$f''(x) = \frac{2}{9}x^{-5/3} = \frac{2}{9x^{5/3}}.$$

The first derivative does not exist at $x = 0$, and 0 is the only critical number for f . Since $f''(0)$ is undefined, the Second Derivative Test is not applicable. However, if $x \neq 0$, then $x^{2/3} > 0$ and $f'(x) = -1/3x^{2/3} < 0$, which means that f is decreasing throughout its domain. Consequently $f(0)$ is not a local extremum.

The fact that f'' is undefined at $x = 0$ leads us to examine whether or not the point $(0, 1)$ on the graph of f is a point of inflection. Let us apply Definition (4.17) with $c = 0$. If $x < 0$, then $x^{5/3} < 0$. Hence

$$f''(x) = \frac{2}{9x^{5/3}} < 0 \quad \text{if } x < 0$$

which implies that the graph of f is concave downward on the interval $(-\infty, 0)$. If $x > 0$, then $x^{5/3} > 0$. Thus

$$f''(x) = \frac{2}{9x^{5/3}} > 0 \quad \text{if } x > 0$$

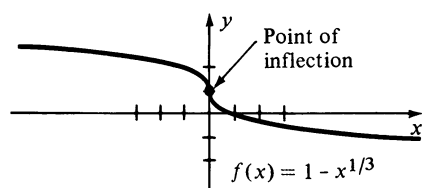


FIGURE 4.26

which means that the graph is concave upward on the interval $(0, \infty)$. By (4.17), the point $(0, 1)$ is a point of inflection. Using this information and plotting several points gives us the sketch in Figure 4.26. ■

We shall conclude this section with an example involving local extrema of trigonometric functions.

Example 5 If $f(x) = 2 \sin x + \cos 2x$, find the local extrema and sketch the graph of f on the interval $[0, 2\pi]$.

Solution Differentiating, we obtain

$$f'(x) = 2 \cos x - 2 \sin 2x$$

$$f''(x) = -2 \sin x - 4 \cos 2x.$$

In Example 4 of Section 4.1 we found that the critical numbers of f in the interval $[0, 2\pi]$ are $\pi/6$, $5\pi/6$, $\pi/2$, and $3\pi/2$. Substituting these critical numbers for x in $f''(x)$ we obtain

$$f''(\pi/6) = -3, \quad f''(5\pi/6) = -3, \quad f''(\pi/2) = 2, \quad f''(3\pi/2) = 6.$$

Applying the Second Derivative Test we see that there are local maxima at $\pi/6$ and $5\pi/6$, and local minima at $\pi/2$ and $3\pi/2$. This information, together with the following table, leads to the sketch in Figure 4.27.

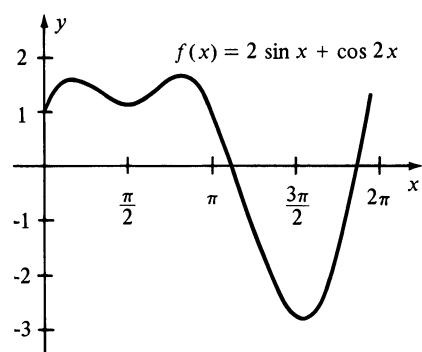


FIGURE 4.27

x	0	$\pi/6$	$\pi/2$	$5\pi/6$	π	$3\pi/2$	2π
$f(x)$	1	$\frac{3}{2}$	1	$\frac{3}{2}$	1	-3	1

4.4 Exercises

In Exercises 1–18, use the Second Derivative Test (whenever applicable) to find the local extrema of f . Discuss concavity, and x -coordinates of points of inflection, and sketch the graph of f .

1 $f(x) = x^3 - 2x^2 + x + 1$

2 $f(x) = x^3 + 10x^2 + 25x - 50$

3 $f(x) = 3x^4 - 4x^3 + 6$

4 $f(x) = 8x^2 - 2x^4$

5 $f(x) = 2x^6 - 6x^4$

6 $f(x) = 3x^5 - 5x^3$

7 $f(x) = (x^2 - 1)^2$

8 $f(x) = x - (16/x)$

9 $f(x) = \sqrt[5]{x} - 1$

10 $f(x) = (x + 4)/\sqrt{x}$

11 $f(x) = x^2 - (27/x^2)$

12 $f(x) = x^{2/3}(1 - x)$

13 $f(x) = x/(x^2 + 1)$

14 $f(x) = x^2/(x^2 + 1)$

15 $f(x) = \sqrt[3]{x^2}(3x + 10)$

16 $f(x) = x^4 - 4x^3 + 10$

17 $f(x) = 8x^{1/3} + x^{4/3}$

18 $f(x) = x\sqrt{4 - x^2}$

In Exercises 19–24, use the Second Derivative Test to find the local extrema of f on the interval $[0, 2\pi]$. (These exercises appeared in the preceding section, where the method of solution involved the First Derivative Test. (See Exercises 17–22 of Section 4.3.)

19 $f(x) = \cos x + \sin x$

20 $f(x) = \cos x - \sin x$

21 $f(x) = \frac{1}{2}x - \sin x$

22 $f(x) = x + 2 \cos x$

23 $f(x) = 2 \cos x + \sin 2x$

24 $f(x) = 2 \cos x + \cos 2x$

In Exercises 25–32 sketch the graph of a continuous function f that satisfies all of the stated conditions.

25 $f(0) = 1; f(2) = 3; f'(0) = f'(2) = 0;$
 $f'(x) < 0$ if $|x - 1| > 1; f'(x) > 0$ if $|x - 1| < 1;$
 $f''(x) > 0$ if $x < 1; f''(x) < 0$ if $x > 1.$

26 $f(0) = 4; f(2) = 2; f(5) = 6; f'(0) = f'(2) = 0;$
 $f'(x) > 0$ if $|x - 1| > 1; f'(x) < 0$ if $|x - 1| < 1;$
 $f''(x) < 0$ if $x < 1$ or if $|x - 4| < 1;$
 $f''(x) > 0$ if $|x - 2| < 1$ or if $x > 5.$

27 $f(0) = 2; f(2) = f(-2) = 1; f'(0) = 0;$
 $f'(x) > 0$ if $x < 0; f'(x) < 0$ if $x > 0;$
 $f''(x) < 0$ if $|x| < 2; f''(x) > 0$ if $|x| > 2.$

28 $f(1) = 4; f'(x) > 0$ if $x < 1; f'(x) < 0$ if $x > 1;$
 $f''(x) > 0$ for all $x \neq 1.$

29 $f(-2) = f(6) = -2; f(0) = f(4) = 0; f(2) = f(8) = 3;$
 f' is undefined at 2 and 6; $f'(0) = 1; f'(x) > 0$ throughout
 $(-\infty, 2)$ and $(6, \infty); f'(x) < 0$ if $|x - 4| < 2; f''(x) < 0$

throughout $(-\infty, 0), (4, 6),$ and $(6, \infty); f''(x) > 0$
 throughout $(0, 2)$ and $(2, 4).$

30 $f(0) = 2; f(2) = 1; f(4) = f(10) = 0; f(6) = -4; f'(2) =$
 $f'(6) = 0; f'(x) < 0$ throughout $(-\infty, 2), (2, 4), (4, 6),$
 and $(10, \infty); f'(x) > 0$ throughout $(6, 10); f'(4)$ and
 $f'(10)$ do not exist; $f''(x) > 0$ throughout $(-\infty, 2),$
 $(4, 10),$ and $(10, \infty); f''(x) < 0$ throughout $(2, 4).$

31 If n is an odd integer, then $f(n) = 1$ and $f'(n) = 0$; if n
 is an even integer, then $f(n) = 0$ and $f'(n)$ does not
 exist; if n is any integer then

(a) $f'(x) > 0$ whenever $2n < x < 2n + 1;$

(b) $f'(x) < 0$ whenever $2n - 1 < x < 2n;$

(c) $f''(x) < 0$ whenever $2n < x < 2n + 2.$

32 $f(x) = x$ if $x = -1, 2, 4,$ or $8; f'(x) = 0$ if $x = -1, 4, 6,$
 or $8; f'(x) < 0$ throughout $(-\infty, -1), (4, 6),$ and $(8, \infty);$
 $f'(x) > 0$ throughout $(-1, 4)$ and $(6, 8); f''(x) > 0$ through-
 out $(-\infty, 0), (2, 3),$ and $(5, 7); f''(x) < 0$ throughout
 $(0, 2), (3, 5),$ and $(7, \infty).$

33 Prove that the graph of a quadratic function has no
 points of inflection. State conditions under which the
 graph is always (a) concave upward; (b) concave down-
 ward. Illustrate with sketches.

34 Prove that the graph of a polynomial function of degree 3
 has exactly one point of inflection. Illustrate this fact with
 sketches.

35 Prove that the graph of a polynomial function of degree
 $n > 2$ has at most $n - 2$ points of inflection.

36 If $f(x) = x^n$, where $n > 1$, prove that the graph of f has
 either one or no points of inflection, according to whether
 n is odd or even. Illustrate with sketches.

4.5 Horizontal and Vertical Asymptotes

Thus far we have used the derivative primarily as a tool for sketching graphs. In particular, we have developed techniques for determining where a graph rises or falls, the high and low points on a graph, concavity, and points of inflection. Before turning to physical applications of derivatives, we shall discuss several other interesting characteristics of some graphs. When applied to the derivative f' , our discussion will provide information about vertical tangent lines to the graph of f' .

Let us begin by considering the graph of $f(x) = 2 + (1/x)$, which is sketched in Figure 4.28, and concentrate on values of $f(x)$ if x is large and positive. For example, $f(100) = 2.01, f(1000) = 2.001, f(10,000) = 2.0001,$

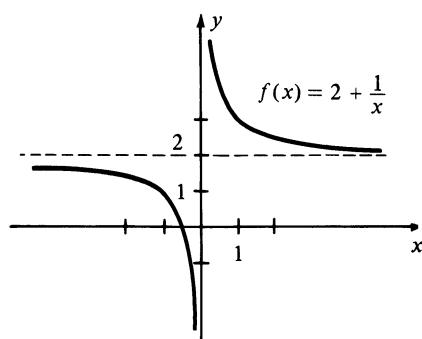


FIGURE 4.28

and $f(100,000) = 2.00001$. It is evident that we can make $f(x)$ as close to 2 as desired or, equivalently, we can make $|f(x) - 2|$ arbitrarily small by choosing x sufficiently large. This behavior of $f(x)$ is expressed by writing

$$\lim_{x \rightarrow \infty} \left(2 + \frac{1}{x} \right) = 2$$

which may be read “the limit of $2 + (1/x)$ as x becomes infinite (or as x increases without bound) is 2.”

The general definition that describes the behavior of $f(x)$ in the preceding illustration may be phrased as follows.

Definition (4.19)

Let f be defined on an interval (c, ∞) . The statement

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that for every $\varepsilon > 0$, there corresponds a positive number N , such that

$$\text{if } x > N, \text{ then } |f(x) - L| < \varepsilon.$$

If $\lim_{x \rightarrow \infty} f(x) = L$ we say that **the limit of $f(x)$ is L as x becomes infinite (or as x increases without bound)**. To emphasize the fact that x is large and positive, we often say that the limit of $f(x)$ is L as x becomes *positively* infinite. It is important to remember that the symbol ∞ does not represent a real number. It is used here to denote that x increases without bound. Later, in Definition (4.22), we shall employ it in a different context.

It is of interest to study the graphical significance of the preceding definition. Suppose $\lim_{x \rightarrow \infty} f(x) = L$, and consider any horizontal lines $y = L \pm \varepsilon$ (see Figure 4.29). According to Definition (4.19), if x is larger than some number N , then all points $P(x, f(x))$ lie between these horizontal lines. Roughly speaking, the graph of f gets closer and closer to the line $y = L$ as x gets larger and larger. The line $y = L$ is called a **horizontal**

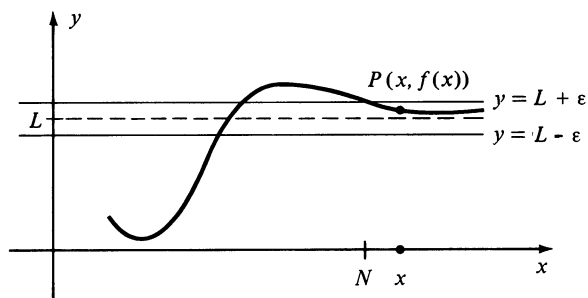


FIGURE 4.29 $\lim_{x \rightarrow \infty} f(x) = L$

asymptote for the graph of f . For example, the line $y = 2$ in Figure 4.28 is a horizontal asymptote for the graph of $f(x) = 2 + (1/x)$.

In Figure 4.29, the graph of f approaches the asymptote $y = L$ from above, that is, with $f(x) > L$. However, we could also have $f(x) < L$ or other variations.

The next definition covers the situation in which $|x|$ is large and x is *negative*.

Definition (4.20)

Let f be defined on an interval $(-\infty, c)$. The statement

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that for every $\varepsilon > 0$, there corresponds a negative number N , such that

$$\text{if } x < N, \text{ then } |f(x) - L| < \varepsilon.$$

If $\lim_{x \rightarrow -\infty} f(x) = L$ we say that **the limit of $f(x)$ is L as x becomes negatively infinite** (or **as x decreases without bound**).

A geometric illustration of (4.20) is given in Figure 4.30. If we consider any lines $y = L \pm \varepsilon$, then all points $P(x, f(x))$ on the graph lie between these lines if x is less than some number N . As before, we refer to the line $y = L$ as a horizontal asymptote for the graph of f .

Limit theorems that are analogous to those in Chapter 2 may be established. In particular, Theorem (2.8) concerning limits of sums, products, and quotients is true for the cases $x \rightarrow \infty$ or $x \rightarrow -\infty$. Similarly, Theorem (2.13) on the limit of $\sqrt[n]{f(x)}$ holds if $x \rightarrow \infty$ or $x \rightarrow -\infty$. Finally, it is trivial to show that

$$\lim_{x \rightarrow \infty} c = c \quad \text{and} \quad \lim_{x \rightarrow -\infty} c = c.$$

The next theorem provides an important tool for calculating specific limits.

Theorem (4.21)

If k is a positive rational number and c is any real number, then

$$\lim_{x \rightarrow \infty} \frac{c}{x^k} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{c}{x^k} = 0$$

provided x^k is always defined.

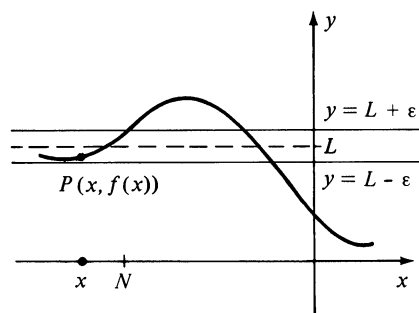


FIGURE 4.30 $\lim_{x \rightarrow -\infty} f(x) = L$

Proof To prove that $\lim_{x \rightarrow \infty} (c/x^k) = 0$ by means of (4.19) we must show that for every $\varepsilon > 0$, there corresponds a number N such that

$$\text{if } x > N, \text{ then } \left| \frac{c}{x^k} - 0 \right| < \varepsilon.$$

If $c = 0$, any $N > 0$ will suffice. If $c \neq 0$, the following four inequalities are equivalent.

$$\left| \frac{c}{x^k} - 0 \right| < \varepsilon; \quad \frac{|x|^k}{|c|} > \frac{1}{\varepsilon}; \quad |x|^k > \frac{|c|}{\varepsilon}; \quad x > \left(\frac{|c|}{\varepsilon} \right)^{1/k}.$$

If we let $N = (|c|/\varepsilon)^{1/k}$, then whenever $x > N$ the fourth, and hence the first, inequality is true, which is what we wished to show. The second part of the theorem may be proved in similar fashion. $\square$

If f is a rational function, then limits as $x \rightarrow \infty$ or $x \rightarrow -\infty$ may be found by first dividing numerator and denominator of $f(x)$ by a suitable power of x and then applying Theorem (4.21). Specifically, suppose $g(x)$ and $h(x)$ are polynomials and we wish to investigate $f(x) = g(x)/h(x)$. If the degree of $g(x)$ is not greater than the degree of $h(x)$, then we divide numerator and denominator by x^k , where k is the degree of $h(x)$. If the degree of $g(x)$ is greater than the degree of $h(x)$, it can be shown that $f(x)$ has no limit as $x \rightarrow \infty$ or as $x \rightarrow -\infty$.

Example 1 Investigate $\lim_{x \rightarrow -\infty} (2x^2 - 5)/(3x^2 + x + 2)$.

Solution Since we are interested only in negative values of x , we may assume that $x \neq 0$. Using the rule stated in the preceding paragraph we first divide numerator and denominator of the given expression by x^2 and then employ the appropriate limit theorems. Thus

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{2x^2 - 5}{3x^2 + x + 2} &= \lim_{x \rightarrow -\infty} \frac{2 - (5/x^2)}{3 + (1/x) + (2/x^2)} \\ &= \frac{\lim_{x \rightarrow -\infty} [2 - (5/x^2)]}{\lim_{x \rightarrow -\infty} [3 + (1/x) + (2/x^2)]} \\ &= \frac{\lim_{x \rightarrow -\infty} 2 - \lim_{x \rightarrow -\infty} (5/x^2)}{\lim_{x \rightarrow -\infty} 3 + \lim_{x \rightarrow -\infty} (1/x) + \lim_{x \rightarrow -\infty} (2/x^2)} = \frac{2 - 0}{3 + 0 + 0} = \frac{2}{3}. \end{aligned}$$

It follows that the line $y = \frac{2}{3}$ is a horizontal asymptote for the graph of f . $\blacksquare$

Example 2 If $f(x) = 4x/(x^2 + 9)$, determine the horizontal asymptotes, find the relative maxima and minima and the points of inflection, discuss concavity, and sketch the graph of f .

Solution We divide numerator and denominator of $f(x)$ by x^2 and use limit theorems as follows:

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{4x}{x^2 + 9} = \lim_{x \rightarrow \infty} \frac{(4/x)}{1 + (9/x^2)} \\ &= \frac{\lim_{x \rightarrow \infty} (4/x)}{\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} (9/x^2)} = \frac{0}{1 + 0} = \frac{0}{1} = 0. \end{aligned}$$

Hence $y = 0$ (the x -axis) is horizontal asymptote for the graph of f . Similarly, $\lim_{x \rightarrow -\infty} f(x) = 0$.

It is left to the reader to show that the first and second derivatives of f are given by

$$f'(x) = \frac{4(9 - x^2)}{(x^2 + 9)^2}; \quad f''(x) = \frac{8x(x^2 - 27)}{(x^2 + 9)^3}.$$

The critical numbers of f are ± 3 . (Why?) Using either the First or Second Derivative Test, we can show that $f(-3)$ is a relative minimum and $f(3)$ is a relative maximum. Points of inflection have x -coordinates 0 and $\pm\sqrt{27} = \pm 3\sqrt{3}$. The reader may verify that the graph is concave downward in the intervals $(-\infty, -3\sqrt{3})$ and $(0, 3\sqrt{3})$ and concave upward in $(-3\sqrt{3}, 0)$ and $(3\sqrt{3}, \infty)$. The graph is sketched in Figure 4.31. ■

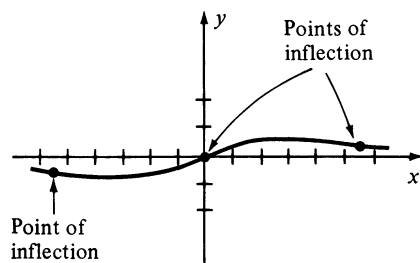


FIGURE 4.31

Example 3 Investigate $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 2}}{3 - 4x}$.

Solution Dividing numerator and denominator of the quotient by x and applying limit theorems gives us

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 2}}{3 - 4x} &= \lim_{x \rightarrow \infty} \frac{\sqrt{9 + (2/x^2)}}{(3/x) - 4} = \frac{\lim_{x \rightarrow \infty} \sqrt{9 + (2/x^2)}}{\lim_{x \rightarrow \infty} [(3/x) - 4]} \\ &= \frac{\sqrt{\lim_{x \rightarrow \infty} (9 + (2/x^2))}}{\lim_{x \rightarrow \infty} (3/x) - \lim_{x \rightarrow \infty} 4} = \frac{\sqrt{\lim_{x \rightarrow \infty} 9 + \lim_{x \rightarrow \infty} (2/x^2)}}{0 - 4} \\ &= \frac{\sqrt{9 + 0}}{-4} = -\frac{3}{4}. \end{aligned}$$

Let us next consider the function f defined by $f(x) = 1/(x - 3)^2$. If x is close to 3 (but $x \neq 3$) the denominator $(x - 3)^2$ is close to zero, and hence $f(x)$ is very large. Indeed, $f(x)$ can be made as large as desired by choosing x sufficiently close to 3. This behavior of $f(x)$ is symbolized by writing

$$\lim_{x \rightarrow 3} \frac{1}{(x - 3)^2} = \infty.$$

In general, we have the following definition.

Definition (4.22)

Let f be defined on an open interval containing a (except possibly at $x = a$). The statement **$f(x)$ becomes infinite** (or **increases without bound**) as x approaches a , written

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that for every positive number M , there corresponds a $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta, \text{ then } f(x) > M.$$

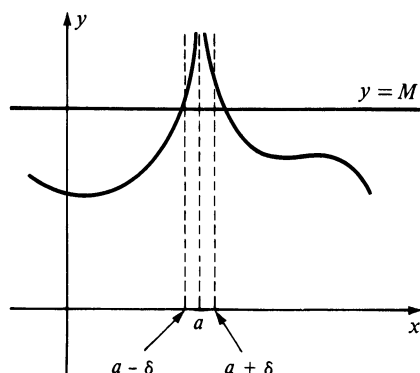


FIGURE 4.32 $\lim_{x \rightarrow a} f(x) = \infty$

In Definition (4.22) we sometimes state that $f(x)$ becomes *positively* infinite as x approaches a . A geometric interpretation in terms of the graph of f is shown in Figure 4.32. If we consider any horizontal line $y = M$, then when x is in a suitable interval $(a - \delta, a + \delta)$, but $x \neq a$, the points on the graph of f lie *above* the horizontal line.

As we mentioned earlier, the symbol ∞ is used as a device for describing certain types of functional behavior. It is *not* a real number. In particular, $\lim_{x \rightarrow a} f(x) = \infty$ does not mean that the limit *exists* as x approaches a . The limit notation is merely used to denote the variation of $f(x)$ that we have described.

The notion of $f(x)$ increasing without bound as x approaches a from the right or left can also be introduced. This is indicated by writing

$$\lim_{x \rightarrow a^+} f(x) = \infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \infty,$$

respectively. Only minor modifications of Definition (4.22) are needed to define these concepts. Thus, for $x \rightarrow a^+$, it is assumed that $f(x)$ exists on some open interval (a, c) , and the inequality $0 < |x - a| < \delta$ in Definition (4.22) is changed to $a < x < a + \delta$; that is, x may only take on values *larger* than a . Similar statements may be made for $x \rightarrow a^-$. Graphs illustrating these ideas are sketched in Figure 4.33.

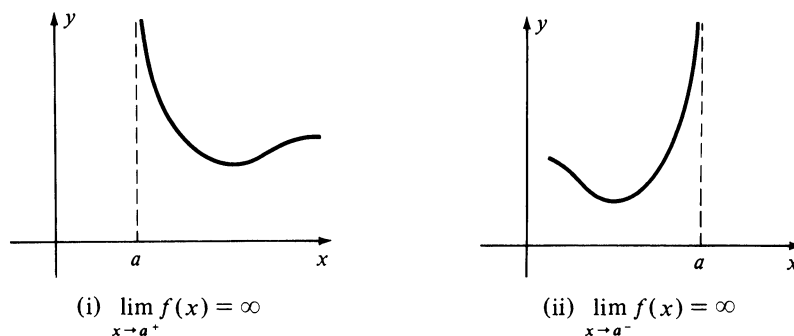


FIGURE 4.33

Definition (4.23)

Let f be defined on an open interval containing a (except possibly at $x = a$). The statement $f(x)$ **becomes negatively infinite** (or **decreases without bound**) as x approaches a , written

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that for every negative number M there corresponds a $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta, \text{ then } f(x) < M.$$

Graphical illustrations of Definition (4.23) together with the cases $x \rightarrow a^+$ and $x \rightarrow a^-$ are sketched in Figure 4.34. If either of Definitions (4.22)

or (4.23) occurs (including the cases of one-sided limits), then the line $x = a$ is called a **vertical asymptote** for the graph of f . We also say that f has an **infinite discontinuity** at $x = a$. The line $x = a$ in Figures (4.32)–(4.34) is a vertical asymptote for the given graph.

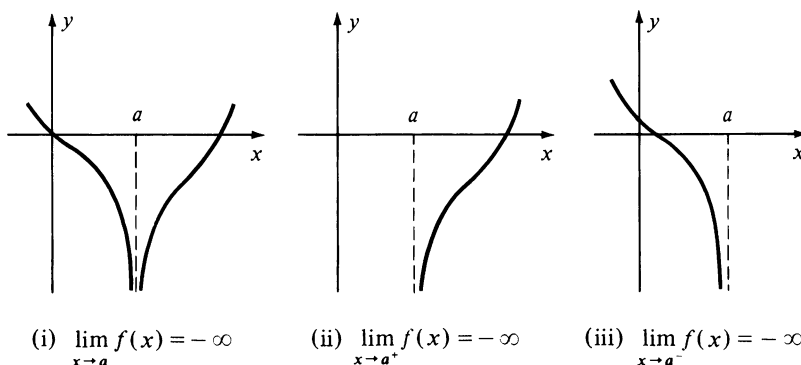


FIGURE 4.34

The next theorem, stated without proof, is useful in the investigation of certain limits.

Theorem (4.24)

(i) If n is an even positive integer, then

$$\lim_{x \rightarrow a} \frac{1}{(x - a)^n} = \infty.$$

(ii) If n is an odd positive integer, then

$$\lim_{x \rightarrow a^+} \frac{1}{(x - a)^n} = \infty \quad \text{and} \quad \lim_{x \rightarrow a^-} \frac{1}{(x - a)^n} = -\infty.$$

It follows that if n is a positive integer and $f(x) = 1/(x - a)^n$, then the graph of f has the line $x = a$ as a vertical asymptote. Note also that since $\lim_{x \rightarrow \infty} 1/(x - a)^n = 0$, the x -axis is a horizontal asymptote.

It is unnecessary to memorize Theorem (4.24), since for any specific problem it is usually possible to determine the answer intuitively. As an illustration, consider $\lim_{x \rightarrow -5^-} 1/(x + 5)^3$. If x is close to -5 and $x < -5$, then $x + 5$ is close to 0 and is *negative*. Consequently $(x + 5)^3$ is close to 0 and negative and, therefore, $1/(x + 5)^3$ is a numerically large negative number. This suggests that

$$\lim_{x \rightarrow -5^-} \frac{1}{(x + 5)^3} = -\infty.$$

To investigate $x \rightarrow -5^+$, we first note that if $x > -5$, then $x + 5 > 0$, and reason that

$$\lim_{x \rightarrow -5^+} \frac{1}{(x + 5)^3} = \infty.$$

Example 4 If $f(x) = 1/(x - 2)^3$, discuss $\lim_{x \rightarrow 2^+} f(x)$, $\lim_{x \rightarrow 2^-} f(x)$, and sketch the graph of f .

Solution If x is very close to 2, but larger than 2, then $x - 2$ is a small positive number and hence $1/(x - 2)^3$ is a large positive number. Thus it is evident that

$$\lim_{x \rightarrow 2^+} \frac{1}{(x - 2)^3} = \infty.$$

This fact also follows from (ii) of Theorem (4.24) with $a = 2$ and $n = 3$.

If x is close to 2 but less than 2, then $x - 2$ is close to 0 and negative. Hence

$$\lim_{x \rightarrow 2^-} \frac{1}{(x - 2)^3} = -\infty.$$

This is also a consequence of (ii) of (4.24). The graph of f is sketched in Figure 4.35.

The line $x = 2$ is a vertical asymptote. The reader should verify that

$$\lim_{x \rightarrow \infty} \frac{1}{(x - 2)^3} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{(x - 2)^3} = 0.$$

Hence $y = 0$ (the x -axis) is a horizontal asymptote. ■

The graphs of the tangent, cotangent, secant and cosecant functions also have vertical asymptotes. These functions will be considered in Chapter 8.

Several properties of sums, products, and quotients of functions which become infinite are stated below. Analogous results hold for the cases $x \rightarrow a^+$ and $x \rightarrow a^-$.

Theorem (4.25)

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = c$ then:

- (i) $\lim_{x \rightarrow a} [g(x) + f(x)] = \infty$.
- (ii) If $c > 0$, then $\lim_{x \rightarrow a} [g(x)f(x)] = \infty$ and $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \infty$.
- (iii) If $c < 0$, then $\lim_{x \rightarrow a} [g(x)f(x)] = -\infty$ and $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = -\infty$.
- (iv) $\lim_{x \rightarrow a} \frac{g(x)}{f(x)} = 0$.

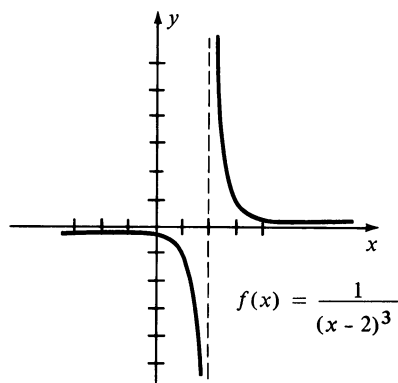


FIGURE 4.35

The conclusions of Theorem (4.25) appear intuitively evident. In (i), for example, if $g(x)$ is close to c and if $f(x)$ increases without bound as x approaches a , then it is reasonable to expect that $g(x) + f(x)$ can be made arbitrarily large by choosing x sufficiently close to a . In (iii), if $\lim_{x \rightarrow a} g(x) = c < 0$, then $g(x)$ is negative when x is close to a . Consequently, if $f(x)$ is large and positive, then when x is close to a the product $g(x)f(x)$ is numerically large and *negative*. This suggests that $\lim_{x \rightarrow a} g(x)f(x) = -\infty$. The reader should supply similar arguments for the remaining parts of Theorem (4.25). We shall not give formal proofs for these results. A similar theorem can be stated for $\lim_{x \rightarrow a} f(x) = -\infty$. Theorems involving more than two functions can also be proved.

Graphs of rational functions frequently have vertical and horizontal asymptotes. Illustrations of this are shown in Figures 4.28, 4.31, and 4.35. Another is given in the following example.

Example 5 If $f(x) = 2x^2/(9 - x^2)$, find the vertical and horizontal asymptotes for the graph of f . Sketch the graph of f .

Solution We begin by writing

$$f(x) = \frac{2x^2}{(3 - x)(3 + x)}.$$

The denominator vanishes at $x = 3$ and $x = -3$, and hence the corresponding lines are candidates for vertical asymptotes. (Why?) Since

$$\lim_{x \rightarrow 3^-} \frac{1}{3 - x} = \infty \quad \text{and} \quad \lim_{x \rightarrow 3^-} \frac{2x^2}{3 + x} = 3$$

we see from (ii) of Theorem (4.25) that

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \left(\frac{1}{3 - x} \right) \left(\frac{2x^2}{3 + x} \right) = \infty;$$

that is, $f(x)$ becomes positively infinite as x approaches 3 from the left. Moreover, since

$$\lim_{x \rightarrow 3^+} \frac{1}{3 - x} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 3^+} \frac{2x^2}{3 + x} = 3 > 0$$

it follows that

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \left(\frac{1}{3 - x} \right) \left(\frac{2x^2}{3 + x} \right) = -\infty.$$

Thus $f(x)$ becomes negatively infinite as x approaches 3 from the right. This variation of $f(x)$ near the vertical asymptote $x = 3$ is illustrated in Figure 4.36.

In like manner, since

$$\lim_{x \rightarrow -3^-} \frac{1}{3 + x} = -\infty \quad \text{and} \quad \lim_{x \rightarrow -3^-} \frac{2x^2}{3 - x} = 3 > 0$$

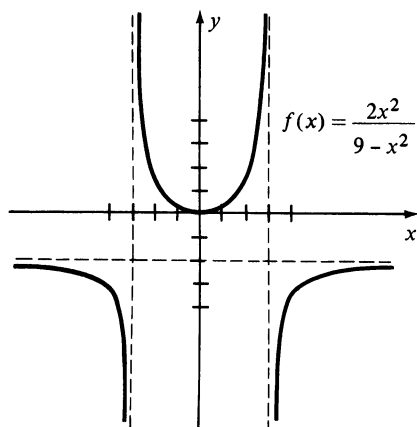


FIGURE 4.36

we conclude that

$$\lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \left(\frac{1}{3+x} \right) \left(\frac{2x^2}{3-x} \right) = -\infty.$$

Similarly,

$$\lim_{x \rightarrow -3^+} \frac{1}{3+x} = \infty \quad \text{implies that} \quad \lim_{x \rightarrow -3^+} f(x) = \infty.$$

This behavior of $f(x)$ near $x = -3$ is illustrated in Figure 4.36.

To find the horizontal asymptotes we consider

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2x^2}{9-x^2} = \lim_{x \rightarrow \infty} \frac{2}{(9/x^2) - 1} = -2.$$

The same limit is obtained if $x \rightarrow -\infty$. Consequently $y = -2$ is a horizontal asymptote. Using this information and plotting several points gives us the sketch in Figure 4.36.

If more detailed information about the graph is desired, we could use the first derivative to show that f is decreasing on the intervals $(-\infty, -3)$ and $(-3, 0]$, and increasing on $[0, 3)$ and $(3, \infty)$, with a minimum occurring at $x = 0$ as shown in the figure. The second derivative could be used to investigate concavity. ■

If we regard a tangent line l as a limiting position of a secant line (see Figure 2.2), then it is possible for l to be vertical. In such cases, as the secant line approaches l its slope increases or decreases without bound. With this in mind we formulate the next definition.

Definition (4.26)

The graph of a function f is said to have a **vertical tangent line** at the point $P(a, f(a))$ if f is continuous at a and

$$\lim_{x \rightarrow a} |f'(x)| = \infty.$$

Example 6 If $f(x) = (x-8)^{1/3} + 1$ and $g(x) = (x-8)^{2/3} + 1$, show that the graphs of f and g have vertical tangent lines at $P(8, 1)$. Sketch the graphs, showing the tangent lines at P .

Solution Both f and g are continuous at 8. Differentiating we obtain

$$f'(x) = \frac{1}{3}(x-8)^{-2/3} = \frac{1}{3(x-8)^{2/3}}$$

$$g'(x) = \frac{2}{3}(x-8)^{-1/3} = \frac{2}{3(x-8)^{1/3}}.$$

Evidently,

$$\lim_{x \rightarrow 8} |f'(x)| = \infty \quad \text{and} \quad \lim_{x \rightarrow 8} |g'(x)| = \infty.$$

The graphs are sketched in Figure 4.37. Note that for f the slope of the tangent line becomes positively infinite whether x approaches 8 from the left or from the right. However, for g the slope becomes negatively infinite if $x \rightarrow 8^-$ and positively infinite if $x \rightarrow 8^+$. ■

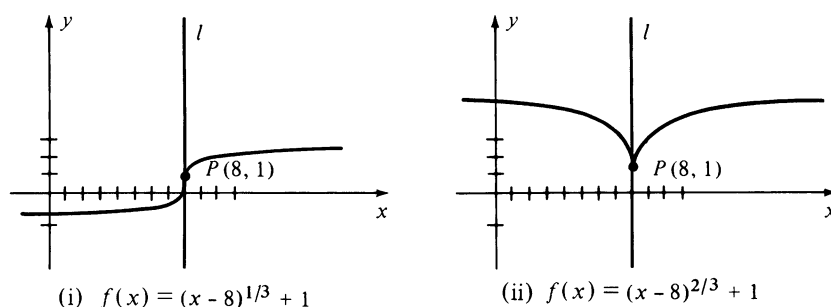


FIGURE 4.37

Definition (4.26) may be modified to include vertical tangent lines at an endpoint of the domain of a function. Thus, if f is continuous on $[a, b]$, but is undefined outside of this interval, then the graph is said to have a vertical tangent line at $P(a, f(a))$ or $Q(b, f(b))$ if

$$\lim_{x \rightarrow a^+} |f'(x)| = \infty \quad \text{or} \quad \lim_{x \rightarrow b^-} |f'(x)| = \infty,$$

respectively. For example, if $f(x) = \sqrt{1 - x^2}$, then the graph of f is the upper half of the unit circle with center at the origin, and there are vertical tangent lines at the points $(-1, 0)$ and $(1, 0)$. (Check this fact.)

4.5 Exercises

Find the limits in Exercises 1–12.

1 $\lim_{x \rightarrow \infty} \frac{5x^2 - 3x + 1}{2x^2 + 4x - 7}$

2 $\lim_{x \rightarrow \infty} \frac{3x^3 - x + 1}{6x^3 + 2x^2 - 7}$

3 $\lim_{x \rightarrow -\infty} \frac{4 - 7x}{2 + 3x}$

4 $\lim_{x \rightarrow -\infty} \frac{(3x + 4)(x - 1)}{(2x + 7)(x + 2)}$

5 $\lim_{x \rightarrow \infty} \sqrt[3]{\frac{8 + x^2}{x(x + 1)}}$

6 $\lim_{x \rightarrow -\infty} \frac{4x - 3}{\sqrt{x^2 + 1}}$

7 $\lim_{x \rightarrow \infty} \frac{\sqrt{4x + 1}}{10 - 3x}$

8 $\lim_{x \rightarrow \infty} \frac{2x^2 - x + 3}{x^3 + 1}$

9 $\lim_{x \rightarrow \infty} \sqrt{x^2 + 5x} - x$

10 $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 - 3x})$

11 $\lim_{x \rightarrow -\infty} \frac{1 + \sqrt[5]{x}}{1 - \sqrt[5]{x}}$

12 $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x}}{x^3 + 1}$

13 Determine the horizontal asymptotes and sketch the graph of $y = 4x^2/(1 + x^2)$.

14 Determine the horizontal asymptotes and sketch the graph of $y = 4/(9 + x^2)$.

15 If f and g are polynomial functions of degree n , find $\lim_{x \rightarrow \infty} f(x)/g(x)$ and $\lim_{x \rightarrow -\infty} f(x)/g(x)$. What is true if the degree of f is less than the degree of g ?

- 16 If f is any polynomial function of degree greater than 0, prove that $\lim_{x \rightarrow \infty} 1/f(x) = 0$.
- 17 Prove, by a direct application of Definition (4.19), that $\lim_{x \rightarrow \infty} [2 + (1/x)] = 2$.
- 18 Define $\lim_{x \rightarrow a^-} f(x) = \infty$ in a manner similar to (4.22).
- 19 If f and g are functions such that $0 < f(x) < g(x)$ for all x , and if $\lim_{x \rightarrow \infty} g(x) = 0$, prove that $\lim_{x \rightarrow \infty} f(x) = 0$.
- 20 Prove that if $f(x)$ has a limit as x becomes infinite, then the limit is unique.

In Exercises 21–30, find $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$. Identify the vertical and horizontal asymptotes, find the local maxima and minima, and sketch the graph of f .

21 $f(x) = \frac{5}{x-4}, a = 4$

22 $f(x) = \frac{5}{4-x}, a = 4$

23 $f(x) = \frac{8}{(2x+5)^3}, a = -\frac{5}{2}$

24 $f(x) = \frac{-4}{7x+3}, a = -\frac{3}{7}$

25 $f(x) = \frac{3x}{(x+8)^2}, a = -8$

26 $f(x) = \frac{3x^2}{(2x-9)^2}, a = \frac{9}{2}$

27 $f(x) = \frac{2x^2}{x^2-x-2}, a = -1, a = 2$

28 $f(x) = \frac{4x}{x^2-4x+3}, a = 1, a = 3$

29 $f(x) = \frac{1}{x(x-3)^2}, a = 0, a = 3$

30 $f(x) = \frac{x^2}{x+1}, a = -1$

37 $f(x) = \frac{x^2 + 3x + 2}{x^2 + 2x - 3}$

38 $f(x) = \frac{x^2 - 5x}{x^2 - 25}$

39 $f(x) = \frac{x+4}{x^2-16}$

40 $f(x) = \frac{\sqrt[3]{16-x^2}}{4-x}$

- 41 If f and g are polynomial functions and the degree of $f(x)$ is greater than the degree of $g(x)$, what is $\lim_{x \rightarrow \infty} f(x)/g(x)$?

- 42 Prove Theorem (4.24).

In Exercises 43–46, find the points where the graph of f has a vertical tangent line.

43 $f(x) = x(x+2)^{3/5}$

44 $f(x) = \sqrt{x+2}$

45 $f(x) = \sqrt{16-9x^2} + 3$

46 $f(x) = \sqrt[3]{x} - 5$

- 47 Prove that the graph of a rational function has no vertical tangent lines.

- 48 Sketch the graph of $y = |x^3 - x|$ and determine where the graph has (a) a horizontal tangent line; (b) a vertical tangent line; (c) no tangent line.

- 49 Prove that if $q(x) = f(x)/g(x)$ where $f(x)$ and $g(x)$ are polynomials, and if the degree of $g(x)$ is n , then the graph of q can have at most n vertical asymptotes. Give examples illustrating that the graph may have anywhere from 0 to n vertical asymptotes.

- 50 Prove that the graph of a function can have at most two different horizontal asymptotes.

- 51 Basic processes governing bacterial growth in a culture may be investigated by utilizing a laboratory device called a *chemostat* (see figure). The rate of bacterial growth is controlled by supplying all nutrients that are essential for excess growth except one, the *rate limiting nutrient*.

Let x denote the concentration of the rate limiting nutrient. If N is the bacterial population and $r(x)$ is the

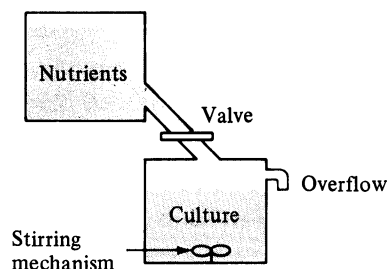


FIGURE FOR EXERCISE 51

In Exercises 31–40 determine the vertical and horizontal asymptotes and sketch the graph of f .

31 $f(x) = \frac{1}{x^2-4}$

32 $f(x) = \frac{5x}{4-x^2}$

33 $f(x) = \frac{2x^4}{x^4+1}$

34 $f(x) = \frac{3x}{x^2+1}$

35 $f(x) = \frac{1}{x^3+x^2-6x}$

36 $f(x) = \frac{x^2-x}{16-x^2}$

rate of growth corresponding to x , then Monod* suggested that

$$r(x) = \frac{axN}{b + x}$$

where a and b are constants. The *relative growth rate* (rate of growth per unit population) $R(x)$ is defined by

$$R(x) = \frac{r(x)}{N} = \frac{ax}{b + x}.$$

The relationship between the constants a and b may be determined by analyzing the graph of the rational function R .

Show that R has no relative extrema, but that the relative growth rate approaches $a = 2R(b)$ as the concentration of the rate limiting nutrient increases without bound. Sketch the graph of R .

* See *Ann. Inst. Pasteur*, Vol. 79 (1950), p. 390.

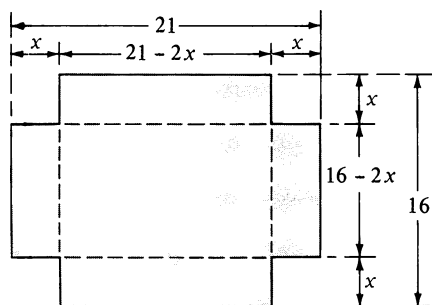
4.6 Applications of Extrema

The theory we have developed for finding extrema of functions can be applied to certain practical problems. These problems may either be described orally, or stated in written words, as is the case in textbooks. To solve them, it is necessary to convert verbal statements into the language of mathematics by introducing formulas, functions, or equations. Since the types of applications are unlimited, it is difficult to state specific rules for finding solutions. However, it is possible to develop a general strategy for attacking such problems. The following guidelines are often helpful.

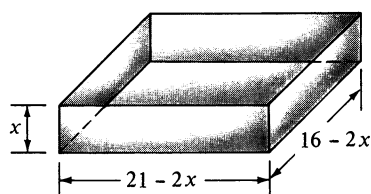
Guidelines for Solving Applied Problems Involving Extrema

- 1 Read the problem carefully several times and think about the given facts, together with the unknown quantities that are to be found.
- 2 If possible, sketch a picture or diagram and label it appropriately, introducing variables for unknown quantities. Words such as “what,” “find,” “how much,” “how far,” or “when” should alert you to the unknown quantities.
- 3 Write down the known facts together with any relationships involving the variables. A relationship may often be described by means of an equation.
- 4 Determine which variable is to be maximized or minimized and express this variable as a function of *one* of the other variables.
- 5 Find the critical numbers of the function obtained in Guideline 4 and test each of them for maxima or minima.
- 6 Check to see whether extrema occur at the endpoints of the domain of the function obtained in Guideline 4.
- 7 Don’t become discouraged if you are unable to solve a given problem. It takes a great deal of effort and practice to become proficient in solving applied problems. Keep trying!

The use of these guidelines is illustrated in the following examples.



(i)



(ii)

FIGURE 4.38

Example 1 An open box with a rectangular base is to be constructed from a rectangular piece of cardboard 16 in. wide and 21 in. long by cutting out a square from each corner and then bending up the sides. Find the size of the corner square which will produce a box having the largest possible volume.

Solution Applying Guideline 2, let us begin by drawing a picture of the cardboard as in (i) of Figure 4.38, where we have introduced a variable x to denote the length of the side of the square to be cut out from each corner. Using Guideline 3, we specify the known facts (the size of the rectangle) by labeling the figure appropriately.

Next (see Guideline 4), we determine that the quantity to be maximized is the volume V of the box to be constructed by folding along the dashed lines (see (ii) of Figure 4.38). Continuing with Guideline 4, let us express V as a function of the variable x . Referring to (ii) of the figure,

$$V = x(16 - 2x)(21 - 2x) = 2(168x - 37x^2 + 2x^3).$$

This equation expresses V as a function of x , and we proceed to find the critical numbers and test them for maxima or minima (see Guideline 5). Differentiating with respect to x ,

$$\begin{aligned} D_x V &= 2(168 - 74x + 6x^2) \\ &= 4(3x^2 - 37x + 84) \\ &= 4(3x - 28)(x - 3). \end{aligned}$$

Thus the possible critical numbers are $\frac{28}{3}$ and 3. Since $\frac{28}{3}$ is outside the domain of x (Why?) the only critical number is 3.

Let us employ the Second Derivative Test for extrema. The second derivative of V is given by

$$D_x^2 V = 2(-74 + 12x) = 4(6x - 37).$$

Substituting the critical number 3 for x we obtain

$$D_x^2 V = 4(18 - 37) = -76 < 0$$

and hence V has a relative maximum at $x = 3$.

Finally, we check for endpoint extrema (see Guideline 6). Since $0 \leq x \leq 8$ (Why?) and since $V = 0$ if either $x = 0$ or $x = 8$, the maximum value of V does not occur at an endpoint of the domain of x . Consequently, a three-inch square should be cut from each corner of the cardboard in order to maximize the volume of the resulting box. ■

In the remaining examples we shall not always point out the guidelines that are employed. The reader should be able to determine specific guidelines by studying the solutions.

Example 2 A circular cylindrical container, open at the top and having a capacity of 24π in.³, is to be manufactured. If the cost of the material used for the bottom of the container is three times that used for the curved part and if there is no waste of material, find the dimensions that will minimize the cost.

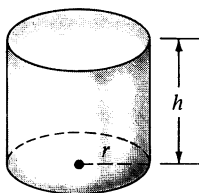


FIGURE 4.39

Solution Let us begin by drawing a picture as in Figure 4.39, letting r denote the length (in inches) of the radius of the base of the container and h the altitude (in inches). Since the volume is 24π in.³, we have

$$\pi r^2 h = 24\pi.$$

This gives us the relationship

$$h = \frac{24}{r^2}$$

between r and h .

Our goal is to minimize the cost C of the material used to manufacture the container.

If a denotes the cost per square inch of the material to be used for the curved part, then the material to be used for the bottom costs $3a$ per in.². The cost of the material for the curved part is, therefore, $a(2\pi rh)$ and the cost for the base is $3a(\pi r^2)$. The total cost C for material is given by

$$C = 3a(\pi r^2) + a(2\pi rh) = \pi(3r^2 + 2rh)$$

or, since $h = 24/r^2$,

$$C = \pi\left(3r^2 + \frac{48}{r}\right).$$

Since a is fixed, this expresses C as a function of one variable r as recommended in Guideline 4. To find the value of r that will lead to the smallest value for C , we find the critical numbers. Differentiating the last equation with respect to r gives us

$$D_r C = \pi\left(6r - \frac{48}{r^2}\right) = 6\pi\left(\frac{r^3 - 8}{r^2}\right).$$

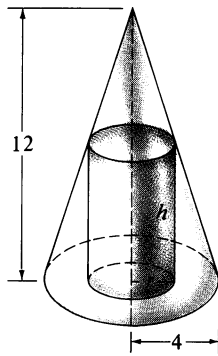
Since $D_r C = 0$ if $r = 2$, we see that 2 is the only critical number. (The number 0 is not a critical number since C is undefined if $r = 0$.) Since $D_r C < 0$ if $r < 2$ and $D_r C > 0$ if $r > 2$, it follows from the First Derivative Test that C has its minimum value if the radius of the cylinder is 2 inches. The corresponding value for the altitude (obtained from $h = 24/r^2$) is $\frac{24}{4}$, or 6 in.

Since the domain of the variable r is the infinite interval $(0, \infty)$, there can be no endpoint extrema. ■

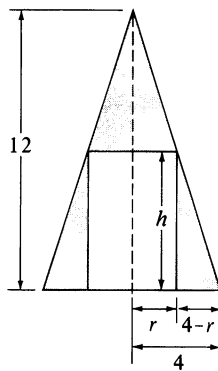
Example 3 Find the maximum volume of a right circular cylinder that can be inscribed in a cone of altitude 12 cm and base radius 4 cm, if the axes of the cylinder and cone coincide.

Solution The problem is illustrated in Figure 4.40 where (ii) represents a cross-section through the axis of the cone and cylinder.

The volume V of the cylinder is given by $V = \pi r^2 h$. In order to express V in terms of one variable (see Guideline 4) we must find a relationship be-



(i)



(ii)

FIGURE 4.40

tween r and h . Referring to (ii) of Figure 4.40 and using similar triangles we see that

$$\frac{h}{4-r} = \frac{12}{4} = 3 \quad \text{or} \quad h = 3(4-r).$$

Consequently,

$$V = \pi r^2 h = \pi r^2 [3(4-r)] = 3\pi(4r^2 - r^3).$$

If either $r = 0$ or $r = 4$ we see that $V = 0$, and hence the maximum volume is not an endpoint extremum. It is sufficient, therefore, to search for relative maximum values. Since

$$D_r V = 3\pi(8r - 3r^2) = 3\pi r(8 - 3r)$$

the critical numbers for V are $r = 0$ and $r = \frac{8}{3}$. Let us apply the Second Derivative Test for $r = \frac{8}{3}$. Differentiating $D_r V$ gives us

$$D_r^2 V = 3\pi(8 - 6r).$$

Substituting $\frac{8}{3}$ for r , we obtain

$$3\pi[8 - 6(\frac{8}{3})] = 3\pi(8 - 16) = -24\pi < 0$$

which means that a maximum occurs. The corresponding value for $h = 3(4-r)$ is

$$h = 3(4 - \frac{8}{3}) = 3(\frac{4}{3}) = 4.$$

Hence the maximum volume of the inscribed cylinder is

$$V = \pi\left(\frac{8}{3}\right)^2(4) = \pi\left(\frac{64}{9}\right)(4) = \frac{256\pi}{9} \text{ cm}^3. \quad \blacksquare$$

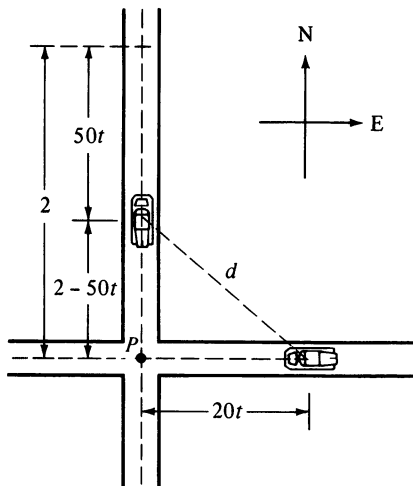


FIGURE 4.41

Example 4 A North-South highway intersects an East-West highway at a point P . An automobile crosses P at 10:00 A.M., traveling east at a constant speed of 20 mph. At that same instant another automobile is 2 mi north of P , traveling south at 50 mph. Find the time at which they are closest to each other and approximate the minimum distance between the automobiles.

Solution Typical positions of the automobiles are illustrated in Figure 4.41. If t denotes the number of hours after 10:00 A.M., then the slower automobile is $20t$ mi east of P , as indicated in the figure. The faster automobile is $50t$ mi south of its position at 10:00 A.M. and hence its distance from P is $2 - 50t$.

By the Pythagorean Theorem, the distance d between the automobiles is given by

$$\begin{aligned} d &= \sqrt{(2 - 50t)^2 + (20t)^2} \\ &= \sqrt{4 - 200t + 2500t^2 + 400t^2} \\ &= \sqrt{4 - 200t + 2900t^2}. \end{aligned}$$

Evidently d has its smallest value when the expression under the radical is minimal. Thus we may simplify our work by letting

$$f(t) = 4 - 200t + 2900t^2$$

and finding the value of t for which f has a minimum. Since

$$f'(t) = -200 + 5800t$$

the only critical number for f is

$$t = \frac{200}{5800} = \frac{1}{29}.$$

Moreover, since $f''(t) = 5800$, the second derivative is always positive and, therefore, f has a relative minimum at $t = \frac{1}{29}$, and $f(\frac{1}{29}) \approx 0.55$. Since the domain of t is $[0, \infty)$ and since $f(0) = 4$, there is no endpoint extremum. Consequently the automobiles will be closest at $\frac{1}{29}$ hours (or approximately 2.07 minutes) after 10:00 A.M. The minimal distance is

$$\sqrt{f(\frac{1}{29})} \approx \sqrt{0.55} \approx 0.74 \text{ mi.}$$

■

Example 5 A man in a rowboat 2 mi from the nearest point on a straight shoreline wishes to reach his house 6 mi further down the shore. If he can row at a rate of 3 mph and run at a rate of 5 mph; how should he proceed in order to arrive at his house in the shortest amount of time?

Solution A diagram of the problem is shown in Figure 4.42 where A denotes the position of the boat, B the nearest point on shore, and C the man's house. As shown in the figure, D is the point at which the boat will reach shore and x denotes the distance between B and D . Thus x is restricted to the interval $[0, 6]$.

By the Pythagorean Theorem, the distance between A and D is $\sqrt{x^2 + 4}$. Using the formula $\text{time} = \text{distance}/\text{rate}$, the time it takes the man to row from A to D is $\sqrt{x^2 + 4}/3$, whereas the time it takes the man to run from D to C is $(6 - x)/5$. Hence the total time T for the trip is given by

$$T = \frac{\sqrt{x^2 + 4}}{3} + \frac{6 - x}{5}$$

or equivalently,

$$T = \frac{1}{3}(x^2 + 4)^{1/2} + \frac{6}{5} - \frac{1}{5}x.$$

We wish to find the minimum value for T . Note that the case $x = 0$ corresponds to the extreme situation in which the man rows directly to B and then runs the entire distance from B to his house. If $x = 6$, then the man rows directly from A to his house. These numbers may be considered as endpoints of the domain of T . If $x = 0$, then from the formula for T ,

$$T = \frac{\sqrt{4}}{3} + \frac{6}{5} - 0 = \frac{28}{15} \approx 1.87$$

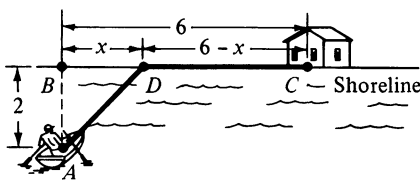


FIGURE 4.42

which is approximately 1 hour and 52 minutes. If $x = 6$, then

$$T = \frac{\sqrt{40}}{3} + \frac{6}{5} - \frac{6}{5} = \frac{2\sqrt{10}}{3} \approx 2.11$$

or approximately 2 hours and 7 minutes.

Differentiating the general formula for T we see that

$$\begin{aligned} D_x T &= \frac{1}{3} \cdot \frac{1}{2} (x^2 + 4)^{-1/2} (2x) - \frac{1}{5} \\ &= \frac{x}{3(x^2 + 4)^{1/2}} - \frac{1}{5}. \end{aligned}$$

To find the critical numbers we let $D_x T = 0$. This leads to the following chain of equations:

$$\begin{aligned} 5x &= 3(x^2 + 4)^{1/2} \\ 25x^2 &= 9(x^2 + 4) \\ 16x^2 &= 36 \\ x^2 &= \frac{36}{16} \\ x &= \frac{6}{4} = \frac{3}{2}. \end{aligned}$$

Thus $\frac{3}{2}$ is the only critical number. We next employ the Second Derivative Test. Applying the quotient rule to the formula for $D_x T$,

$$\begin{aligned} D_x^2 T &= \frac{3(x^2 + 4)^{1/2} - x \cdot 3(\frac{1}{2})(x^2 + 4)^{-1/2}(2x)}{9(x^2 + 4)} \\ &= \frac{3(x^2 + 4) - 3x^2}{9(x^2 + 4)^{3/2}} \\ &= \frac{4}{3(x^2 + 4)^{3/2}} \end{aligned}$$

which is positive if $x = \frac{3}{2}$. Consequently T has a local minimum at $x = \frac{3}{2}$. The time T that corresponds to $x = \frac{3}{2}$ is

$$T = \frac{1}{3} \left(\frac{9}{4} + 4 \right)^{1/2} + \frac{6}{5} - \frac{3}{10} = \frac{26}{15}$$

or, equivalently, 1 hour and 44 minutes.

We have already examined the values of T at the endpoints of the domain, obtaining approximately 1 hour 52 minutes and over 2 hours, respectively. Hence the minimum value of T occurs at $x = \frac{3}{2}$ and, therefore, the boat should land between B and C , $1\frac{1}{2}$ mi from B . For a similar problem, but one in which the endpoints of the domain lead to minimum time, see Exercise 22. ■

Example 6 A wire 60 in. long is to be cut into two pieces. One of the pieces will be bent into the shape of a circle and the other into the shape of an equilateral triangle. Where should the wire be cut so that the sum of the areas of the circle and triangle is (a) a maximum; (b) minimum?

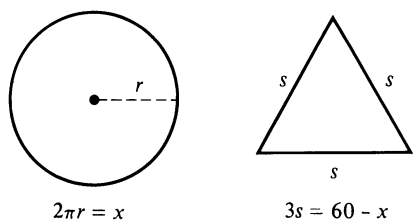


FIGURE 4.43

Solution If the wire is cut and x denotes the length of one of the pieces of wire, then the length of the other piece is $60 - x$. Let the piece of length x be bent to form a circle of radius r so that $2\pi r = x$. If the remaining piece is bent into an equilateral triangle of side s , then $3s = 60 - x$ (see Figure 4.43).

The area of the circle is

$$\pi r^2 = \pi \left(\frac{x}{2\pi} \right)^2 = \left(\frac{1}{4\pi} \right) x^2$$

and the area of the triangle is

$$\frac{\sqrt{3}}{4} s^2 = \frac{\sqrt{3}}{4} \left(\frac{60 - x}{3} \right)^2.$$

Consequently, the sum A of the areas can be expressed in terms of one variable x as follows:

$$A = \left(\frac{1}{4\pi} \right) x^2 + \left(\frac{\sqrt{3}}{36} \right) (60 - x)^2.$$

We now find the critical numbers. Differentiating,

$$\begin{aligned} D_x A &= \left(\frac{1}{2\pi} \right) x - \left(\frac{\sqrt{3}}{18} \right) (60 - x) \\ &= \left(\frac{1}{2\pi} + \frac{\sqrt{3}}{18} \right) x - \frac{10\sqrt{3}}{3} \end{aligned}$$

and, therefore, $D_x A = 0$ if and only if

$$x = \frac{10\sqrt{3}/3}{(1/2\pi) + (\sqrt{3}/18)} \approx 22.61.$$

The second derivative

$$D_x^2 A = \frac{1}{2\pi} + \frac{\sqrt{3}}{18}$$

is always positive and hence the indicated critical number will yield a minimum value for A . This minimum value is approximated by

$$A \approx \frac{1}{4\pi} (22.61)^2 + \frac{\sqrt{3}}{36} (60 - 22.61)^2 \approx 107.94.$$

Since there are no other critical numbers, the maximum value of A must be an endpoint extremum. If $x = 0$, then all the wire is used to form a triangle and

$$A = \frac{\sqrt{3}}{36} (60)^2 \approx 173.21.$$

If $x = 60$, then all the wire is used for the circle, and

$$A = \frac{1}{4\pi} (60)^2 \approx 268.48.$$

Thus the maximum value of A occurs if the wire is not cut and the entire length of wire is bent into the shape of a circle. ■

4.6 Exercises

- 1 Find two real numbers whose difference is 40 and whose product is a minimum.
- 2 Find two positive real numbers whose sum is 40 and whose product is a maximum.
- 3 If a box with a square base and open top is to have a volume of 4 ft^3 , find the dimensions that require the least material (neglect the thickness of the material and waste in construction).
- 4 Work Exercise 3 if the box has a closed top.
- 5 A fence 8 ft tall stands on level ground and runs parallel to a tall building (see figure). If the fence is 1 ft from the building, find the shortest ladder that will extend from the ground over the fence to the wall of the building.

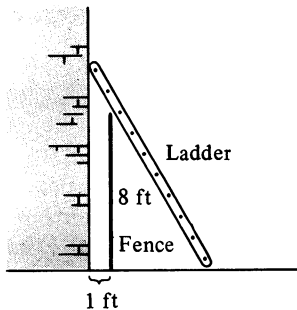


FIGURE FOR EXERCISE 5

- 6 A page of a book is to have an area of 90 in.^2 , with 1-inch margins at the bottom and sides and a $\frac{1}{2}$ -inch margin at the top. Find the dimensions of the page that will allow the largest printed area.
- 7 Find the dimensions of the rectangle of maximum area that can be inscribed in a semicircle of radius a , if two vertices lie on the diameter.
- 8 Find the dimensions of the rectangle of maximum area that can be inscribed in an equilateral triangle of side a , if two vertices of the rectangle lie on one of the sides of the triangle.
- 9 Of all possible right circular cones which can be inscribed in a sphere of radius a , find the volume of the one that has maximum volume.
- 10 Find the dimensions of the right circular cylinder of maximum volume that can be inscribed in a sphere of radius a .

- 11 A metal cylindrical container with an open top is to hold one cubic foot. If there is no waste in construction, find the dimensions that require the least amount of material.
- 12 If the circular base of the container in Exercise 11 is cut from a square sheet and the remaining metal is discarded, find the dimensions that require the least amount of material.
- 13 A long rectangular sheet of metal, 12 in. wide, is to be made into a rain gutter by turning up two sides at right angles to the sheet. How many inches should be turned up to give the gutter its greatest capacity?
- 14 Work Exercise 13 if the sides of the gutter make an angle of 120° with the base.
- 15 Prove that the rectangle of largest area having a given perimeter p is a square.
- 16 A right circular cylinder is generated by rotating a rectangle of perimeter p about one of its sides. What dimensions of the rectangle will generate the cylinder of maximum volume?
- 17 The strength of a rectangular beam varies jointly as the product of its width and the square of the depth of a cross section. Find the dimensions of the strongest beam that can be cut from a cylindrical log of radius a (see figure).

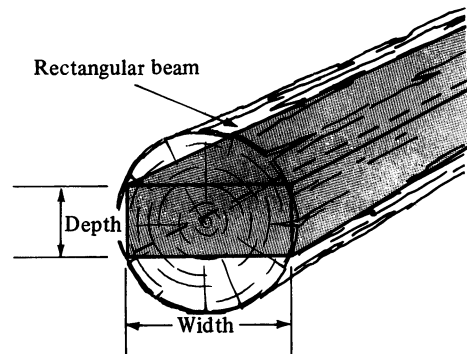


FIGURE FOR EXERCISE 17

- 18 A window has the shape of a rectangle surmounted by a semicircle. If the perimeter of the window is 15 ft, find the dimensions that will allow the maximum amount of light to enter.
- 19 Find the point on the graph of $y = x^2 + 1$ that is closest to the point $(3, 1)$.
- 20 Find the x -coordinate of the point on the graph of $y = x^3$ that is closest to the point $(4, 0)$.

- 21 A company sells running shoes to dealers at a rate of \$20 per pair if less than 50 pairs are ordered. If 50 or more pairs are ordered (up to 600), the price per pair is reduced at a rate of 2 cents times the number ordered. What size order will produce the maximum amount of money for the company?
- 22 Refer to Example 5 of this section. If the man is in a motorboat which can travel at an average rate of 15 mph, how should he proceed in order to arrive at his house in the least time?
- 23 The illumination from a light source is directly proportional to the strength of the source and inversely proportional to the square of the distance from the source. If two light sources of strengths S_1 and S_2 are d units apart, at what point on the line segment joining the two sources is the illumination minimal?
- 24 At 1:00 P.M. ship A is 30 miles due south of ship B and is sailing north at a rate of 15 mph. If ship B is sailing west at a rate of 10 mph, find the time at which the distance between the ships is minimal (see figure).

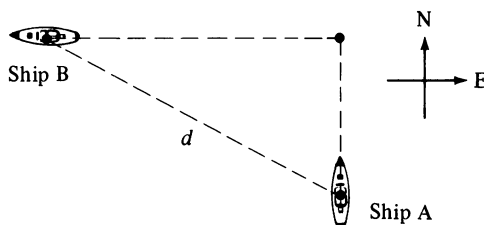


FIGURE FOR EXERCISE 24

- 25 A veterinarian has 100 ft of fencing and wishes to construct six dog kennels by first building a fence around a rectangular region, and then subdividing that region into six smaller rectangles by placing five fences parallel to one of the sides. What dimensions of the region will maximize the total area?
- 26 A paper cup having the shape of a right circular cone is to be constructed. If the volume desired is $36\pi \text{ in.}^3$, find the dimensions that require the least amount of paper (neglect any waste that may occur in the construction).
- 27 A steel storage tank for propane gas is to be constructed in the shape of a right circular cylinder with a hemisphere at each end (see figure). The cost per square foot of constructing the end pieces is twice that of the cylindrical piece. If the desired capacity of $10\pi \text{ ft}^3$, what dimensions will minimize the cost of construction?

Propane gas tank

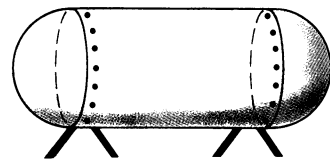


FIGURE FOR EXERCISE 27

- 28 A pipeline for transporting oil will connect two points A and B that are 3 miles apart and on opposite banks of a straight river one mile wide (see figure). Part of the pipeline will run under water from A to a point C on the opposite bank, and then above ground from C to B . If the cost per mile of running the pipeline under water is four times the cost per mile of running it above ground, find the location of C that will minimize the cost (ignore the slope of the river bed).

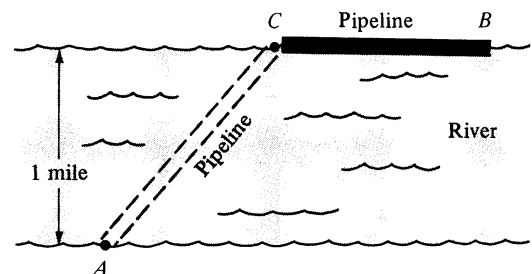


FIGURE FOR EXERCISE 28

- 29 A wire 36 cm long is to be cut into two pieces. One of the pieces will be bent into the shape of an equilateral triangle and the other into the shape of a rectangle whose length is twice its width. Where should the wire be cut if the combined area of the triangle and rectangle is (a) a minimum; (b) a maximum?
- 30 An isosceles triangle has base b and equal sides of length a . Find the dimensions of the rectangle of maximum area that can be inscribed in the triangle if one side of the rectangle lies on the base of the triangle.
- 31 Two vertical poles of lengths 6 ft and 8 ft stand on level ground, with their bases 10 ft apart. Approximate the minimal length of cable that can reach from the top of one pole to some point on the ground between the poles, and then to the top of the other pole.
- 32 Find the dimensions of the rectangle of maximum area having two vertices on the x -axis and two vertices above the x -axis, on the graph of $y = 4 - x^2$.
- 33 A window has the shape of a rectangle surmounted by an equilateral triangle. If the perimeter of the window is

- 12 ft, find the dimensions of the rectangle that will produce the largest area for the window.
- 34 If a trapezoid has three nonparallel sides of length 8, find the length of the fourth side which will maximize the area.
- 35 The owner of an apple orchard estimates that if 24 trees are planted per acre, then each mature tree will yield 600 apples per year. For each additional tree planted per acre the number of apples produced by each tree decreases by 12 per year. How many trees should be planted per acre to obtain the most apples?
- 36 A real estate company owns 180 apartments which are fully occupied when the rent is \$300 per month. The company estimates that for each \$10 increase in rent, five apartments will become unoccupied. What rent should be charged in order to obtain the largest gross income?
- 37 A package can be sent by parcel post only if the sum of its length and girth (the perimeter of the base) is not more than 96 in. Find the dimensions of the box of maximum volume that can be sent if the base of the box is a square.
- 38 A North-South highway A and an East-West highway B intersect at a point P . At 10:00 A.M. an automobile crosses P traveling north on highway A at a speed of 50 mph. At that same instant, an airplane flying east at a speed of 200 mph and an altitude of 26,400 ft is directly above the point on highway B that is 100 miles west of P . If the automobile and airplane maintain the same speed and direction, at what time will they be closest to one another?
- 39 Use the First Derivative Test to prove that the shortest distance from a point (x_1, y_1) to the line $ax + by + c = 0$ is given by

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}.$$

- 40 Prove that the shortest distance from a point (x_1, y_1) to the graph of a differentiable function f is measured along a normal line to the graph, that is, a line perpendicular to the tangent line.
- 41 A water cup in the shape of a right circular cone is to be constructed by removing a circular sector from a circular sheet of paper and then joining the two straight edges of the remaining paper. Find the central angle of the sector that will maximize the volume of the cup.

- 42 An object of weight W is pulled along a horizontal plane by a force applied to a rope that is attached to the object. If the rope makes an angle θ with the horizontal, then the magnitude of the force is given by $F = \mu W / (\mu \sin \theta + \cos \theta)$ where μ is a constant called the *coefficient of friction*. If $\mu = \sqrt{3}/3$, find the angle that will minimize F .
- 43 Two corridors 3 ft and 4 ft wide, respectively, meet at right angles. Find the length of the longest thin straight rod that will pass horizontally around the corner (see figure).

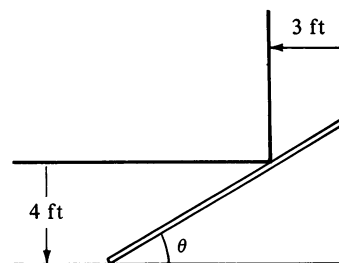


FIGURE FOR EXERCISE 43

- 44 When light travels from one point to another it takes the path that requires the least amount of time. Suppose that light has velocity v_1 in air and v_2 in water, where $v_1 > v_2$. If light travels from a point P in air to a point Q in water (see figure), show that the path requiring the least amount of time occurs if

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}.$$

(This is an example of **Snell's Law of Refraction**.)

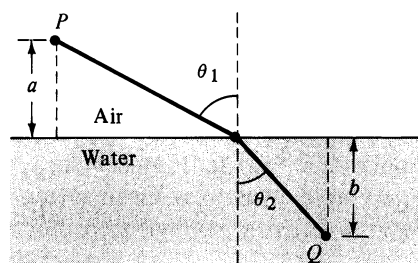


FIGURE FOR EXERCISE 44

4.7 The Derivative as a Rate of Change

Most quantities encountered in everyday life change with time. This is especially evident in scientific investigations. For example, a chemist may be interested in the rate at which a certain substance dissolves in water. An electrical engineer may wish to know the rate of change of current in part of an electrical circuit. A biologist may be concerned with the rate at which the bacteria in a culture increases or decreases. Numerous other examples could be cited, including many from fields other than the natural sciences. Let us consider the following general situation, which can be applied to all of the preceding examples.

Suppose a variable w is a function of time such that at time t , w is given by $w = g(t)$, where g is a differentiable function. The difference between the initial and final values of w in the time interval $[t, t + h]$ is $g(t + h) - g(t)$. As in our development of the velocity concept, we formulate the following definition.

Definition (4.27)

Let $w = g(t)$, where g is a differentiable function and t represents time.

(i) The **average rate of change** of $w = g(t)$ in the interval $[t, t + h]$ is

$$\frac{g(t + h) - g(t)}{h}.$$

(ii) The **rate of change** of $w = g(t)$ with respect to t is

$$\frac{dw}{dt} = g'(t) = \lim_{h \rightarrow 0} \frac{g(t + h) - g(t)}{h}.$$

The units to be used in Definition (4.27) depend on the nature of the quantity represented by w . Sometimes dw/dt is referred to as the **instantaneous rate of change** of w with respect to t .

Example 1 A scientist finds that if a certain substance is heated, the Celsius temperature after t minutes, where $0 \leq t \leq 5$, is given by the formula $g(t) = 30t + 6\sqrt{t} + 8$.

- Find the average rate of change of $g(t)$ during the time interval $[4, 4.41]$.
- Find the rate of change of $g(t)$ at $t = 4$.

Solution

- Letting $t = 4$ and $h = 0.41$ in (i) of Definition (4.27), the average rate of change of g in $[4, 4.41]$ is

$$\begin{aligned} \frac{g(4.41) - g(4)}{0.41} &= \frac{[30(4.41) + 6\sqrt{4.41} + 8] - [120 + 6\sqrt{4} + 8]}{0.41} \\ &= \frac{12.9}{0.41} \approx 31.46^\circ\text{C/min.} \end{aligned}$$

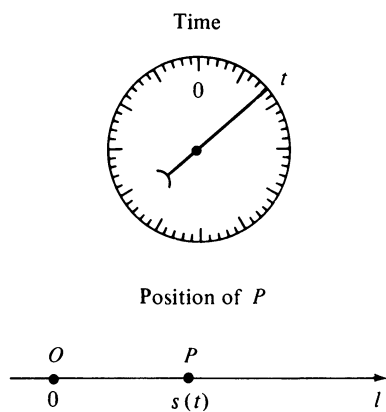


FIGURE 4.44

(b) From (ii) of Definition (4.27) the rate of change of g at time t is given by $g'(t) = 30 + 3/\sqrt{t}$. In particular,

$$g'(4) = 30 + \frac{3}{\sqrt{4}} = 31.5^\circ\text{C/min.}$$

The velocity of a point P that moves on a coordinate line l was defined in Chapter 3. In the future we shall often use the letter s to denote functions whose values are distances. If the coordinate of P on l is $s(t)$, as illustrated in Figure 4.44, then s is called the **position function** of P . By Definition (3.3), the velocity $v(t)$ of P at time t is $s'(t)$. Note that in terms of Definition (4.27), the velocity is the rate of change of $s(t)$ with respect to time. The next definition summarizes this discussion and introduces some new terminology.

Definition (4.28)

Let the position of a point P on a coordinate line l at time t be given by $s(t)$, where s is a differentiable function.

- (i) The **velocity** $v(t)$ of P is $v(t) = s'(t)$.
- (ii) The **speed** of P at time t is $|v(t)|$.
- (iii) The **acceleration** $a(t)$ of $P(t)$ is $a(t) = v'(t) = s''(t)$.

We shall call v the **velocity function** of P and a the **acceleration function** of P . Using another notation,

$$v = \frac{ds}{dt} \quad \text{and} \quad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

If t is measured in seconds and $s(t)$ in centimeters, then the unit for $v(t)$ is cm/sec (read “centimeters per second”). Similarly, if t is measured in hours and $s(t)$ in miles, then $v(t)$ is in mi/hr (miles per hour), etc. By Definition (4.27), the acceleration $a(t)$ is the rate of change of velocity of P at time t . The units for $a(t)$ are cm/sec² (centimeters per second per second), mi/hr² (miles per hour per hour), etc.

The results of Section 4.3 may be used to determine the direction of motion of P . If the velocity $v(t)$ is positive in a time interval I , then $s'(t) > 0$ and hence, by Theorem (4.13), $s(t)$ is increasing; that is, P is moving in the positive direction on l . Similarly, if $v(t)$ is negative, then P is moving in the negative direction. The speed $|v(t)|$ indicates the rate at which P is moving without specifying the direction of motion.

Since $v(t) = s'(t)$, the times at which the velocity is zero are critical numbers for the position function s and hence lead to possible local maxima or minima for s . If a local maximum or minimum for s occurs at t_1 , then t_1 is usually a time at which P reverses direction.

Applying the theory of increasing and decreasing functions developed in Section 4.3, it follows that if $a(t) > 0$, then the velocity is increasing at time t . If $a(t) < 0$, the velocity is decreasing. Note that if both $a(t) < 0$ and $v(t) < 0$,

then the velocity is decreasing in the sense of becoming more negative. In this event, even though the velocity is decreasing, the speed $|v(t)|$ is increasing. An illustration of this occurs in Example 2.

Since the acceleration function a is the second derivative of the position function s , it can sometimes be used in conjunction with the Second Derivative Test to find the local extrema of s . For example, if $v(t_1) = 0$ and $a(t_1) < 0$, then $s(t_1)$ is a local maximum for s and hence P changes direction from right to left at time t_1 . Similarly, if $v(t_1) = 0$ and $a(t_1) > 0$, then $s(t_1)$ is a local minimum for s and P changes direction from left to right at time t_1 .

Example 2 The position function s of a point P on a coordinate line is given by

$$s(t) = t^3 - 12t^2 + 36t - 20,$$

where t is measured in seconds and $s(t)$ in centimeters. Describe the motion of P during the time interval $[-1, 9]$.

Solution Differentiating, we obtain

$$v(t) = s'(t) = 3t^2 - 24t + 36 = 3(t - 2)(t - 6),$$

$$a(t) = v'(t) = 6t - 24 = 6(t - 4).$$

Consequently, the velocity is 0 at $t = 2$ and $t = 6$. These critical numbers for s lead us to examine the time intervals $(-1, 2)$, $(2, 6)$ and $(6, 9)$. The following table describes the direction of motion of P . The signs of $v(t)$ may be determined by using test values.

Time Interval	Sign of $v(t)$	Direction of Motion
$(-1, 2)$	+	right
$(2, 6)$	-	left
$(6, 9)$	+	right

The table shows that the function s has a local maximum at time $t = 2$ and a local minimum at $t = 6$. This can also be verified by noting that $a(2) = -12 < 0$ and $a(6) = 12 > 0$. The next table displays the values of the position, velocity, and acceleration functions at important times, namely the endpoints of the time interval $[-1, 9]$, and the times at which the velocity or acceleration is zero.

t	-1	2	4	6	9
$s(t)$	-69	12	-4	-20	61
$v(t)$	63	0	-12	0	63
$a(t)$	-30	-12	0	12	30

It is convenient to represent the motion of P schematically as in Figure 4.45. The curve above the coordinate line is not the path of the point, but is

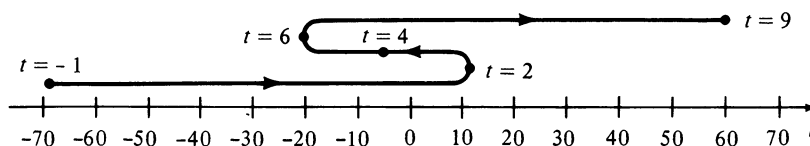


FIGURE 4.45

intended to show the manner in which P moves on l . As indicated by the table and Figure 4.45, at $t = -1$ the point is 69 cm to the left of the origin and is moving to the right with a velocity of 63 cm/sec. The negative acceleration -30 cm/sec^2 indicates that the velocity is decreasing at a rate of 30 cm per second each second. The point continues to move to the right and to slow down until it has zero velocity at $t = 2$, 12 cm to the right of the origin. The point P then reverses direction and moves until, at $t = 6$, it is 20 cm to the left of the origin. It then again reverses direction and moves to the right for the remainder of the time interval, with increasing velocity. The direction of motion is indicated by the arrows on the curve in Figure 4.45.

The time $t = 4$, at which the acceleration is zero, is a critical number for the velocity function. (Why?) The significance of this time is clearer if we consider the speed $|v(t)| = |3(t - 2)(t - 6)|$. In the time interval $[2, 6]$, the speed increases from 0 at $t = 2$ to a local maximum $|-12| = 12$ at $t = 4$ and then decreases to 0 at $t = 6$. ■

Example 3 A projectile is fired straight upward with a velocity of 400 ft/sec. Its distance above the ground t seconds after being fired is given by $s(t) = -16t^2 + 400t$. Find the time and the velocity at which the projectile hits the ground. What is the maximum altitude achieved by the projectile? What is the acceleration at any time t ?

Solution The path of the projectile is on a vertical coordinate line with origin at ground level and positive direction upward. The projectile will be on the ground when $-16t^2 + 400t = 0$, that is, when $-16t(t - 25) = 0$. Hence the projectile hits the ground after 25 sec. The velocity at time t is given by

$$v(t) = s'(t) = -32t + 400.$$

In particular, at $t = 25$,

$$v(25) = -32(25) + 400 = -400 \text{ ft/sec.}$$

The maximum altitude occurs when $s'(t) = 0$, that is, when $-32t + 400 = 0$. Solving for t gives us $t = \frac{400}{32} = \frac{25}{2}$ and hence its maximum altitude is

$$s\left(\frac{25}{2}\right) = -16\left(\frac{25}{2}\right)^2 + 400\left(\frac{25}{2}\right) = 2500 \text{ feet.}$$

Finally, the acceleration at any time t is $a(t) = v'(t) = -32 \text{ ft/sec}^2$. ■

One of the most important types of rectilinear motion is described in the following definition.

Definition (4.29)

A point P that moves on a coordinate line l such that its distance $s(t)$ from the origin at time t is given by either

$$s(t) = a \cos(\omega t + b) \quad \text{or} \quad s(t) = a \sin(\omega t + b)$$

where a, b , and $\omega > 0$ are real numbers, is said to be in **simple harmonic motion**.

Another way of defining simple harmonic motion is to demand that the point P moves on l such that the acceleration $s''(t)$ satisfies the condition

$$s''(t) = -\omega^2 s(t)$$

for all t . It can be shown that this condition is equivalent to Definition (4.29).

In simple harmonic motion, as defined in (4.29), the point P oscillates between the points on l with coordinates $-a$ and a . The **amplitude** of the motion is the maximum displacement $|a|$ of the point from the origin. The **period** is the time $2\pi/\omega$ required for one complete oscillation. The **frequency** $\omega/2\pi$ is the number of oscillations per unit of time.

Simple harmonic motion takes place in many different types of wave motion, such as water waves, sound waves, radio waves, light waves, and distortional waves that are present in vibrating bodies. Another example of simple harmonic motion can be obtained by considering a spring with an attached weight, which is oscillating vertically relative to a coordinate line, as illustrated in Figure 4.46. The number $s(t)$ represents the coordinate of a fixed point P in the weight, and we assume that the amplitude a of the motion is constant. In this case, there is no frictional force retarding the motion. If friction is present, then the amplitude decreases with time, and the motion is said to be *damped*.

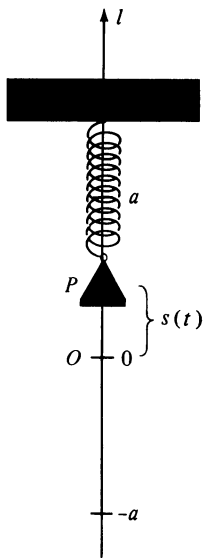


FIGURE 4.46

Example 4 Suppose the weight shown in Figure 4.46 is oscillating according to the law

$$s(t) = 10 \cos\left(\frac{\pi}{6} t\right)$$

where t is measured in seconds and $s(t)$ in centimeters. Discuss the motion of the weight.

Solution By Definition (4.29), the motion is simple harmonic. The amplitude a is 10 cm. Since $\omega = \pi/6$, the period is $2\pi/\omega = 2\pi/(\pi/6) = 12$. Thus, it requires 12 seconds for one complete oscillation. The frequency is $\omega/2\pi = (\pi/6)/2\pi = \frac{1}{12}$, that is, $\frac{1}{12}$ of an oscillation takes place each second.

Let us examine the motion during the time interval $[0, 12]$. The velocity and acceleration functions are given by

$$v(t) = s'(t) = -10 \sin\left(\frac{\pi}{6}t\right) \cdot \left(\frac{\pi}{6}\right) = -\frac{5\pi}{3} \sin\left(\frac{\pi}{6}t\right),$$

$$a(t) = v'(t) = -\frac{5\pi}{3} \cos\left(\frac{\pi}{6}t\right) \cdot \left(\frac{\pi}{6}\right) = -\frac{5\pi^2}{18} \cos\left(\frac{\pi}{6}t\right).$$

The velocity is 0 at $t = 0$, $t = 6$ and $t = 12$, since $\sin((\pi/6)t) = 0$ for these values of t . The acceleration is 0 at $t = 3$ and $t = 9$, since in these cases $\cos((\pi/6)t) = 0$. The times at which the velocity and acceleration are zero lead us to examine the time intervals $(0, 3)$, $(3, 6)$, $(6, 9)$, and $(9, 12)$. The following table displays the main characteristics of the motion.

Time Interval	Sign of $v(t)$	Direction of Motion	Sign of $a(t)$	Variation of $v(t)$	Speed $ v(t) $
$(0, 3)$	—	Downward	—	Decreasing	Increasing
$(3, 6)$	—	Downward	+	Increasing	Decreasing
$(6, 9)$	+	Upward	+	Increasing	Increasing
$(9, 12)$	+	Upward	—	Decreasing	Decreasing

The reader should verify all entries in the table. Note that if $0 < t < 3$, the velocity $v(t)$ is negative and decreasing. This implies that the speed $|v(t)|$ is *increasing*. If $3 < t < 6$, the velocity is negative and increasing. In this case the speed of P is *decreasing* in the time interval $(3, 6)$. Similar remarks can be made for the intervals $(6, 9)$ and $(9, 12)$.

We may summarize the motion of P as follows. At $t = 0$, $s(0) = 10$ and the point P is 10 cm above the origin O . It then moves downward, gaining speed until it reaches O at $t = 3$. It then slows down until it reaches a point 10 cm below O at the end of 6 sec. The direction of motion is then reversed, and the weight moves upward, gaining speed until it reaches O at $t = 9$, after which it slows down until it returns to its original position at the end of 12 sec. The direction of motion is then reversed again, and the same pattern is repeated indefinitely. ■

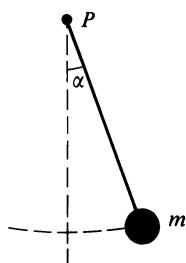


FIGURE 4.47

The definition of simple harmonic motion is usually extended to include situations where $s(t)$ is *any* mathematical or physical quantity (not necessarily a distance). In the next illustration $s(t)$ is itself an angle.

Historically, the first type of harmonic motion to be scientifically investigated involved pendulums. Figure 4.47 illustrates a *simple pendulum* consisting of a bob of mass m attached to one end of a string, with the other end of the string attached to a fixed point P .

In the ideal case it is assumed that the string is weightless, there is no air resistance, and the only force acting on the bob is gravity. If the bob is displaced sideways and released, the pendulum oscillates back and forth in a vertical plane. Let α denote the *angular displacement* at time t (see Figure

4.47). If the bob moves through a small arc (say $|\alpha| < 5^\circ$), then it can be shown, by using physical laws, that

$$\alpha = \beta \cos(\omega t + \alpha_0)$$

where α_0 is the initial displacement, ω is the frequency of oscillation of the angle α , and β is the (angular) amplitude of oscillation. This means that the angle α is in simple harmonic motion and hence we refer to the motion as *angular simple harmonic motion*.

As a final illustration, in electrical circuits an alternating electromotive force (emf) and the current may vary harmonically. For example, the emf e (measured in volts) at time t may be given by

$$e = E \sin \omega t$$

where E is the maximum value of e . If an emf of this type is impressed on a circuit containing only a resistance R , then by Ohm's Law, the current i at time t is

$$i = \frac{e}{R} = \frac{E}{R} \sin \omega t = I \sin \omega t$$

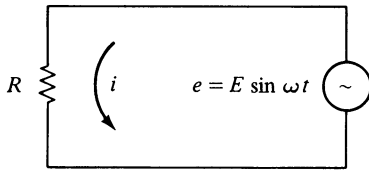


FIGURE 4.48

where $I = E/R$. A schematic drawing of an electric circuit of this type is illustrated in Figure 4.48.

In this case, the maximum value I of i occurs at the same time as the maximum value E of e . In other situations, these maximum values may occur at different times, in which case we say there is a **phase difference** between e and i . For example, if $e = E \sin \omega t$, we could have either

$$(a) \ i = I \sin(\omega t - \varphi) \quad \text{or} \quad (b) \ i = I \sin(\omega t + \varphi)$$

where $\varphi > 0$. In case (a), the current is said to **lag** the emf by an amount φ/ω , and the graph of i can be obtained by shifting the graph of $i = I \sin \omega t$ an amount φ/ω to the *right*. In case (b), we shift the graph to the *left*, and the current is said to **lead** the emf by an amount φ/ω .

We may investigate rates of change with respect to variables other than time, as indicated by the next definition.

Definition (4.30)

Let $y = f(x)$, where x is any variable.

- (i) The **average rate of change of y with respect to x in the interval $[x, x + h]$** is the quotient

$$\frac{f(x + h) - f(x)}{h}.$$

- (ii) The **rate of change of y with respect to x** is the limit of the average rate of change as $h \rightarrow 0$, that is, dy/dx .

By (ii) of (4.30), if the variable x changes, then y changes at a rate of dy/dx units per unit change in x . To illustrate, suppose a quantity of gas is

enclosed in a spherical balloon. If the gas is heated or cooled and if the pressure remains constant, then the balloon expands or contracts and its volume V is a function of the temperature T . The derivative dV/dT gives us the rate of change of volume with respect to temperature.

Example 5 The current I (in amperes) in a certain electrical circuit is given by $I = 100/R$, where R denotes resistance (in ohms). Find the rate of change of I with respect to R when the resistance is 20 ohms.

Solution Since $dI/dR = -100/R^2$, we see that if $R = 20$ then $dI/dR = -\frac{100}{400} = -\frac{1}{4}$. Thus, if R is increasing, then when $R = 20$, the current I is decreasing at a rate of $\frac{1}{4}$ amperes per ohm. ■

4.7 Exercises

- 1 As a spherical balloon is being inflated, its radius (in centimeters) after t minutes is given by $r(t) = 3\sqrt[3]{t+8}$, where $0 \leq t \leq 10$. What is the rate of change with respect to t of each of the following at $t = 8$?

- (a) $r(t)$
- (b) The volume of the balloon
- (c) The surface area

- 2 The volume of a spherical hot air balloon (in ft^3) t hours after 1:00 P.M. is given by $V(t) = \frac{4}{3}\pi(9-2t)^3$, where $0 \leq t \leq 4$. What is the rate of change with respect to t of each of the following at 4:00 P.M.?

- (a) $V(t)$
- (b) The radius of the balloon
- (c) The surface area

- 3 Suppose that t seconds after he starts to run, an individual's pulse rate (in beats/min) is given by $P(t) = 56 + 2t^2 - t$, where $0 \leq t \leq 7$. Find the rate of change of $P(t)$ with respect to t at (a) $t = 2$, (b) $t = 4$, and (c) $t = 6$.

- 4 The temperature T (degrees Celsius) of a solution at time t (minutes) is given by $T(t) = 10 + 4t + 3/(t+1)$, where $1 \leq t \leq 10$. Find the rate of change of $T(t)$ with respect to t at (a) $t = 2$, (b) $t = 5$, and (c) $t = 9$.

- 5 A stone is dropped into a pond, causing water waves that form concentric circles. If, after t seconds, the radius of one of the waves is $40t$ cm, find the rate of change with respect to t of the area of the circle caused by the wave at (a) $t = 1$, (b) $t = 2$, and (c) $t = 3$.

- 6 Boyle's Law for gases states that $pv = c$, where p denotes the pressure, v the volume, and c is a constant. Suppose that at time t (minutes) the pressure is $20 + 2t$ gm/cm²,

where $0 \leq t \leq 10$. If the volume is 60 cm^3 at $t = 0$, find the rate at which the volume is changing with respect to t at $t = 5$.

Position functions of points moving rectilinearly are defined in Exercises 7–16. Find the velocity and acceleration at time t and describe the motion of the point during the indicated time interval. Illustrate the motion by means of a diagram of the type shown in Figure 4.45.

- 7 $s(t) = 3t^2 - 12t + 1$, $[0, 5]$
- 8 $s(t) = t^2 + 3t - 6$, $[-2, 2]$
- 9 $s(t) = t^3 - 9t + 1$, $[-3, 3]$
- 10 $s(t) = 24 + 6t - t^3$, $[-2, 3]$
- 11 $s(t) = t + (4/t)$, $[1, 4]$
- 12 $s(t) = 2\sqrt{t} + (1/\sqrt{t})$, $[1, 4]$
- 13 $s(t) = 2t^4 - 6t^2$, $[-2, 2]$
- 14 $s(t) = 2t^3 - 6t^5$, $[-1, 1]$
- 15 $s(t) = t^3 + 1$, $[0, 4]$
- 16 $s(t) = \sqrt[3]{t}$, $[-8, 0]$

- 17 A projectile is fired directly upward with a velocity of 144 ft/sec. Its height $s(t)$ in feet above the ground after t sec is given by $s(t) = 144t - 16t^2$. What are the velocity and acceleration after t sec? After 3 sec? What is the maximum height? When does it strike the ground?

- 18 A ball rolls down an inclined plane such that the distance $s(t)$ (in inches) it rolls in t seconds is given by $s(t) = 5t^2 + 2$. What is the velocity after 1 sec? 2 sec? When will the velocity be 28 in./sec?

- 19 The position function s of a point moving rectilinearly is given by $s(t) = 2t^3 - 15t^2 + 48t - 10$ where t is in seconds and $s(t)$ is in meters. Find the acceleration when the velocity is 12 m/sec. Find the velocity when the acceleration is 10 m/sec².

- 20 Work Exercise 19 if $s(t) = t^5 - (5/3)t^3 - 48t$.

In Exercises 21 and 22, s is the position function of a particle that is moving rectilinearly. Find (a) the velocity and acceleration at the indicated time t , and (b) the subintervals of $[0, 2\pi]$ in which the particle moves in the positive direction.

21 $s(t) = \sqrt{3}t + \cos 2t, t = \pi/4$

22 $s(t) = 2 \sin^2 t - \sin 2t, t = \pi/6$

In Exercises 23–26, s is the position function of a particle moving in simple harmonic motion, and t is the time in seconds. Find the amplitude, period, and frequency, and describe the motion of the particle during one complete oscillation.

23 $s(t) = 5 \cos (\pi/4)t$

24 $s(t) = 4 \sin \pi t$

25 $s(t) = 6 \sin (2\pi/3)t$

26 $s(t) = 3 \cos 2t$

- 27 The illumination from a light source is directly proportional to the strength of the source and inversely proportional to the square of the distance s from the source. At a distance of 2 ft from a glowing fire, a photographer's light meter registers 120 units. If the photographer backs away from the fire, find the rate of change of the meter reading with respect to s when the photographer is 20 ft from the fire.

- 28 Show that the rate of change of the volume of a sphere with respect to its radius is numerically equal to the surface area of the sphere.

- 29 Show that the rate of change of the radius of a circle with respect to its circumference is independent of the size of the circle. Illustrate this fact by using great circles on two spheres—one the size of a basketball and the other the size of the Earth.

- 30 The formula for the adiabatic expansion of air is $pv^{1.4} = c$, where p is the pressure, v the volume, and c is a constant.

Find a formula for the rate of change of pressure with respect to volume.

- 31 The relationship between the temperature F on the Fahrenheit scale and the temperature C on the Celsius scale is given by $C = \frac{5}{9}(F - 32)$. What is the rate of change of F with respect to C ?

- 32 If a sum of money P_0 is invested at an interest rate of 100% per cent per year, compounded monthly, then the principal P at the end of one year is $P = P_0(1 + r/12)^{12}$. If r is regarded as a real number, find the rate of change of P with respect to r when $P_0 = \$1,000$ and $r = 0.12$.

- 33 The period T of a simple pendulum, that is, the time required for one complete oscillation, varies directly as the square root of its length l . What can be said about the rate of change of T with respect to l ?

- 34 The electrical resistance R of a copper wire of fixed length is inversely proportional to the square of its diameter d . What is the rate of change of R with respect to d ?

- 35 An open box is to be constructed from a rectangular piece of cardboard 40 cm wide and 60 cm long by cutting out a square of side s cm from each corner and then bending up the cardboard. Express the volume V of the box as a function of s and find the rate of change of V with respect to s .

- 36 The formula $1/f = (1/p) + (1/q)$ is used in optics, where f is the focal length of a convex lens, and p and q are the distances from the lens to the object and image, respectively. If f is fixed, find a general formula for the rate of change of q with respect to p .

- 37 A point $P(x, y)$ is moving at a constant rate around a circle that has the equation $x^2 + y^2 = a^2$. Prove that the projection $Q(x, 0)$ of P on the x -axis is in simple harmonic motion.

- 38 If a point P moves on a coordinate line such that

$$s(t) = a \cos \omega t + b \sin \omega t,$$

show that P is in simple harmonic motion by

(a) using the remark following Definition (4.29).

(b) using only trigonometric methods. (*Hint*: Show that $s(t) = a \cos (\omega t - c)$ for some c .)

4.8 Related Rates

In applications it is not unusual to encounter two variables x and y that are differentiable functions of time t , say $x = f(t)$ and $y = g(t)$. In addition, x and y may be related by means of some equation such as

$$x^2 - y^3 - 2x + 7y^2 - 2 = 0.$$

Differentiating with respect to t and using the Chain Rule produces an equation involving the rates of change dx/dt and dy/dt . As an illustration, the previous equation leads to

$$2x \frac{dx}{dt} - 3y^2 \frac{dy}{dt} - 2 \frac{dx}{dt} + 14y \frac{dy}{dt} = 0.$$

The derivatives dx/dt and dy/dt in this equation are called **related rates**, since they are related by means of the equation. The last equation can be used to find one of the rates when the other is known. This observation has many practical applications. The following examples give several illustrations.

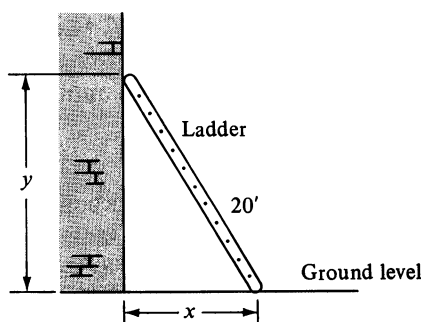


FIGURE 4.49

Example 1 A ladder 20 ft long leans against a vertical building. If the bottom of the ladder slides away from the building horizontally at a rate of 2 ft/sec, how fast is the ladder sliding down the building when the top of the ladder is 12 ft above the ground?

Solution We begin by representing a general position of the ladder schematically, as in Figure 4.49, introducing a variable x to denote the distance from the base of the building to the bottom of the ladder, and a variable y to denote the distance from the ground to the top of the ladder.

Since x is changing at a rate of 2 ft/sec, we may write

$$\frac{dx}{dt} = 2 \text{ ft/sec.}$$

Our objective is to find dy/dt , the rate at which the top of the ladder is sliding down the building, at the instant that $y = 12$ ft.

The relationship between x and y may be determined by applying the Pythagorean Theorem to the triangle in the figure. This gives us

$$x^2 + y^2 = 400.$$

Differentiating with respect to t leads to

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

from which it follows that

$$\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}.$$

The last equation is a *general* formula relating the two rates of change under consideration. Let us now consider the special case $y = 12$. The corresponding value of x may be determined from

$$x^2 + 144 = 400, \quad \text{or} \quad x^2 = 400 - 144 = 256.$$

Thus $x = \sqrt{256} = 16$ when $y = 12$. Substituting in the general formula for dy/dt we obtain

$$\frac{dy}{dt} = -\frac{16}{12}(2) = -\frac{8}{3} \text{ ft/sec.} \quad \blacksquare$$

Some of the Guidelines listed in Section 4.6 for applications of extrema are useful when attempting to solve a related rate problem. In particular, the first three Guidelines—(1) read the problem carefully, (2) sketch and label a suitable diagram, and (3) write down known facts and relationships—are strongly recommended. After following these steps, it is essential to formulate a *general* equation that relates the variables involved in the problem, as we did in Example 1. *A common error is to introduce specific values for the rates and variable quantities too early in the solution.* Always remember to obtain a *general* formula that involves the rates of change at *any* time t . Specific values should not be substituted for variables until the final steps of the solution.

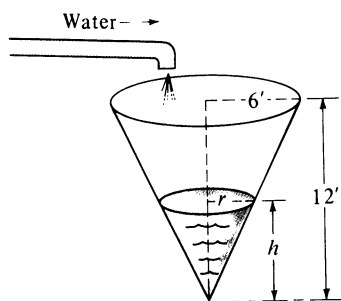


FIGURE 4.50

Example 2 A water tank has the shape of an inverted right circular cone of altitude 12 ft and base radius 6 ft. If water is being pumped into the tank at a rate of 10 gal/min, approximate the rate at which the water level is rising when it is 3 ft deep (1 gal ≈ 0.1337 ft³).

Solution We begin by sketching the tank as in Figure 4.50, letting r denote the radius of the surface of the water when the depth is h . Note that r and h are functions of time t . The volume V of water in the tank corresponding to depth h is given by

$$V = \frac{1}{3}\pi r^2 h.$$

Let us express V in terms of one variable. Referring to similar triangles in Figure 4.50 we see that r and h are related by the equations

$$\frac{r}{h} = \frac{6}{12}, \quad \text{or} \quad r = \frac{h}{2}.$$

Consequently, at depth h ,

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{1}{12}\pi h^3.$$

Differentiating with respect to t gives us the following general relationship between the rates of change of V and h at any time:

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt}.$$

Another method of solution is to write $x = 16t$, $y = 20t$, and

$$z = [x^2 + (25 - y)^2]^{1/2} = [256t^2 + (25 - 20t)^2]^{1/2}.$$

The derivative dz/dt may then be found, and substitution of $\frac{1}{2}$ for t produces the desired rate of change. ■

4.8 Exercises

- 1 A ladder 20 ft long leans against a vertical building. If the bottom of the ladder slides away from the building horizontally at a rate of 3 ft/sec, how fast is the ladder sliding down the building when the top of the ladder is 8 ft from the ground?
- 2 As a circular metal griddle is being heated, its diameter changes at a rate of 0.01 cm/min. When the diameter is 30 cm at what rate is the area of one side changing?
- 3 Gas is being pumped into a spherical balloon at a rate of $5 \text{ ft}^3/\text{min}$. If the pressure is constant, find the rate at which the radius is changing when the diameter is 18 in.
- 4 A girl starts at a point A and runs east at a rate of 10 ft/sec. One minute later, another girl starts at A and runs north at a rate of 8 ft/sec. At what rate is the distance between them changing 1 minute after the second girl starts?
- 5 A light is at the top of a 16-ft pole. A boy 5 ft tall walks away from the pole at a rate of 4 ft/sec (see figure). At what rate is the tip of his shadow moving when he is 18 ft from the pole? At what rate is the length of his shadow increasing?

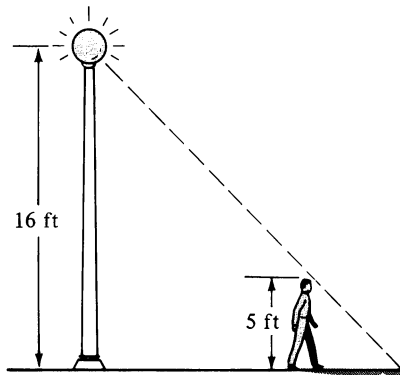


FIGURE FOR EXERCISE 5

- 6 A man on a dock is pulling in a boat by means of a rope attached to the bow of the boat 1 ft above water level and passing through a simple pulley located on the dock 8 ft above water level (see figure). If he pulls in the rope at a rate of 2 ft/sec, how fast is the boat approaching the dock

when the bow of the boat is 25 ft from a point that is 7 ft directly below the pulley?

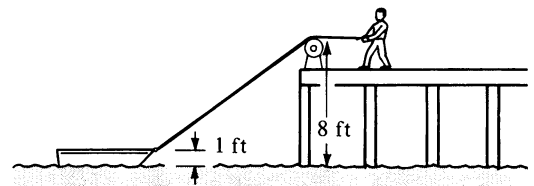


FIGURE FOR EXERCISE 6

- 7 The top of a silo has the shape of a hemisphere of diameter 20 ft. If it is coated uniformly with a layer of ice, and if the thickness is decreasing at a rate of $\frac{1}{4}$ in./hr, how fast is the volume of ice changing when the ice is 2 in. thick?
- 8 As sand leaks out of a hole in a container, it forms a conical pile whose altitude is always the same as its radius. If the height of the pile is increasing at a rate of 6 in./min, find the rate at which the sand is leaking out when the altitude is 10 in.
- 9 A boy flying a kite holds the string 5 ft above ground level and pays out string at a rate of 2 ft/sec as the kite moves horizontally at an altitude of 105 ft. (see figure). Assuming there is no sag in the string, find the rate at which the kite is moving when 125 ft of string has been paid out.

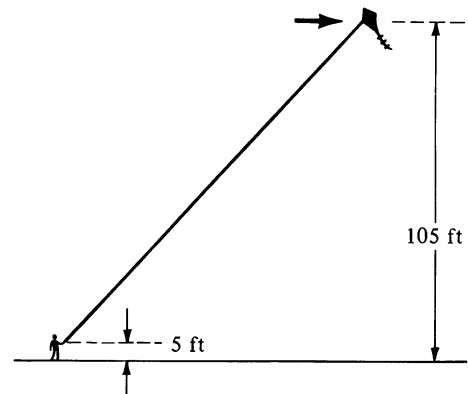


FIGURE FOR EXERCISE 9

- 10 A weather balloon is rising vertically at a rate of 2 ft/sec. An observer is situated 100 yd from a point on the ground directly below the balloon. At what rate is the distance between the balloon and the observer changing when the altitude of the balloon is 500 ft?
- 11 Boyle's Law for gases states that $p v = c$, where p denotes the pressure, v the volume, and c is a constant. At a certain instant the volume is 75 in.³, the pressure is 30 lb/in.², and the pressure is decreasing at a rate of 2 lb/in.² every minute. At what rate is the volume changing at this instant?
- 12 Suppose a spherical snowball is melting and the radius is decreasing at a constant rate, changing from 12 in. to 8 in. in 45 minutes. How fast was the volume changing when the radius was 10 in.?
- 13 The ends of a water trough 8 ft long are equilateral triangles whose sides are 2 ft long (see figure). If water is being pumped into the trough at a rate of 5 ft³/min, find the rate at which the water level is rising when the depth is 8 in.

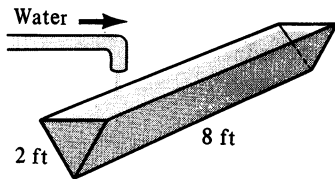


FIGURE FOR EXERCISE 13

- 14 Work Exercise 13 if the ends of the trough have the shape of the graph of $y = 2|x|$ between the points $(-1, 2)$ and $(1, 2)$.
- 15 A point $P(x, y)$ moves on the graph of the equation $y = x^3 + x^2 + 1$, the x -coordinate changing at a rate of 2 units/sec. How fast is the y -coordinate changing at the point $(1, 3)$?
- 16 A point $P(x, y)$ moves on the graph of $y^2 = x^2 - 9$ such that $dx/dt = 1/x$. Find dy/dt at the point $(5, 4)$.
- 17 The area of an equilateral triangle is decreasing at a rate of 4 cm²/min. Find the rate at which the length of a side is changing when the area of the triangle is 200 cm².
- 18 Gas is escaping from a spherical balloon at a rate of 10 ft³/hr. At what rate is the radius changing when the volume is 400 ft³?
- 19 A stone is dropped into a lake, causing circular waves whose radii increase at a constant rate of 0.5 m/sec. At what rate is the circumference of a wave changing when its radius is 4 m?

- 20 A softball diamond has the shape of a square with sides 60 ft long. If a player is running from second base to third at a speed of 24 ft/sec, at what rate is her distance from home plate changing when she is 20 ft from third?
- 21 When two electrical resistances R_1 and R_2 are connected in parallel (see figure), the total resistance R is given by $1/R = (1/R_1) + (1/R_2)$. If R_1 and R_2 are increasing at rates of 0.01 ohm/sec and 0.02 ohm/sec, respectively, at what rate is R changing at the instant that $R_1 = 30$ ohms and $R_2 = 90$ ohms?

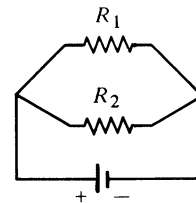


FIGURE FOR EXERCISE 21

- 22 The formula for the adiabatic expansion of air is $p v^{1.4} = c$, where p denotes pressure, v denotes volume, and c is a constant. At a certain instant the pressure is 40 dynes/cm² and is increasing at a rate of 3 dynes/cm² per second. If, at that same instant, the volume is 60 cm³, find the rate at which the volume is changing.
- 23 If a spherical tank of radius a contains water which has a maximum depth h , then the volume V of water in the tank is given by $V = \frac{1}{3}\pi h^2(3a - h)$. Suppose a spherical tank of radius 16 ft is being filled at a rate of 100 gal/min. Approximate the rate at which the water level is rising when $h = 4$ ft (1 gal ≈ 0.1337 ft³).
- 24 A spherical water storage tank for a small community is coated uniformly with a 2-inch layer of ice. If the volume of ice is melting at a rate which is directly proportional to its surface area, show that the outside diameter is decreasing at a constant rate.
- 25 From the edge of a cliff which overlooks a lake 200 ft below, a boy drops a stone and then, two seconds later, drops another stone from exactly the same position. Discuss the rate at which the distance between the two stones is changing during the next second. (Assume that the distance an object falls in t seconds is $16t^2$ ft.)
- 26 A metal rod has the shape of a right circular cylinder. As it is being heated, its length is increasing at a rate of 0.005 cm/min and its diameter is increasing at 0.002 cm/min. At what rate is the volume changing when the rod has length 40 cm and diameter 3 cm?

- 27 An airplane, flying at a constant speed of 360 mi/hr and climbing at an angle of 45° , passes over an air traffic control tower on the ground at an altitude of 10,560 ft. Find the rate at which its distance from the tower is changing one minute later (neglect the height of the tower).
- 28 Refer to Exercise 38 of Section 4.6. At what rate is the distance between the airplane and the automobile changing at 10:15 A.M.?
- 29 A paper cup containing water has the shape of a frustum of a right circular cone of altitude 6 in. and lower and upper base radii 1 in. and 2 in., respectively. If water is leaking out of the cup at a rate of $3 \text{ in.}^3/\text{hr}$, at what rate is the water level decreasing when its depth is 4 in.? (Note: The volume V of a frustum of a right circular cone of altitude h and base radii a and b is given by $V = \frac{1}{3}\pi h(a^2 + b^2 + ab)$.)
- 30 The top part of a swimming pool is a rectangle of length 60 ft and width 30 ft. The depth of the pool varies uniformly from 4 ft to 9 ft through a horizontal distance of 40 ft and then is level for the remaining 20 ft, as illustrated by the cross-sectional view in the accompanying figure. If the pool is being filled with water at a rate of 500 gal/min, approximate the rate at which the water level is rising when the depth at the deep end is 4 ft ($1 \text{ gal} \approx 0.1337 \text{ ft}^3$).

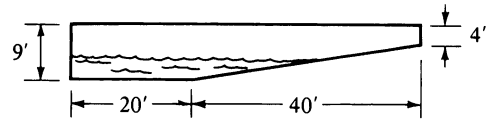


FIGURE FOR EXERCISE 30

- 31 An airplane at an altitude of 10,000 ft is flying at a constant speed on a line which will take it directly over an observer on the ground. If, at a given instant, the observer notes that the angle of elevation of the airplane is 60° and is increasing at a rate of 1° per second, find the speed of the airplane. (Hint: $\tan \theta = \sin \theta / \cos \theta$.)
- 32 In Exercise 6, let θ be the angle that the rope makes with the horizontal. Find the rate at which θ is changing at the instant that $\theta = 30^\circ$.
- 33 An isosceles triangle has equal sides 6 in. long. If the angle θ between the equal sides is changing at a rate of 2° per minute, how fast is the area of the triangle changing when $\theta = 30^\circ$?
- 34 A ladder 20 ft long leans against a vertical building. If the bottom of the ladder slides away from the building horizontally at a rate of 2 ft/sec, at what rate is the angle between the ladder and the ground changing when the top of the ladder is 12 ft above the ground?

4.9 Antiderivatives

In Chapter 3 certain problems were stated in the form “Given a function g , find the derivative g' .” We shall now consider the converse problem, namely, “Given the derivative g' , find the function g .” An equivalent way of stating this converse problem is, “Given a function f , find a function F such that $F' = f$.” As a simple illustration, suppose $f(x) = 8x^3$. In this case it is easy to find a function F that has f as its derivative. We know that differentiating a power of x reduces the exponent by 1 and therefore to obtain F we must increase the given exponent by 1. Thus $F(x) = ax^4$ for some number a . Differentiating, we obtain $F'(x) = 4ax^3$ and, in order for this to equal $f(x)$, a must equal 2. Consequently, the function F defined by $F(x) = 2x^4$ has the desired property. According to the next definition, F is called an *antiderivative* of f .

Definition (4.31)

A function F is an **antiderivative** of a function f if $F' = f$.

For convenience, we shall use the phrase “ $F(x)$ is an antiderivative of $f(x)$ ” synonymously with “ F is an antiderivative of f .” The domain of f

antiderivative will not usually be specified. It follows from the Fundamental Theorem of Calculus (5.21) in the next chapter that a suitable domain is any closed interval $[a, b]$ on which f is continuous.

Antiderivatives are never unique. Indeed, since the derivative of a constant is zero, it follows that if F is an antiderivative of f , so is the function G defined by $G(x) = F(x) + C$, for every number C . For example, if $f(x) = 8x^3$, then functions defined by expressions such as $2x^4 + 7$, $2x^4 - \sqrt{3}$, and $2x^4 + \frac{2}{5}$ are antiderivatives of f . The next theorem brings out the fact that functions of this type are the only possible antiderivatives of f .

Theorem (4.32)

If F_1 and F_2 are differentiable functions such that $F_1'(x) = F_2'(x)$ for all x in a closed interval $[a, b]$, then $F_2(x) = F_1(x) + C$ for some number C and all x in $[a, b]$.

Proof If we define the function g by

$$g(x) = F_2(x) - F_1(x)$$

then
$$g'(x) = F_2'(x) - F_1'(x) = 0$$

for all x in $[a, b]$. If x is any number such that $a < x \leq b$, then applying the Mean Value Theorem (4.12) to the function g and the closed interval $[a, x]$, there exists a number z in the open interval (a, x) such that

$$g(x) - g(a) = g'(z)(x - a) = 0 \cdot (x - a) = 0.$$

Hence $g(x) = g(a)$ for all x in $[a, b]$. Substitution in the first equation stated in the proof gives us

$$g(a) = F_2(x) - F_1(x).$$

Adding $F_1(x)$ to both sides we obtain the desired conclusion, with $C = g(a)$. □

If F_1 and F_2 are antiderivatives of the same function f , then $F_1' = f = F_2'$ and hence, by the theorem just proved, $F_2(x) = F_1(x) + C$ for some C . In other words, if $F(x)$ is an antiderivative of $f(x)$, then every other antiderivative has the form $F(x) + C$ where C is an arbitrary constant (that is, an unspecified real number). We shall refer to $F(x) + C$ as the **most general antiderivative** of $f(x)$.

Theorem (4.33)

If $f'(x) = 0$ for all x in $[a, b]$, then f is a constant function on $[a, b]$.

Proof Denote f by F_2 and let the function F_1 be defined by $F_1(x) = 0$ for all x . Since $F_1'(x) = 0$ and $F_2'(x) = f'(x) = 0$, we see that $F_1'(x) = F_2'(x)$ for all x in $[a, b]$. Applying Theorem (4.32), there is a number C such that $F_2(x) = F_1(x) + C$; that is, $f(x) = C$ for all x in $[a, b]$. This completes the proof. $\square$

Rules for derivatives may be used to obtain formulas for antiderivatives, as in the proof of the following important result.

Power Rule for Antidifferentiation
(4.34)

Let a be any real number, r any rational number different from -1 , and C an arbitrary constant.

$$\text{If } f(x) = ax^r, \text{ then } F(x) = \left(\frac{a}{r+1}\right)x^{r+1} + C$$

is the most general antiderivative of $f(x)$.

Proof It is sufficient to show that $F'(x) = f(x)$. This fact follows readily from the Power Rule for Derivatives (3.32), since

$$F'(x) = \left(\frac{a}{r+1}\right)(r+1)x^r = ax^r = f(x). \quad \square$$

Example 1 Find the most general antiderivative of

(a) $4x^5$ (b) $7/x^3$ (c) $\sqrt[3]{x^2}$.

Solution

(a) Using the Power Rule (4.34) with $a = 4$ and $r = 5$ gives us the antiderivative

$$\frac{4}{6}x^6 + C, \text{ or } \frac{2}{3}x^6 + C.$$

(b) Writing $7/x^3$ as $7x^{-3}$ and using Rule (4.34) with $a = 7$ and $r = -3$ leads to

$$\frac{7}{-2}x^{-2} + C, \text{ or } -\frac{7}{2x^2} + C.$$

(c) Since $\sqrt[3]{x^2} = x^{2/3}$ we may apply the Power Rule with $a = 1$ and $r = \frac{2}{3}$, obtaining

$$\frac{1}{\frac{5}{3}}x^{5/3} + C, \text{ or } \frac{3}{5}x^{5/3} + C. \quad \blacksquare$$

To avoid algebraic errors, it is *highly recommended* to check solutions of problems involving antidifferentiation by differentiating the final antiderivative. In each case the given expression should be obtained.

Theorem (4.35)

If $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$, respectively, then

- (i) $F(x) + G(x)$ is an antiderivative of $f(x) + g(x)$.
- (ii) $cF(x)$ is an antiderivative of $cf(x)$, where c is any real number.

Proof By hypothesis, $D_x[F(x)] = f(x)$ and $D_x[G(x)] = g(x)$. Hence

$$D_x[F(x) + G(x)] = D_x[F(x)] + D_x[G(x)] = f(x) + g(x).$$

This proves (i). To prove (ii) we merely note that

$$D_x[cF(x)] = cD_x[F(x)] = cf(x). \quad \square$$

Part (i) of the preceding theorem can be extended to any finite sum of functions. This fact may be stated as follows: “*An antiderivative of a sum is the sum of the antiderivatives.*” As usual, when working with several functions we assume that the domain is restricted to the intersection of the domains of the individual functions. A similar result is true for differences.

Example 2 Find the most general antiderivative $F(x)$ of

$$f(x) = 3x^4 - x + 4 + (5/x^3).$$

Solution Writing the last term of $f(x)$ as $5x^{-3}$ and applying the Power Rule for Antidifferentiation to each term gives us

$$F(x) = \frac{3}{5}x^5 - \frac{1}{2}x^2 + 4x - \frac{5}{2}x^{-2} + C.$$

It is unnecessary to introduce an arbitrary constant for each of the four antiderivatives, since they could be added together to produce the one constant C . ■

The next theorem provides formulas for antiderivatives of the sine and cosine functions.

Theorem (4.36)

In each of the following statements, $F(x)$ is the most general antiderivative of $f(x)$, and k is any nonzero real number.

- (i) If $f(x) = \sin kx$, then $F(x) = -\frac{1}{k} \cos kx + C$.
- (ii) If $f(x) = \cos kx$, then $F(x) = \frac{1}{k} \sin kx + C$.